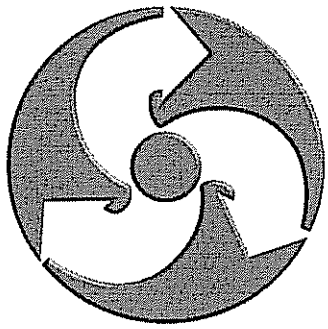


American National Standard for

Centrifugal Pumps

for Installation, Operation and Maintenance

ANSI/HI 1.4-2000

Hydraulic Institute

9 Sylvan Way
Parsippany, New Jersey
07054-3802
www.pumps.org

This page intentionally blank.

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

American National Standard for

Centrifugal Pumps

for Installation, Operation and Maintenance

Sponsor
Hydraulic Institute
www.pumps.org

Approved August 11, 1999
American National Standards Institute, Inc.



Recycled
paper

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

American National Standard

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus and other criteria for approval have been met by the standards developer.

Consensus is established when, in the judgement of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether he has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation of any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised or withdrawn at any time. The procedures of the American National Standards Institute require that action be taken periodically to reaffirm, revise, or withdraw this standard. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Published By

Hydraulic Institute
9 Sylvan Way, Parsippany, NJ 07054-3802

www.pumps.org

Copyright © 2000 Hydraulic Institute
All rights reserved.

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without prior written permission of the publisher

Printed in the United States of America

ISBN 1-880952-29-7

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Contents

		Page
Foreword		v
1.4	Installation, operation and maintenance	1
1.4.1	Scope	1
1.4.2	Pre-installation instructions	1
1.4.2.1	Unloading/receiving inspection	1
1.4.2.2	Storing equipment at site	1
1.4.2.3	Handling equipment and tools for installation	1
1.4.2.4	Manufacturer's instructions	1
1.4.2.5	Use of manufacturer's personnel	1
1.4.2.6	Site preparation	1
1.4.3	Installation – horizontal pumps	2
1.4.3.1	Alignment steps	2
1.4.3.2	Grouting	5
1.4.3.3	Pre-run stuffing-box steps	5
1.4.3.4	Final alignment – doweling	6
1.4.3.5	Suction and discharge piping - general comments	7
1.4.3.6	Suction piping requirements	7
1.4.3.7	Discharge piping requirements	7
1.4.3.8	Pre-run lubrication, pump and driver	7
1.4.3.9	Controls and alarms	8
1.4.4	Installation – vertical volute pumps	8
1.4.4.1	Configurations	8
1.4.4.2	Pump leveling/plumbness	8
1.4.4.3	Grouting	8
1.4.4.4	Suction piping requirements	8
1.4.4.5	Discharge piping requirements	8
1.4.4.6	Mounting driver/coupling and alignment	9
1.4.4.7	Pre-run stuffing-box steps (see Paragraph 1.4.3.3)	10
1.4.5	Operation of centrifugal pumps	10
1.4.5.1	System preparation	10
1.4.5.2	Bearing lubrication	11
1.4.5.3	Start-up	12
1.4.5.4	Operation considerations	13
1.4.6	Maintenance of centrifugal pumps	15
1.4.6.1	Wear/parts replacement	15
1.4.6.2	Noise	15

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

1.4.6.3 Troubleshooting 15
1.4.7 Pump vibration 17

Appendix A References 18

Appendix B Index 19

Figures

1.99 — Typical foundation bolt design 2
1.100 — Method of leveling 2
1.101 — Checking angular alignment 3
1.102 — Checking parallel alignment 3
1.103 — Dial indicator method of alignment 4
1.104 — Alignment of gear type coupling 5
1.105 — Alignment of spacer type couplings 5
1.106 — Vertical in-line centrifugal pump 8
1.107 — Vertical separately coupled clear liquid or non-clog pump 8
1.108 — Vertical clear liquid or non-clog flexible shafting driven pump 9
1.109 — Vertical wet pit submerged bearing or wet pit cantilever – clear liquid or non-clog pump 9
1.110 — Instrument locations 11
1.111 — Temperature versus time 12
1.112A — Reverse runaway speed ratio versus specific speed when head equals pump head at BEP (metric) 14
1.112B — Reverse runaway speed ratio versus specific speed when head equals pump head at BEP (US units) 15

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Foreword (Not part of Standard)

Purpose and aims of the Hydraulic Institute

The purpose and aims of the Institute are to promote the continued growth and well-being of pump manufacturers and further the interests of the public in such matters as are involved in manufacturing, engineering, distribution, safety, transportation and other problems of the industry, and to this end, among other things:

a) To develop and publish standards for pumps;

b) To collect and disseminate information of value to its members and to the public;

c) To appear for its members before governmental departments and agencies and other bodies in regard to matters affecting the industry;

d) To increase the amount and to improve the quality of pump service to the public;

e) To support educational and research activities;

f) To promote the business interests of its members but not to engage in business of the kind ordinarily carried on for profit or to perform particular services for its members or individual persons as distinguished from activities to improve the business conditions and lawful interests of all of its members.

Purpose of Standards

1) Hydraulic Institute Standards are adopted in the public interest and are designed to help eliminate misunderstandings between the manufacturer, the purchaser and/or the user and to assist the purchaser in selecting and obtaining the proper product for a particular need.

2) Use of Hydraulic Institute Standards is completely voluntary. Existence of Hydraulic Institute Standards does not in any respect preclude a member from manufacturing or selling products not conforming to the Standards.

Definition of a Standard of the Hydraulic Institute

Quoting from Article XV, Standards, of the By-Laws of the Institute, Section B:

"An Institute Standard defines the product, material, process or procedure with reference to one or more of the following: nomenclature, composition, construction, dimensions, tolerances, safety, operating characteristics, performance, quality, rating, testing and service for which designed."

Comments from users

Comments from users of this Standard will be appreciated, to help the Hydraulic Institute prepare even more useful future editions. Questions arising from the content of this Standard may be directed to the Hydraulic Institute. It will direct all such questions to the appropriate technical committee for provision of a suitable answer.

If a dispute arises regarding contents of an Institute publication or an answer provided by the Institute to a question such as indicated above, the point in question shall be referred to the Executive Committee of the Hydraulic Institute, which then shall act as a Board of Appeals.

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Revisions

The Standards of the Hydraulic Institute are subject to constant review, and revisions are undertaken whenever it is found necessary because of new developments and progress in the art. If no revisions are made for five years, the standards are reaffirmed using the ANSI canvass procedure.

Units of Measurement

Metric units of measurement are used; and corresponding US units appear in brackets. Charts, graphs and sample calculations are also shown in both metric and US units.

Since values given in metric units are not exact equivalents to values given in US units, it is important that the selected units of measure to be applied be stated in reference to this standard. If no such statement is provided, metric units shall govern.

Consensus for this standard was achieved by use of the Canvass Method

The following organizations, recognized as having an interest in the standardization of centrifugal pumps were contacted prior to the approval of this revision of the standard. Inclusion in this list does not necessarily imply that the organization concurred with the submittal of the proposed standard to ANSI.

A.R. Wilfley & Sons
ANSIMAG Inc.
Bechtel Corp.
Black & Veatch
Brown & Caldwell
Camp Dresser & McKee, Inc.
Carver Pump Company
Cheng Fluid Systems, Inc.
Crane Company, Chempump Div.
Cuma S.A.
Dean Pump Div., Metpro Corp.
DeWante & Stowell
Dow Chemical
EnviroTech Pumpsystems
Essco Pump Division
Exeter Energy Ltd. Partnership
Fairbanks Morse Pump Corp.
Fluid Sealing Association
Franklin Electric
GKO Engineering
Grundfos Pumps Corp.
Illinois Dept. of Transportation
IMC - Agrico Chemical Corp.
Ingersoll-Dresser Pump Company
ITT Fluid Handling (B & G)
ITT Fluid Technology
ITT Industrial Pump Group
Iwaki Walchem Corp.
J.P. Messina Pump & Hydr. Cons.
John Crane, Inc.
Krebs Consulting Service
KSB, Inc.
M.W. Kellogg Company
Malcolm Pirnie, Inc.
Marine Machinery Association
Marley Pump Company
Marshall Engineered Products Company
Montana State University
MWI, Moving Water Industries
Oxy Chem
Pacer Pumps
Paco Pumps, Inc.
Pinellas Cty, Gen. Serv. Dept.
The Process Group, LLC
Raytheon Engineers & Constructors
Reddy-Buffaloes Pump, Inc.
Robert Bein, Wm. Frost & Assoc.
Scott Process Equipment Corp.
Settler Supply Company
Skidmore
South Florida Water Mgmt. Dist.
Sta-Rite Industries, Inc.
Sterling Fluid Systems (USA), Inc.
Stone & Webster Engineering Corp.
Sulzer Bingham Pumps, Inc.
Summers Engineering, Inc.
Systecon, Inc.
Val-Matic Valve & Mfg. Corp.
Yeomans Chicago Corp.
Zoeller Engineered Products

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

1.4 Installation, operation and maintenance

1.4.1 Scope

This standard is for centrifugal and regenerative turbine pumps of all industrial/commercial types except vertical single and multistage diffuser types. It includes installation, operation and maintenance.

1.4.2 Pre-installation instructions

1.4.2.1 Unloading/receiving inspection

Immediately upon receipt of pump equipment, check carefully to see that all equipment has been received and is in good condition. Report any shortage or damage to the transportation company handling the shipment, noting the extent of damage or shortage on the freight bill and bill of lading. This should be done at once. Do not leave the unit exposed to construction or weather hazards where it may be damaged mechanically or environmentally.

1.4.2.2 Storing equipment at site

1.4.2.2.1 Short term

The pump and equipment, as shipped, have adequate protection for short-term (60 days) storage in a covered, dry and ventilated location prior to installation and start-up.

1.4.2.2.2 Long term

If it is anticipated that the equipment will be subject to extended storage, over 60 days, prior to installation, the manufacturer should be advised so that special protection can be provided for the equipment. Treatment of bearings, seals and machined surfaces with preservatives may be required. Periodic rotation of the pump and driver shaft is recommended.

1.4.2.3 Handling equipment and tools for installation

Overhead handling equipment, with proper slings or chains for setting pump and equipment, may be required. Lifting equipment should be carefully selected for safety with consideration for load carrying ability and compatibility with pump manufacturer's installation recommendations. Safety slings, wire rope or chain should be placed only at specified lift points and should not contact other points of the unit. A bubble level is needed for pump and driver alignment as well as for proper pump orientation.

1.4.2.4 Manufacturer's instructions

The service manual provided should be read thoroughly before installing or operating the equipment. These instructions should be retained for reference.

1.4.2.5 Use of manufacturer's personnel

It is recommended that the services of a manufacturer's erecting engineer be employed in installing and starting pump equipment which is of appreciable value or of a precision type. This is to ensure that the equipment is properly installed. The purchaser then is also afforded the opportunity of receiving adequate and authoritative instructions and seeing them implemented.

1.4.2.6 Site preparation

1.4.2.6.1 Protection against elements/environment

If there is any possibility of freezing when the pump is not running, the pump casing should be drained by removing the bottom drain plug. Another approach is to wrap or trace the pump with heating coils or electrical heating wire and cover the outside with insulating material. Provide adequate protection against other elements such as rain, dust, sand, sun, etc.

1.4.2.6.2 Foundation requirements (forces and mass requirements)

The foundation should be sufficiently substantial to absorb vibration (e.g., at least five times the weight of the pump unit) and to form a permanent, rigid support for the base plate. This is important in maintaining the alignment for a flexibly coupled unit. A concrete foundation on a solid base should be satisfactory. Foundation bolts of the proper size should be embedded in the concrete, located by a drawing or template. A pipe sleeve larger in diameter than the bolt should be used to allow movement for final positioning of the bolts (see Figure 1.99).

1.4.2.6.3 Access for maintenance

Pumps should have adequate access and working room for maintenance operations. Adequate overhead space for lifting devices and working clearance must be provided.

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

1.4.2.6.4 Location of unit

Suction and discharge pipes should be short and direct to minimize friction losses (see Section 1.4.3.5 – Suction and discharge piping).

1.4.3 Installation – horizontal pumps

1.4.3.1 Alignment steps

1.4.3.1.1 Alignment general

The following discussion of alignment applies primarily to horizontal, general servic

e, centrifugal pumps driven by an independent driver through a flexible coupling and with pump and driver mounted on a common base plate.

Pumps and drivers that are received from the factory with both machines mounted on a common base plate were aligned or checked for alignability before shipment. All base plates are flexible to some extent and, therefore, must not be relied upon to maintain the factory alignment. Realignment is necessary after the complete unit has been leveled, the grout has set and foundation bolts have been tightened. The alignment must be rechecked after the unit is piped and rechecked periodically as outlined in the following paragraphs. To facilitate field alignment, most manufacturers do not dowel the pump or drivers on the base plates before shipment, or at most, dowel the pump only.

When the drive is to be mounted at the place of installation, the pump is positioned and bolted to the base at the factory, but the holes for fastening the driver may not be provided.

1.4.3.1.2 Leveling pump/driver

When the unit is received with the pump and the driver mounted on the base plate, it should be placed on the foundation and the coupling halves disconnected. The coupling should not be reconnected until the alignment operations have been completed. The base plate should be supported on rectangular metal blocks and shims or on metal wedges having a small taper. The support pieces should be placed close to the foundation bolts (see Figure 1.100). On large units (e.g., over 3 m [10 ft] long), small jacks made of cap screws and nuts are very convenient.

In each case, the supports should be directly under the part of the base plate carrying the greatest weight and spaced closely enough to give uniform support. A spacing of 610 mm (24 inches) is suggested on medium size units (e.g., over 1.5 m [5 ft]). A gap of about 20 to 40 mm (0.75 to 1.5 inches) should be allowed between the base plate and the foundation for grouting.

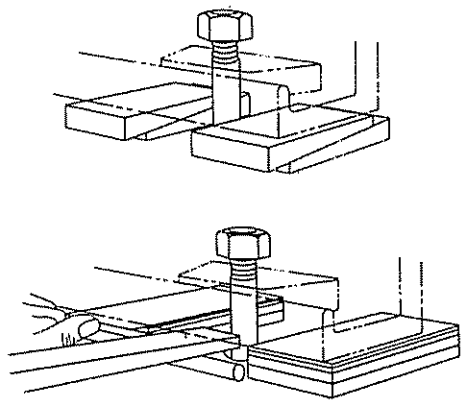


Figure 1.100 — Method of leveling

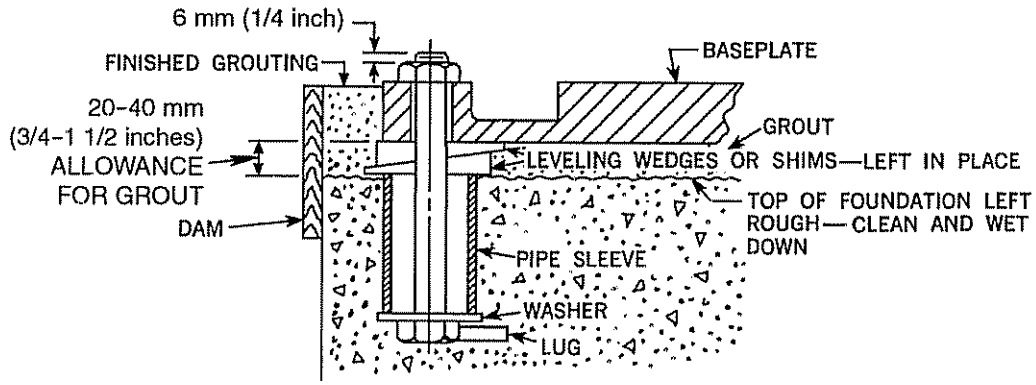


Figure 1.99 — Typical foundation bolt design

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Adjust the metal supports or wedges until the shafts of the pump and driver are level. Check the coupling faces as well as the suction and discharge flanges of the pump for horizontal or vertical position by means of a level. Make corrections if necessary by adjusting the supports or wedges under the base plate.

1.4.3.1.3 Shaft/coupling alignment

A flexible coupling is used to compensate for minor misalignment of the pump and driver shafts (refer to pump manufacturers' recommendations).

The main purpose of the flexible coupling is to compensate for minor temperature changes and to permit end movement of the shaft without interference with each other while transmitting power from the driver to the pump. A "hot" alignment may be required for hot pumpage, steam turbines, etc.

There are two forms of misalignment between the pump shaft and the driver shaft, as follows: Angular Misalignment - shafts with axis concentric but not parallel; and Parallel Misalignment - shafts with axes parallel but not concentric.

1.4.3.1.4 Straightedge method of alignment

The necessary tools for checking the alignment of a flexible coupling are a straightedge and a taper gauge or a set of feeler gauges.

The faces of the coupling halves should be spaced far enough apart so that they cannot strike each other when the driver rotor is moved axially toward the pump as far as it will go. A minimum dimension for the separation of the coupling halves and misalignment limits are specified by the manufacturer.

Proceed with checks for angular and parallel alignment by the following method only if satisfied that face and outside diameters of the coupling halves are square and concentric with the coupling bores. If this condition does not exist, the Alternate Method of Alignment described below is recommended. A check for angular alignment is made by inserting the taper gauge or feelers between the coupling faces at 90° intervals (see Figure 1.101).

The unit will be in angular alignment when the measurements show that the coupling faces are the same distance apart at all points.

A check for parallel alignment is made by placing a straightedge across both coupling rims at the top, bottom and at both sides. The unit will be in parallel alignment when the straightedge rests evenly across both coupling rims at all positions (see Figure 1.102).

Allowance may be necessary for coupling halves that are not of the same outside diameter.

Angular and parallel misalignment are corrected by means of shims under the motor mounting feet. After each change, it is necessary to re-check the alignment.

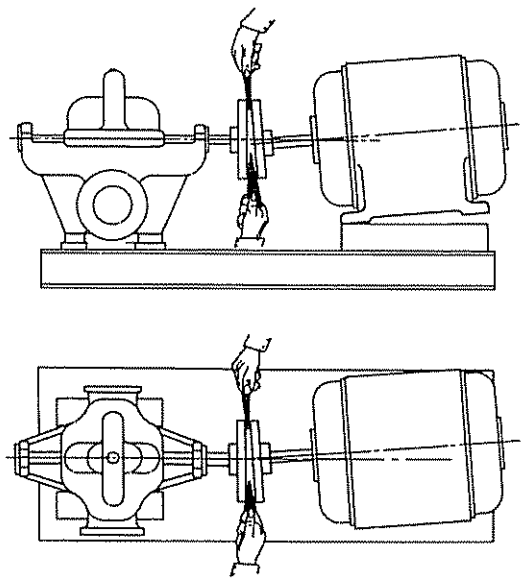


Figure 1.101 — Checking angular alignment

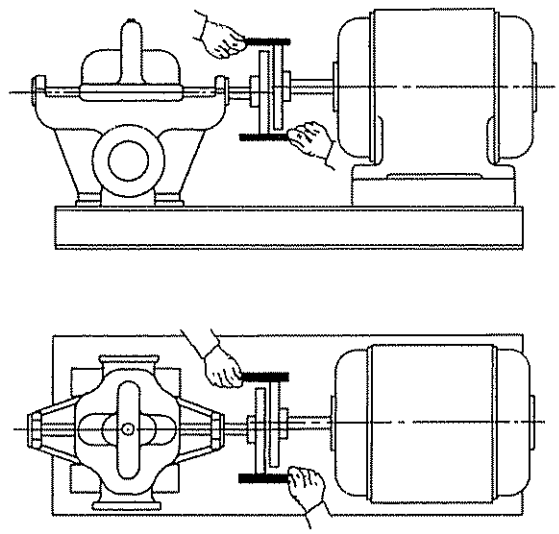


Figure 1.102 — Checking parallel alignment

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Adjustment in one direction may disturb adjustments already made in another direction. It is wise to start with shims under all motor feet so it can be raised or lowered during initial or subsequent aligning procedures.

When the driver is to be mounted on the base plate in the field, it is necessary to place the base plate with pump on the foundation, to level the pump shaft, to check the coupling faces, suction and discharge flanges for horizontal or vertical position, and to make any necessary corrective adjustments. Pads, if provided on the base plate for the driver, should be coated with chalk to facilitate marking the location of the bolt holes. Place the driver on the base plate so that the distance between the coupling halves is correct. The alignment of pump and driver coupling halves should then be checked and corrected. If the base is not pre-drilled, scribe on the base plate pads the circumference of the bolt holes in the driver feet. Remove the driver and drill and tap as required for bolts, allowing clearance for subsequent alignments. Replace driver on the base plate, check motor rotation, insert the bolts and align the driver before tightening. The subsequent procedures are the same as for factory-mounted units.

When units are aligned cold, it may be necessary to make allowance for the vertical rise of the driver and/or pump caused by heating. Finally adjust at operating temperature. Refer to instructions supplied by manufacturer for specific couplings; i.e., rubber shear types that the above instructions do not apply to.

1.4.3.1.5 Dial indicator method of alignment

A dial indicator can be used to attain more accurate coupling alignment. First rough align by using a straightedge, tapered gauge or feelers using the procedure indicated previously.

Fasten the indicator to the pump half of the coupling, with the indicator button resting on the other half coupling periphery (see Figure 1.103). Set the dial to zero, and chalk mark the coupling half beside where the button rests. Rotate both shafts by the same amount (i.e., all readings on the dial must be made with button beside the chalk mark).

The dial readings will indicate whether the driver has to be raised or lowered or moved to either side. After each adjustment, re-check both parallel and angular alignments. With this method, accurate alignment of shaft centers can be obtained, even where faces or outside diameters of the coupling halves are not square or concentric with the bores, provided all measurements for angular alignment are made between the same two points on the outside diameters. For angular alignment, change the indicator so it bears against the face of the same coupling half and proceed as described for parallel alignment. Gross deviations in squareness or concentricity, however, may cause problems due to coupling unbalance or abnormal coupling wear and may need to be corrected for reasons other than accomplishment of shaft alignment.

Example: If the dial reading at the starting point (either top or one side) is set to zero and the diametrically opposite reading at the bottom or other side shows a plus or minus reading of .5 mm (.020 inch), the driver must be raised or lowered by the use of suitable shims, or moved to one side or the other by half of this reading.

NOTE: Keep both shafts pressed radially to one side when taking concentricity readings and push both shaft ends as far apart as possible when checking for angular alignment.

1.4.3.1.6 Laser method of alignment

Laser detector systems are used to determine the extent of shaft misalignment by measuring the movement of a laser beam across the surface of a detector plate as the shafts are rotated. Several different systems of lasers and dectors are used, and the procedure for alignment is provided by the laser system's producer.

1.4.3.1.7 Alignment of gear type couplings

Gear type couplings are aligned in the same manner as outlined above. However, the coupling covers must be moved back out of the way and measurements made on the coupling hubs as shown on Figure 1.104.

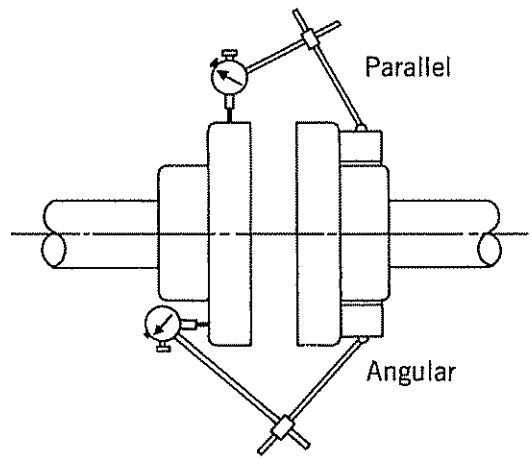


Figure 1.103 — Dial indicator method of alignment

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

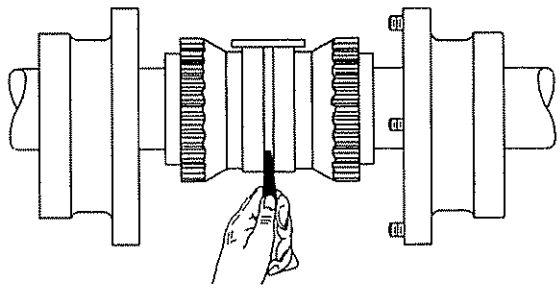


Figure 1.104 — Alignment of gear type coupling

1.4.3.1.8 Alignment of spacer type couplings

To align units with spacer coupling, remove the spacer between the pump and driver. Make a bracket, as shown in Figure 1.105, which can be fastened to one of the coupling halves and which is long enough to reach the other coupling half. Fasten this bracket to one coupling half and a dial indicator to the bracket arm so that the indicator button is in contact with the other coupling half as shown at A, Figure 1.105. Make a chalk mark on the coupling half beside where the button rests and set the dial to zero. To check for parallel alignment, rotate both shafts by the same amount (i.e., all readings are made with the button beside the chalk mark).

After parallel alignment has been obtained, change the indicator so it bears against the face of the same coupling half and follow the same procedure to check for angular alignment that was used for parallel alignment. If the shafts have end play, it is preferable to make this check of angular alignment by using inside micrometers as shown at B, Figure 1.105.

After final alignment is obtained, replace the spacer.

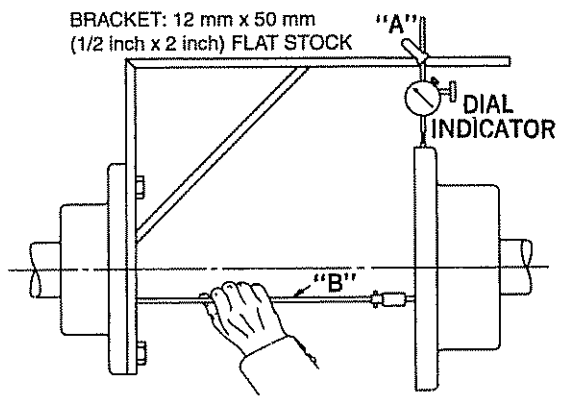


Figure 1.105 — Alignment of spacer type couplings

1.4.3.1.9 Special couplings

NOTE: On certain large units, limited end float couplings are used, and the instruction book furnished with such units should be consulted for the special alignment instructions that apply.

1.4.3.1.10 V-belt drive

Good alignment must be maintained for full power transmission, minimum vibration and long life. Parallel and angular alignment is verified by placing a straightedge or a string across the faces of the sheaves. Regardless of belt section used, the belt should never be allowed to bottom in the groove. This will cause the belts to lose the wedging action, and slippage can occur. Maintain proper belt tension. Excess tension can cause belt fatigue and hot bearings. Keep the belts clean. Belt dressing is not recommended, since it has only a temporary effect.

1.4.3.1.11 Coupling guard

Before proceeding, after alignment is complete, make sure that the coupling guard provided by the manufacturer is properly reinstalled.

1.4.3.2 Grouting

When the alignment is correct, the foundation bolts should be tightened evenly but not fully. The unit can then be grouted to the foundation. The base plate should be completely filled with grout, and it is desirable to grout the leveling pieces, shims or wedges in place. Vent holes are normally provided on all but the smaller bases to allow the air to be pushed out. Grout should come up to these vent holes. Foundation bolts should not be fully tightened until the grout is hardened, usually about 48 hours after pouring.

1.4.3.3 Pre-run stuffing-box steps

1.4.3.3.1 Packing

The stuffing-box may or may not be filled with packing before shipment. Instructions may be found with the box of packing. If not, the following may be used as a guide:

Carefully clean the stuffing-box. Make sure the packing rings are of proper section and length. When installed, the rings should butt tightly but not overlap at the joints. The joints should be staggered.

Where a lantern ring is required, be sure that sufficient packing is placed in back of the lantern ring so that the

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

liquid for sealing is brought in at the lantern ring and not at the packing.

The pipe supplying the sealing liquid should be fitted tightly so that no air enters. On suction lifts, a small quantity of air entering the pump at this point may result in loss of suction.

If the liquid to be pumped is dirty or gritty, clean sealing liquid should be piped to the stuffing-boxes in order to prevent damage to the packing and shaft sleeves. Clear sealing liquid is also required if the stuffing-box materials are not completely compatible with the pumpage. Sealing liquid should be at a pressure sufficient to ensure flow of clean liquid into the pump but not so high as to require excessive tightening of the packing.

When a pump is first put into operation, the packing should be left quite loose. After the pump has been found to operate properly, the stuffing-box gland may be tightened very slowly if the leakage is excessive. A leakage of about 60 drops per minute from the stuffing-box is necessary to provide lubrication and cooling.

When the leakage can no longer be controlled by adjusting the gland, all rings of packing should be replaced. The addition of a single ring to restore gland adjustment is not recommended.

If the pump is to be left idle for a long period of time, it is recommended that the packing be removed from the stuffing-box.

1.4.3.3.2 Mechanical seals

A mechanical seal consists of a rotating element and a stationary element. The sealing faces are highly lapped surfaces on materials selected for their low coefficient of friction and their resistance to corrosion by the liquid being pumped. The faces run with a very thin film of liquid between them. In addition, there must be a means of loading the seal. This is accomplished either with a spring (or springs) or with a flexible member of some organic material.

Since mechanical seals are made in a wide variety of designs, the instructions for the specific seal must be carefully studied and followed. A mechanical seal is a precision device and must be treated accordingly.

Mechanical seals normally require no adjustment during operation. Except for possible slight initial leakage, the seal should operate with negligible leakage. They should not be run dry. Seals may require a continuous supply of flush and/or cooling liquid. Where seal damage due to system uncleanliness is expected, it may be advisable to operate the pump with packing or temporary seals and sleeves until the system is clean and start-up problems are resolved. Packing or temporary seals are normally used on systems where the start-up pumpage is different from the final process pumpage and are replaced once the process pumpage is introduced.

1.4.3.3.3 Bushings

For applications where the consequences of leaking pumpage due to mechanical seal failure are particularly severe (e.g., flammable or toxic pumpage), the mechanical seal gland is fitted with a throttle bushing, the function of which is to minimize leakage upon complete failure of the seal. This bushing is non-sparking and may be either pressed into the gland or floating depending upon clearance requirements. Prior to pump start-up, the bushing should be checked to be sure it is free floating and is not rubbing on the shaft. If rapid thermal transients are expected during start-up, a cooling flush should be applied to the gland to prevent rubbing due to thermal conduction of the bushing.

1.4.3.4 Final alignment – doweling

1.4.3.4.1 Empty versus full pump

The introduction of pumpage into a piping system which is not well-designed or adjusted may cause strain on the pump, which can lead to misalignment or even impeller rub. For this reason, final alignment checks should be done with the system full.

1.4.3.4.2 Final check of alignment

After the grout has set and the foundation bolts have been properly tightened, the unit should be checked for parallel and angular alignment and, if necessary, corrective action taken. After the piping of the unit has been connected and the system filled, the alignment should be checked again.

The direction of rotation of the driver should be checked to make certain that it matches that of the pump. The direction of rotation of the pump is usually indicated by a direction arrow on the pump casing, or it may be obvious by the shape of the volute.

1.4.3.4.3 Hot versus cold liquid

In cases where the operating temperature of a pump and driver is expected to be considerably different, a misalignment can exist between the pump and driver

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

due to unequal thermal expansion. In most cases, this can be compensated by setting the hot running unit lower than the other. The pump manufacturer should be consulted for recommendations about the appropriate setting. In cases where exact alignment is critical, an alignment check should be repeated after both units have reached operating temperature. In cases where large swings in unit operating temperatures occur, some misalignment between pump and driver under some operating conditions is inevitable and should, therefore, be anticipated in the coupling selection as well as the alignment process.

1.4.3.5 Suction and discharge piping - general comments

1.4.3.5.1 Pipe support/anchors

Suction and discharge piping must be anchored, supported and restrained near the pump to avoid application of forces and moments to the pump except in certain cases, such as API 610 pumps, which are designed to absorb forces and moments. In calculating forces and moments, the weights of the pipe, contained fluid and insulation, as well as thermal expansion and contraction, must be considered.

1.4.3.5.2 Expansion joints and couplings

If an expansion joint is installed in the piping between the pump and the nearest anchor in the piping, a force equal to the area of the maximum ID of the expansion joint, times the pressure in the pipe, will be transmitted to the pump. Pipe couplings which are not axially rigid have the same effect. This force may be larger than can be safely absorbed by the pump or its support system. The Fluid Sealing Association "Technical Handbook, Non-Metallic Expansion Joints and Flexible Pipe Connectors" shows information on the design of expansion joints and the calculation of thrust.

The allowable forces and moments values that various pump types can withstand are found in ANSI/HI 9.6.2-2001, Centrifugal and Vertical Pumps - Allowable Nozzle Loads.

If it is necessary to use an expansion joint or non-rigid coupling, it is recommended that a pipe anchor be located between it and the pump. Note that an anchor provides axial restraint, whereas a pipe support or guide does not.

If a pipe anchor cannot be used, acceptable installations can also be obtained using tie rods across the expansion joint or flexible pipe coupling, provided careful attention is given to the design of the tie rods. The total axial rigidity of the tie rods, including their supporting brackets, shall equal that of the pipe, or as an alternate limit axial deflection to .125 mm (0.005 inches) when subjected to the maximum working pressure in the system. Many tie rod designs are inadequate for use near pumps because they are based on maximum allowable stress only, and deflection is not considered. In fact, some tie rod designs result in very high deflection values due to the use of high strength steel in the tie rods which allow high stress values. Since deflection is proportional to stress, these high allowable stresses result in high deflections.

1.4.3.5.3 Flat faced flanges

Cast iron and non-metallic pump flanges are usually made with flat faces. To avoid breaking the flange when tightening the bolting, mating pipe flanges should also have flat faces, and a full-face or scalloped gasket should be used.

1.4.3.6 Suction piping requirements

See ANSI/HI 9.8 - 1998, Pump Intake Design, Section 9.8.4, for an in-depth discussion of this subject.

1.4.3.7 Discharge piping requirements

A check valve and a shut-off valve should be installed in the discharge line. The check valve, placed between the pump and the shut-off valve, is for protecting the pump from reverse flow and excessive back pressure. The shut-off valve is used in priming and starting or stopping the pump for maintenance.

Except for axial flow and mixed flow pumps, it is advisable to close the shut-off valve before stopping the pump. This is especially important if there is no discharge check valve and the pump is operated against a high static head. If increasers are used on the discharge piping, they should be placed between the check valve and pump. If expansion joints are used, they should be placed on the pump side of the check valve.

1.4.3.8 Pre-run lubrication, pump and driver

Before starting, the pump and driver should be checked to see if:

a) Grease-lubricated bearings have been properly greased with manufacturer's recommended grease;

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

b) Oil-lubricated bearing reservoirs on pumps, drivers and/or gear boxes have been filled to required level with manufacturer's recommended oil;

c) Mechanical seal reservoirs are filled with proper isolating liquid;

d) Couplings are lubricated in accordance with vendor's instructions.

1.4.3.9 Controls and alarms

All control and alarm systems should be checked for proper installation, in accordance with the manufacturer's installation instructions. All alarm point settings should be verified.

1.4.4 Installation - vertical volute pumps

1.4.4.1 Configurations

There are four basic configurations of vertical volute pumps: In-Line (see Figure 1.106), Separately Coupled (see Figure 1.107), Flexible or Line Shaft (see Figure 1.108), and Wet Pit (see Figure 1.109).

1.4.4.2 Pump leveling/plumbness

After setting the pump feet, pit cover, or sole plate approximately 25 mm (1 inch) above the rough foundation, the pump should be leveled and/or checked for vertical plumbness. The pump and driver on Flexible or Line Shaft pumps (see Figure 1.108) should be aligned relative to each other, in accordance with the manufacturer's recommendations. On large pumps, the sole plates may be installed and grouted separately. The pump base is often used for sole plate alignment.

1.4.4.3 Grouting

After leveling and alignment, the pumps should be grouted following the manufacturer's instructions.

1.4.4.4 Suction piping requirements

See ANSI/HI 9.8-1998, Pump Intake Design.

1.4.4.5 Discharge piping requirements

See Section 1.4.3.7.

Vertical wet pit volute casing pumps (see Figure 1.109), due to their long overhang, are more sensitive to misalignment because of pipe strain, unless the discharge pipe is attached to the pit cover.

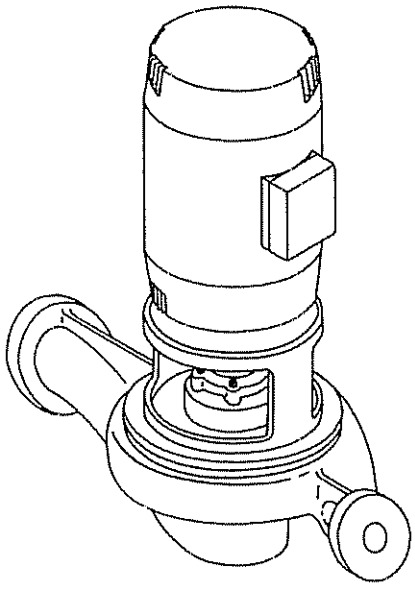


Figure 1.106 — Vertical in-line centrifugal pump

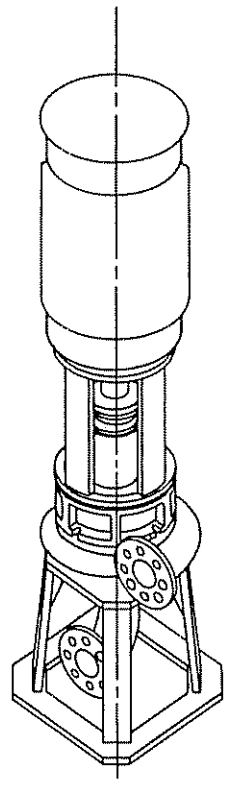


Figure 1.107 — Vertical separately coupled clear liquid or non-clog pump

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

1.4.4.6 Mounting driver/coupling and alignment

The driver may be factory or field mounted to a suitable support structure of sufficient strength and rigidity to carry the load and prevent excessive deflection as well as undesirable vibration.

A registered fit or other means, like radial jacking screws, will position and hold the motor in proper alignment. Dowel pins may be installed after alignment to fix the position.

Axially flexible couplings are provided for pumps with thrust bearings. Solid shaft couplings with adequate axial load capability are required where the axial (thrust) load is supported by the driver bearings.

1.4.4.6.1 Alignment

The alignment of vertical pumps is essentially the same as for horizontal pumps, when the pump is equipped with a thrust bearing and a flexible coupling is used. The pump has to be properly supported with all the anchor bolts tightened before checking the final alignment. (See Section 1.4.3.4).

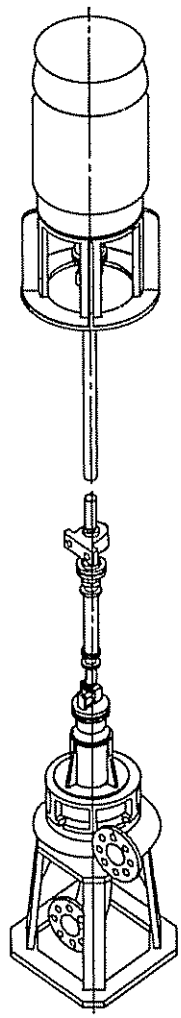


Figure 1.108 — Vertical clear liquid or non-clog flexible shafting driven pump

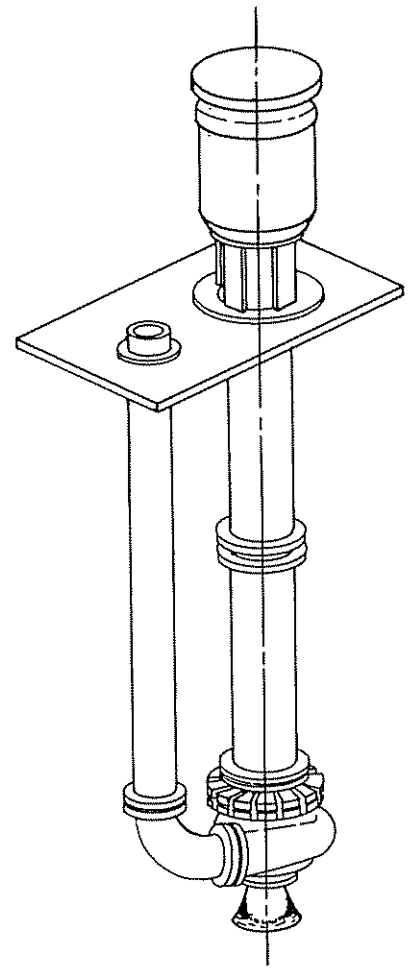


Figure 1.109 — Vertical wet pit submerged bearing or wet pit cantilever – clear liquid or non-clog pump

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

1.4.4.6.2 Solid shaft coupling

Before mounting the motor on the driver stand, check the rabbet fit (if furnished) and the mounting face on the motor for acceptable tolerance on run out and squareness, respectively, using a dial indicator mounted on the motor shaft. Next check the squareness of the face of the drive half coupling, usually mounted on the motor shaft with a light shrink fit and seated against a split ring, using a dial indicator on a firm base.

1.4.4.6.3 V-belt drive (see Section 1.4.3.1.10)

1.4.4.7 Pre-run stuffing-box steps (see Section 1.4.3.3)

1.4.5 Operation of centrifugal pumps

1.4.5.1 System preparation

1.4.5.1.1 Flushing

New and old systems should be flushed to eliminate all foreign matter. Heavy scale, welding spatter and wire, or other large foreign matter can clog the pump impeller, thereby reducing the rate of flow of the pump and causing cavitation or excessive vibration. Small size foreign matter will either clog tight clearances or erode them. Initially, the system should be flushed to waste; then a temporary strainer with a finer mesh than the permanent strainer should be put in place for additional flushing. When it appears that the flushing has adequately eliminated foreign matter, then a permanent strainer of a size recommended by the pump manufacturer should be put in place.

1.4.5.1.2 Filling

Vents should be located at the highest point, so entrained gases and air can escape. However, if the gases are flammable, toxic, or corrosive, they should be vented to an appropriate place to prevent harm to personnel or other parts of the system. Pipe hangers and anchors should be checked to make sure they are properly set to take the additional weight of the pumpage.

All drains should be closed when filling the system. Filling should be done slowly so that excessive velocities do not cause rotation of the pumping elements which may cause damage to the pump or its driver. The adequacy of the anchors and hangers may be checked by mounting a dial indicator off of any rigid structure not tied to the piping and setting the indicator button on the pump flange in the axial direction of the nozzle. If the indicator moves as the filling proceeds, the anchors and supports are not adequate or set properly and should be corrected.

1.4.5.1.3 Priming

The pump must not be run unless it is completely filled with liquid or, in the case of a vertical wet pit pump (see Figure 1.109), it is provided with the minimum required submergence, otherwise there is danger of damaging some of the pump components. Typically, wearing rings, bushings, seals or packings, and internal sleeve bearings depend on liquid for their lubrication and may seize if the pump is run dry. When required, pumps may be primed by one of the following methods:

1.4.5.1.3.1 Priming by ejector or exhauster

When steam, pressurized water, or compressed air is available, the pump may be primed by attaching an air ejector to the highest points in the pump casing. The ejector will remove the air from the pump and suction line, provided a tight valve is located in the discharge line close to the pump.

As soon as the air- or steam-driven ejector waste pipe exhausts water continuously, the pumps may be started. After starting, a steady stream of water from the waste pipe indicates that the pump is primed. If this stream of water is not obtained, the pump must be stopped at once and the process of priming repeated. A foot valve is unnecessary when this kind of device is used.

1.4.5.1.3.2 Priming with foot valve

When it is not practical to prime by ejector or exhauster, a foot valve in the suction inlet will prevent liquid from running out the suction inlet, and the pump can be completely filled with liquid from some outside source. Vents on top of the pump should be opened during filling to allow the air to escape. A tight foot valve will keep the pump constantly primed so that the pump may be used for automatic operation. The valve must be inspected frequently, however, to see that it does not develop leaks and thus allow the pump to be started dry.

1.4.5.1.3.3 Priming by vacuum pumps

The pump may also be primed by the use of a vacuum pump to exhaust the air from the pump casing and suction line. A wet vacuum pump is preferable, as it

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

will not be injured if water enters it. When a dry vacuum pump is to be used, the installation must be such as to prevent liquid being taken into the air pump. The manufacturer's instructions should be followed.

NOTE: Careful attention to the priming method at the time of installation may save later annoyance because of improper equipment or procedure.

1.4.5.1.4 Pre-filling

A self-priming pump must be pre-filled before it is started for the first time.

1.4.5.2 Bearing lubrication

1.4.5.2.1 Sleeve and tilting pad bearings

Before starting the pump, make certain that bearings and bearing housings are free of dirt and foreign substances which may have entered during shipment or installation. The bearing should then be filled with the lubricant, as recommended by the manufacturer. The lubricant should be changed when it becomes dirty or oxidized, or at recommended intervals, and the bearing cleaned out at the same time. Bearings should be examined periodically for wear.

When the pump is first started, the operator should make sure that the oil rings (where used) are turning freely. They may be inspected through the oil holes in the bearing caps in some designs.

If the pump is equipped with a forced-feed lubrication system, check the sight glasses to ensure oil is flowing. The bearings should be checked for overheating.

For special instructions, see the manufacturer's instructions book.

1.4.5.2.2 Rolling element bearings

Bearings should be lubricated at the time intervals and with the lubricant recommended by the manufacturer.

Heating of rolling element bearings often is caused by too much grease or oil, and careful inspection to determine the cause of trouble should be made before more lubricant is added.

Rolling element bearings should be cleaned by flushing with a low-volatility petroleum solvent.

1.4.5.2.3 Measurement of operating temperature of ball bearings

One of the following types of instruments: pyrometer, thermometer, or thermocouple, shall be placed on the outer surface perpendicular to the shaft centerline, over the center of the bearing(s) being recorded (see Figure 1.110). On pumps with horizontal shafts, the instrument shall be placed as close as possible to a vertical position. The instrument shall be placed between structural ribbing when ribbing is part of the design.

The pump shall be operated at rated conditions. When there are differences in specific gravity or viscosity between test and job site liquid, adjustment to test bearing temperatures must be agreed to by all parties prior to testing. Cooling plans should be installed and be operational if necessary to duplicate field conditions. This should be agreed to by all parties.

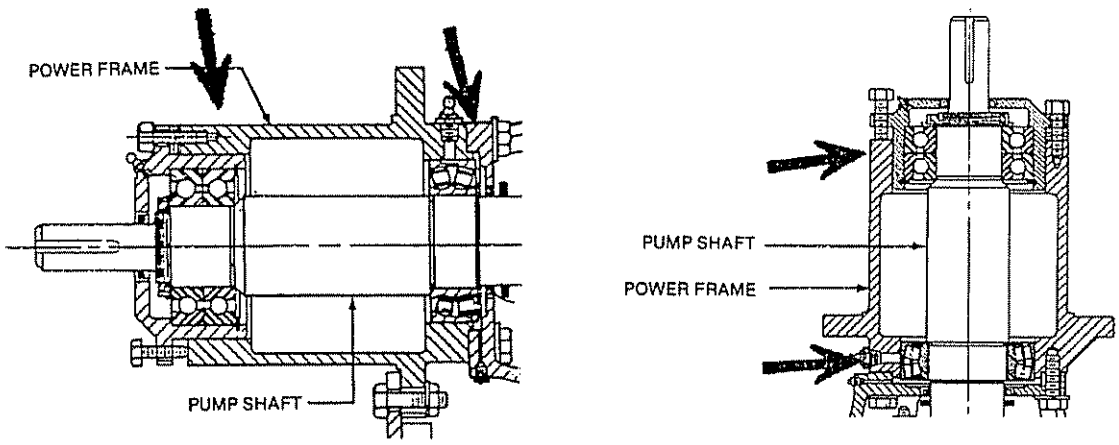


Figure 1.110 — Instrument locations

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Temperature readings shall be taken every 10 minutes for the first hour and every 15 minutes until stabilization. (Basic temperature stabilization usually occurs after the first 45 minutes. However, some bearings take up to 24 hours to stabilize and should be noted by all parties before the start of the test.) Stabilization is defined as three consecutively recorded readings taken over intervals of at least 15 minutes that fall within a 2°C (3°F) band when adjusted for a change in ambient temperature, if it occurs.

When testing with a TEFC motor, the air flow from the motor should be blocked from the bearing housing where testing is being conducted. Tests have shown that the motor air flow can cause as much as a 11°C (20°F) false temperature reading.

Similarly, the ambient air must be still. Circulating fans and opened windows can cause false readings.

1.4.5.2.3.1 Plotting data

A curve of temperature versus time can be plotted as shown in Figure 1.111, for analysis of the temperature stabilization.

1.4.5.2.3.2 Acceptance

The stabilization temperature is to be compared with the manufacturer's stated standards or that agreed upon by the customer and manufacturer. The manufacturer's standard will be based on experience with the type of pump, bearing material, bearing housing, construction pump materials, lubricant, speed and environmental application conditions.

The temperature limit with a mineral oil such as ISO VG 100 or mineral oil-based grease is dependent on the pump manufacturer. A maximum bearing housing skin temperature of 80°C (180°F) can be expected. Maximum temperature limits of synthetic grease- or synthetic oil-lubricated bearings may be higher,

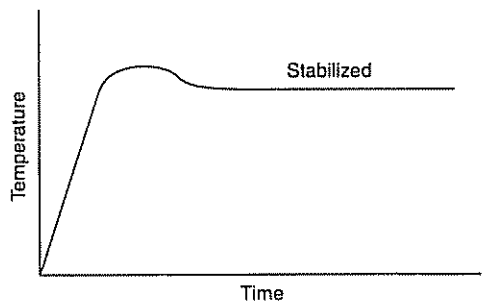


Figure 1.111 — Temperature versus time

depending on lubricant properties, construction and materials of the bearing and housing.

1.4.5.2.4 Sleeve bearings (for wet pit pump Figure 1.109)

There are several types of sleeve bearings used, all of which must be supplied with clean lubricant. For solids-laden pumpage, the lubricant also helps keep the solids out of the bearing. Unlike anti-friction bearings, sleeve bearings do not overheat from excess lubricant.

Fluted marine rubber bearings are commonly used and require a copious amount of clean water. The friction between dry rubber and the shaft is high compared to other dry bearing materials, so this bearing is more dependent on lubrication. Slurry applications require an external water source.

An electric solenoid valve can be used to start the water flow automatically before the pump starts and then shut off the water after the pump has stopped.

When metal sleeve bearings are grease-lubricated, automatic greasing methods are often used. However, care must be taken to avoid the use of grease when sleeve bearings are made of non-metallic, heat-retaining material such as rubber, teflon or carbon.

1.4.5.3 Start-up

1.4.5.3.1 Discharge valve position

A low or medium specific speed centrifugal pump (below values of 7000 [6000]), when primed and operated at full speed with discharge shutoff valve closed, requires much less power input than when it is operated at its rated flow rate and head with the valve open. For this reason, it is advantageous to close (or nearly close) the discharge valve when the pump is being started.

The input power required at shutoff on higher specific speed pumps (values above 7000 [6000]) may equal or exceed the power required with the discharge valve open. Starting with the discharge valve closed is therefore not recommended.

Brief shut-off operation of most centrifugal pumps is often necessitated by system start-up or shut-down requirements. Prolonged operation at shut-off is harmful because of:

a) Increased vibration level affecting the bearings, stuffing-boxes, or mechanical seals;

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

b) Increased radial thrust and resultant stresses in the shafts and bearings of centrifugal volute type pumps;

c) Heat buildup resulting in a dangerous temperature rise of the liquid being handled and of pump elements in contact with it;

d) Excessive cavitation and accompanying damage resulting from internal recirculation.

CAUTION: Operation of a centrifugal pump with the suction valve closed (discharge valve open) may cause serious damage and should not be attempted.

WARNING: Operation of a centrifugal pump with both valves closed for even brief periods of time is an unacceptable and dangerous practice. It can rapidly lead to a violent pump failure.

1.4.5.3.2 Rotation

Before starting, check the direction of rotation. The proper direction is usually indicated by a direction arrow on the pump casing or bearing housing. The proper rotation is also easily determined by observing the direction of the casing scroll and the position of the discharge nozzle. When electric motors are used as drivers, the rotation should be checked with the coupling disconnected. Check the manufacturer's start-up instructions (see Section 1.1.5.7).

CAUTION: Before starting, make sure adequate submergence is provided and the pump is primed.

1.4.5.3.3 Speed-torque curves

A plot of speed versus torque requirements during the starting phase of a centrifugal pump is sometimes checked against the speed versus torque curve of the driving motor. The driver must be capable of supplying more torque at each speed than required by the pump to bring the pump to rated speed. This condition is generally easily attainable with standard induction or synchronous motors but, under certain conditions, such as high specific speed pumps or reduced voltage starting, a motor with high pull-in torque may be required (see the Design and Application section, Section 1.3.4.1.9).

1.4.5.3.4 Motor starting considerations

See Section 1.3.4.1.5 Start-up and Shut-down.

1.4.5.3.5 Checking speed, rate of flow, pressure, power, vibration and leaks

Once the unit is energized, check operating speed, rate of flow, suction and discharge pressure, power input and vibration. While it may not be possible to repeat the factory performance exactly, initial field test data becomes a valuable baseline for future checking to determine possible wear and need for maintenance. Auxiliary piping and gasketed joints should be checked for leaks and proper make-up.

1.4.5.3.6 Final alignment check/factors causing misalignment

1.4.5.3.6.1 Dowelling (when desired or recommended by the manufacturer)

After the unit has been running for about one week, the coupling halves should be given a final check for misalignment caused by pipe strains or temperature strains and corrections made, if necessary. When the alignment is correct, both pump and driver may be dowelled to the base plate.

1.4.5.3.6.2 Misalignment causes

If the unit does not stay in alignment after being properly installed, the following are possible causes:

a) Setting, seasoning or springing of the foundation;

b) Pipe strains distorting or shifting the machine.

1.4.5.4 Operation considerations

1.4.5.4.1 Reduced flow/min. flow/bypass

Operation at low flows may result in any or all of the harmful effects listed in HI 1.3–2000, Section 1.3.4.2.3. If it becomes necessary to operate a pump for long periods at flows below the minimum continuous rate specified by the manufacturer, a bypass line should be installed from the pump discharge to the suction source. The bypass line should be sized so that the system flow plus the bypass flow is equal to or larger than the manufacturer's specified minimum.

1.4.5.4.2 Water hammer

See Section 1.3.4.1.4 in the Design and Application section.

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

1.4.5.4.3 Parallel and series operation

Pumps should not be operated in series or parallel unless specifically procured for this purpose, since serious equipment damage may occur.

For parallel operation, the pumps must have approximately matching head characteristics. Otherwise, the system operating head may exceed the shut-off head of one or more pumps, and result in the latter operating with zero output flow. This result would have the same effect as operating against a closed discharge valve.

In series operation, the pumps must have approximately the same flow characteristics. Since each pump will take suction from the preceding pumps, the stuffing-boxes and casing must be designed for the higher pressure, and the thrust bearing requirements may also increase (see HI 1.3–2000, Section 1.3.4.2.1).

1.4.5.4.4 Stopping unit/reverse runaway speed

A sudden power and check valve failure during pump operation against a static head will result in reverse pump rotation.

Vertical pump drivers can be equipped with non-reverse ratchets to prevent reverse rotation. However, their application is not always desirable, and a review should always be made with the manufacturer.

If the pump is driven by a prime mover offering little resistance while running backwards, the reverse speed may approach its maximum consistent with zero torque. This speed is called "reverse runaway speed." If the head, under which such operation may occur, is equal to or greater than that developed by the pump at its best efficiency point during normal operation, the runaway speed will exceed that corresponding to normal pump operation. This excess speed may impose high mechanical stresses on the rotating parts both of the pump and the prime mover and, therefore, knowledge of this speed is essential to safeguard the equipment from possible damage.

It has been found practical to express the runaway speed as a percentage of that during normal operation. The head consistent with the runaway speed is in this case assumed to be equal to that developed by the pump at the best efficiency point.

The ratio of runaway speed (n_{ro}) to normal speed (n) for single and double suction pumps varies with specific speed. This relationship is shown by Figure 1.112 A and B. The data shown should be used as a guide, since it is recognized that variations can be experienced with individual designs.

Transient conditions during which runaway speed may take place often result in considerable head variations due to surging in the pressure line. Because most pumping units have relatively little inertia, surging can

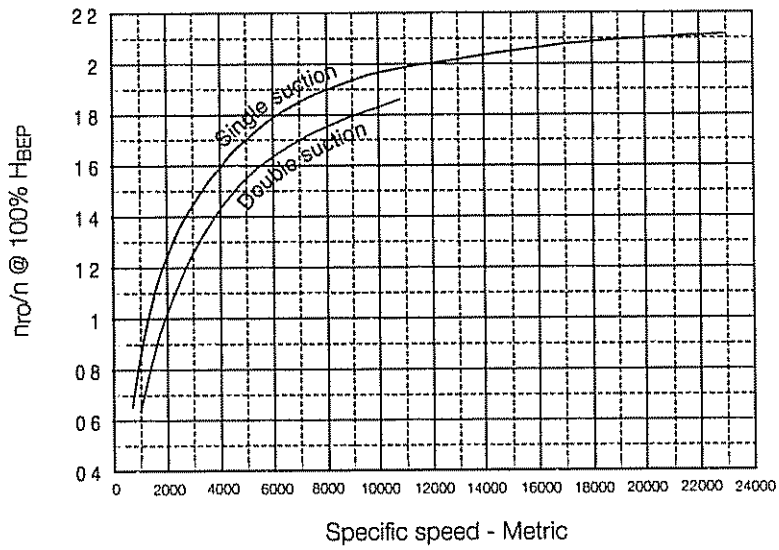


Figure 1.112A — Reverse runaway speed ratio versus specific speed when head equals pump head at BEP (metric)

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

cause rapid speed fluctuations. The runaway speed may, in such a case, be consistent with the highest head resulting from surging. Therefore, knowledge of the surging characteristic of the pipeline is essential for determining the runaway speed and this is particularly important in case of long lines.

1.4.6 Maintenance of centrifugal pumps

1.4.6.1 Wear/parts replacement

1.4.6.1.1 Wear rings

Pumps with shrouded (enclosed) impellers are commonly fitted with wear rings in the casing and possibly on the impellers. These wear rings make it possible to restore running clearances to reduce the quantity of liquid leaking from the high pressure side to the suction side. These rings depend on the liquid in the pump for lubrication. They will eventually wear so that the clearance becomes greater and more liquid passes into the suction. This rate of wear depends on the character of the liquid pumped. Badly worn wearing rings will result in severe degradation of pump performance, particularly on small pumps. See HI 1.3–2000, Section 1.3.4.3.

1.4.6.1.2 Wear plates

Pumps with open impellers on erosive type service are often equipped with wear plates fitted to the casing or suction cover. They perform the same function as wear rings. Some means of axial adjustment is usually provided in the pump design, so close running clearances can be maintained. However, when extreme or uneven wear takes place, the wear plate must be replaced.

1.4.6.2 Noise

Noise is undesired sound energy. A vibrating structure will excite the air surrounding it, resulting in noise. For example, a vibrating steel plate can be felt as a vibration and heard as a noise. Many cures for vibration problems likewise cure a noise problem.

Windage noise is another problem. Fans, couplings or any rotating elements are sources of windage noise.

Still another noise source is the liquid flow. The interaction of the liquid with the pump casing or piping will cause them to vibrate and, in turn, excite the air surrounding them. The more turbulent the flow, the greater the vibratory excitation and the louder the noise.

Further discussion on noise can be found in HI 1.3–2000, Section 1.3.4.5.

1.4.6.3 Troubleshooting

When investigating pump trouble at the job site, every effort must first be made to eliminate all outside influences. If the performance is suspect, the correct use

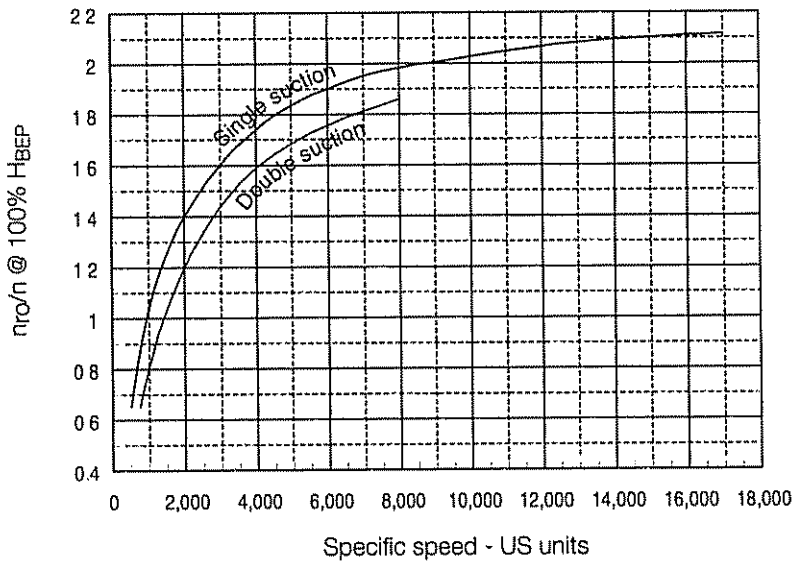


Figure 1.112B — Reverse runaway speed ratio versus specific speed when head equals pump head at BEP (US units)

Copyright © 2000 By Hydraulic Institute, All Rights Reserved

and accuracy of instruments should first be checked. In addition, note that pump performance is substantially affected by such fluid characteristics as temperature, specific gravity and viscosity.

1.4.6.3.1 Little or no discharge flow

Little or no discharge from a pump may be caused by any of the following conditions:

— pump not primed;

— speed too low;

— system head too high;

— suction lift higher than that for which pump is designed;

— impeller completely plugged;

— impeller installed backward;

— wrong direction of rotation;

— air leak in the suction line;

— air leak through stuffing-box;

— well draw-down below minimum submergence;

— pump damaged during installation;

— broken line shaft or coupling;

— impeller loose on shaft;

— closed suction or discharge valve.

1.4.6.3.2 Insufficient discharge flow or pressure

Insufficient discharge from a pump may be caused by any of the following conditions:

— air leaks in suction and stuffing-boxes;

— speed too low;

— system head higher than anticipated;

— insufficient NPSHA;

— foot valve too small;

— wearing rings worn;

— impeller damage;

— impeller(s) loose on shaft;

— vortex at suction supply;

— suction or discharge valve partially closed;

— impeller installed backwards;

— wrong direction rotation.

1.4.6.3.3 Loss of suction

Loss of suction may be caused by any of the following conditions:

— leaky suction line;

— water line to seal plugged;

— suction lift too high or insufficient NPSHA;

— air or gas in liquid;

— suction flange gasket defective;

— clogged strainer;

— excessive well draw-down.

1.4.6.3.4 Excessive power consumption

Excessive power consumption may be caused by any of the following conditions:

— speed too high;

— system head lower than rating, pumps too much liquid (radial and mixed flow pumps);

— system head higher than rating, pumps too little liquid (axial flow pumps);

— specific gravity or viscosity of liquid pumped is too high;

— shaft bent;

— rotating element binds;

— stuffing-boxes too tight;

— wearing rings worn;

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

— undersized motor cable;

— incorrect lubrication;

— mechanical seal power consumption;

— pump and motor operating in reverse direction;

— impeller mounted on shaft with inverted orientation.

1.4.7 Pump vibration

See ANSI/HI 9.6.4-2000, Centrifugal and Vertical Pumps–Allowable Vibration Levels, for an in-depth discussion of this subject.

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Appendix A

References

This appendix is not part of this standard, but is presented to help the user in considering factors beyond the standard sump design.

API-American Petroleum Institute

API Standard 610, *Centrifugal Pumps for General Refinery Service*

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Appendix B

Index

This appendix is not part of this standard, but is presented to help the user in considering factors beyond this standard.

Note: an *f.* indicates a figure, a *t.* indicates a table.

Alignment (horizontal pumps)
  angular, 3, 3*f.*
  and coupling guard, 5
  dial indicator method, 4, 4*f.*
  final, 6
  of full pump, 6
  of gear type couplings, 4, 5*f.*
  laser method, 4
  leveling pump and driver, 2
  misalignment causes, 13
  parallel, 3, 3*f.*
  shaft and coupling, 3
  of spacer type couplings, 5, 5*f.*
  of special couplings, 5
  straightedge method, 3
  and thermal expansion, 7
  of v-belt drive, 5
Alignment (vertical pumps), 9
  misalignment causes, 13

Bearing lubrication
  comparison of stabilization temperature with manufacturer's standards, 12
  measurement of operating temperature, 11, 12*f.*
  rolling element bearings, 11
  sleeve and tilting pad bearings, 11
  sleeve bearings, 12
  temperature vs. time, 12
Bushings, 6
Bypass, 13

Centrifugal pumps, 1
  horizontal pump installation, 2–8
  maintenance, 15
  operation, 10–15
  vertical volute pump installation, 8–10

Discharge valve position, 12
Dowelling, 13

Flow rate check, 13
Force and mass requirements, 1

Foundation
  bolts, 1, 2*f.*
  requirements, 1

Grouting
  horizontal pumps, 5
  vertical volute pumps, 8

Handling equipment, 1
Horizontal pump installation
  alignment, 2
  alignment of gear type couplings, 4, 5*f.*
  alignment of spacer type couplings, 5, 5*f.*
  alignment of special couplings, 5
  angular alignment, 3, 3*f.*
  controls and alarms, 8
  coupling guard, 5
  dial indicator method of alignment, 4, 4*f.*
  final alignment, 6
  final alignment check, 6
  full pump alignment, 6
  grouting, 5
  laser method of alignment, 4
  leveling pump and driver, 2, 2*f.*
  parallel alignment, 3, 3*f.*
  pre-run lubrication, 7
  shaft and coupling alignment, 3
  straightedge method of alignment, 3
  stuffing-box bushings, 6
  stuffing-box mechanical seals, 6
  stuffing-box packing, 5
  stuffing-box steps, 5
  suction and discharge pipes, 7
  thermal expansion and alignment, 7
  v-belt drive, 5

Installation
  horizontal pumps, 2–8
  tools, 1
  vertical volute pump, 8–10

Leak check, 13
Location of unit, 2

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

Long-term storage, 1

Maintenance
  access, 1
  excessive power consumption, 16
  insufficient discharge flow or pressure, 16
  little or no discharge flow, 16
  loss of suction, 16
  noise, 15
  troubleshooting, 15
  wear plates, 15
  wear rings, 15
Manufacturer's erecting engineer, 1
Manufacturer's instructions, 1
Mechanical seals, 6

Noise, 15

Operation
  bearing lubrication, 11
  bypass, 13
  filling, 10
  flushing, 10
  minimum flow, 13
  parallel, 14
  pre-filling, 11
  priming, 10
  reduced flow, 13
  reverse runaway speed, 14, 14*f*
  series, 14
  start-up, 12
  system preparation, 10
  water hammer, 13

Parallel operation, 14
Power check, 13
Pre-installation
  foundation bolts, 1, 2*f*
  foundation requirements, 1
  handling equipment, 1
  installation tools, 1
  location of unit, 2
  long-term storage, 1
  maintenance and repair access, 1
  manufacturer's erecting engineer, 1
  manufacturer's instructions, 1
  protection against elements and environment, 1
  receiving inspection, 1
  short-term storage, 1
  site preparation, 1
  suction and discharge pipes, 2
Pressure check, 13
Priming, 10
  by ejector or exhauster, 10
  with foot valve, 10
  by vacuum pumps, 10
Pump vibration, 17

Receiving inspection, 1
Regenerative turbine pumps, 1
Reverse runaway speed, 14, 14*f*
Rotation, 13

Series operation, 14
Short-term storage, 1
Site preparation
  foundation bolts, 1, 2*f*
  foundation requirements, 1
  location of unit, 2
  maintenance access, 1
  protection against elements and environment, 1
  suction and discharge pipes, 2
Speed check, 13
Speed-torque curves, 13
Start, 12
Start-up
  discharge valve position, 12
  dowelling, 13
  final alignment check, 13
  flow rate check, 13
  leak check, 13
  misalignment causes, 13
  motor, 13
  power check, 13
  pressure check, 13
  rotation, 13
  speed check, 13
  speed-torque curves, 13
  vibration check, 13
Storage, 1
Stuffing box
  bushings, 6
  mechanical seals, 6
  packing, 5
Suction and discharge pipes, 2
  expansion joints and couplings, 7
  flat faced flanges, 7
  pipe support and anchors, 7
  requirements, 7, 8
System preparation
  filling, 10
  flushing, 10
  pre-filling, 11
  priming, 10

Temperature
  limits, 12
  measurement, 11
  vs. time, 12, 12*f*

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.

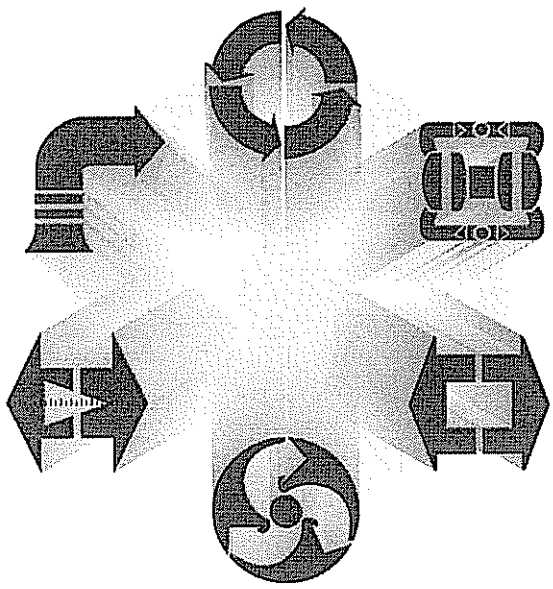
Troubleshooting
  excessive power consumption, 16
  insufficient discharge flow or pressure, 16
  little or no discharge flow, 16
  loss of suction, 16

Vertical diffuser pumps (excluded), 1
Vertical volute pump installation
  alignment, 9
  configurations, 8
  couplings, 9, 10
  discharge piping requirements, 8
  flexible or line shaft configuration, 8
  grouting, 8
  in-line configuration, 8
  mounting to support structure, 9
  pump leveling and plumbness, 8
  separately coupled configuration, 8
  solid shaft coupling, 10
  stuffing-box steps, 10
  suction piping requirements, 8
  v-belt drive, 10
  wet pit configuration, 8
Vibration, 17
Vibration check, 13

Water hammer, 13
Wear plates, 15
Wear rings, 15

Copyright © 2000 By Hydraulic Institute, All Rights Reserved

Copyright © 2000 By Hydraulic Institute, All Rights Reserved.



Master Index

for Complete Set:
ANSI/HI Pump Standards
2002 Release

Hydraulic INSTITUTE

9 Sylvan Way
Parsippany, New Jersey
07054-3802
www.pumps.org

This page intentionally blank.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Hydraulic Institute Standards

Index of Complete Set: 2002 Release

This index is not part of any standard, but is presented to help the user in considering factors beyond the standards.

Note: Bold numbers indicate the standard number, non-bold numbers indicate the page number; an f. indiactes a figure, a t. indicates a table.

Abrasion, **9.1-9.5:** 11
  severe, **9.1-9.5:** 15
Abrasion resistant cast irons, **9.1-9.5:** 19
Acceleration head, **6.1-6.5:** 25–27, **8.1-8.5:** 12
Acceleration pressure, **6.1-6.5:** 25–27, **8.1-8.5:** 12
Accessory equipment, **3.1-3.5:** 41–44
Accumulator, **9.1-9.5:** 3
Acoustical calibration, **9.1-9.5:** 50
Actuating mechanism *See* Valve gear
Additives in liquid, **9.6.1:** 4
Adhesives, **9.1-9.5:** 26
Adjustment factors for alternate designs, **3.1-3.5:** 42t.
Affinity laws, **1.6:** 16, **11.6:** 28
Air entrainment, **4.1-4.6:** 20
Air gap, **4.1-4.6:** 7, **5.1-5.6:** 12
Airborne noise, **3.1-3.5:** 28
Airborne sound measurement, **9.1-9.5:** 50
  6 dB drop-off, **9.1-9.5:** 50
  acoustical calibration, **9.1-9.5:** 50
  averaging of readings, **9.1-9.5:** 52
  A-weighted sound level, **9.1-9.5:** 50, 51, 52
  background sound level and corrections, **9.1-9.5:** 52, 54f.
  calculation and interpretation of readings, **9.1-9.5:** 52
  caution (extraneous noise), **9.1-9.5:** 51
  data presentation, **9.1-9.5:** 52
  graphic plot, **9.1-9.5:** 52
  instrumentation, **9.1-9.5:** 50
  measurements and technique, **9.1-9.5:** 51
  microphone locations, **9.1-9.5:** 50, 51,54f.–60f.
  microphone system, **9.1-9.5:** 50
  octave-band analyzer, **9.1-9.5:** 50
  octave-band sound pressure levels, **9.1-9.5:** 50, 51, 52
  operation of pumping equipment, **9.1-9.5:** 50
  primary microphone location, **9.1-9.5:** 51
  recorders, **9.1-9.5:** 50
  reference sound source, **9.1-9.5:** 50
  sound level meter, **9.1-9.5:** 50
  test data tabulation, **9.1-9.5:** 52
  test environment, **9.1-9.5:** 50
  test reports, **9.1-9.5:** 52, 53f.
Alarm limit (defined), **9.6.5:** 2
Alignment, **3.1-3.5:** 36, 37f.
  and elevated temperatures, **3.1-3.5:** 38
Alignment (horizontal pumps)
  angular, **1.4:** 3, 3f.
  and coupling guard, **1.4:** 5
  dial indicator method, **1.4:** 4, 4f.
  final, **1.4:** 6
  of full pump, **1.4:** 6
  of gear type couplings, **1.4:** 4, 5f.
  laser method, **1.4:** 4
  leveling pump and driver, **1.4:** 2
  misalignment causes, **1.4:** 13
  parallel, **1.4:** 3, 3f.
  shaft and coupling, **1.4:** 3
  of spacer type couplings, **1.4:** 5, 5f.
  of special couplings, **1.4:** 5
  straightedge method, **1.4:** 3
  and thermal expansion, **1.4:** 7
  of v-belt drive, **1.4:** 5
Alignment (vertical pumps), **1.4:** 9
  misalignment causes, **1.4:** 13
All bronze pumps, **9.1-9.5:** 16, 17
All iron pumps, **9.1-9.5:** 16, 17
All stainless steel pumps, **9.1-9.5:** 16, 17
Alleviator, **9.1-9.5:** 3
Allowable operating range, **1.1-1.2:** 58, **2.1-2.2:** 22
Allowable operating region, **9.6.3:** 1
  centrifugal pumps, **9.6.3:** 5, 5f., 6f., 7f.
  factors affecting, **9.6.3:** 1
  large boiler feed pumps, **9.6.3:** 8
  vertical turbine pumps, **9.6.3:** 8, 8t.
Alnico, **4.1-4.6:** 8, **5.1-5.6:** 14
Aluminum and aluminum alloys, **9.1-9.5:** 22
Aluminum bronze, **9.1-9.5:** 21
American National Metric Council, **9.1-9.5:** 7
American Society for Testing and Materials, **9.1-9.5:** 11
Angular misalignment, **3.1-3.5:** 36, 37, 37f., 38
ANSI/ASME B73.1M, **9.6.2:** 1, 3, 4, 5t., 6t., 7t.
  1.5x1-8 CF8M (Type 316) pump
    combined axis deflection evaluation, **9.6.2:** 25
    derating loads, **9.6.2:** 22
    individual nozzle load evaluation, **9.6.2:** 22

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

ANSI/ASME B73 (continued)
individual nozzle load evaluation (new loads), **9.6.2:** 23
nozzle stress, bolt stress and pump slippage, **9.6.2:** 23
nozzle stress, bolt stress and pump slippage on baseplate evaluation (new loads), **9.6.2:** 24
Y-axis deflection evaluation (new loads), **9.6.2:** 24
Z-axis deflection evaluation (new loads), **9.6.2:** 24
3x1.5-13 Alloy 20 pump
combined axis deflection evaluation, **9.6.2:** 27
derating loads, **9.6.2:** 25
nozzle stress, bolt stress and pump slippage, **9.6.2:** 26
Y-axis deflection evaluation, **9.6.2:** 27
Z-axis deflection evaluation, **9.6.2:** 27
ANSI/ASME B73.2M, **9.6.2:** 11
ANSI/ASME B73.3M, **9.6.2:** 1, 3, 4
ANSI/ASME B73.5M, **9.6.2:** 1, 3
1.5x1-8 pump
derating loads, **9.6.2:** 28
individual nozzle load evaluation, **9.6.2:** 29
AOR *See* Allowable operating region
Apparent viscosity, **3.1-3.5:** 19, **6.1-6.5:** 27, **9.1-9.5:** 5
Application guidelines, **5.1-5.6:** 23–26, **8.1-8.5:** 12
Applications, **4.1-4.6:** 11
factors in selecting rotary sealless pumps, **4.1-4.6:** 12–16
stripping, **4.1-4.6:** 15
Approach pipe lining, **9.8:** 60
ASME B73.2M
4030/28 Alloy 20 pump
derating loads, **9.6.2:** 31
individual nozzle load evaluation, **9.6.2:** 31
size 2015/17 CF8M (Type 316) pump
derating loads, **9.6.2:** 30
nozzle load evaluation, **9.6.2:** 30
ASTM *See* American Society for Testing and Materials
Atmospheric head, **1.1-1.2:** 57, **1.6:** 5, **2.1-2.2:** 22, **2.6:** 6, **11.6:** 5
Austenitic ductile iron, **9.1-9.5:** 19
Austenitic gray cast iron, **9.1-9.5:** 18
Auxiliary drive (steam) valve, **8.1-8.5:** 4
Auxiliary piping, **5.1-5.6:** 22
A-weighted sound level, **9.1-9.5:** 50, 51, 52
Axial flow impellers, **2.1-2.2:** 3, 11f.
Axial flow pumps, **1.1-1.2:** 4, 4f.
impeller between bearings–separately coupled–single stage axial (horizontal) split case, **1.1-1.2:** 46f.
impeller between bearings–separately coupled–single stage axial (horizontal) split case pump on base plate, **1.1-1.2:** 45f.
separately coupled single stage–(horizontal) split case, **1.1-1.2:** 16f.
separately coupled single stage–horizontal, **1.1-1.2:** 15f.
separately coupled–mulitstage–(horizontal) split case, **1.1-1.2:** 18f.
Axial load, **5.1-5.6:** 13
Axial split case pumps
casing hold-down bolts, **9.6.2:** 15
coordinate system, **9.6.2:** 16f.
driver and pump, **9.6.2:** 15
limiting factors, **9.6.2:** 15
nozzle loads, **9.6.2:** 15, 16f.
Axial thrust
calculation, **2.3:** 41f., 41
terminology, **2.3:** 40
vs. rate of flow, **2.3:** 42, 43f.
with various impeller and shaft configurations, **2.3:** 38, 38f., 39f., 40f.
Axial thrust (for enclosed impellers for volute pump), **1.3:** 60–63

Balancing *See* Rotor balancing
Bare rotor
multistage, axially split, single or double suction centrifugal pumps, **1.1-1.2:** 25
single stage, axially (horizontally) split, single or double suction centrifugal pump, **1.1-1.2:** 25
Barometric pressure, **6.1-6.5:** 22, 23t., **8.1-8.5:** 9
and altitude, **8.1-8.5:** 9, 10t.
Barrel or can (lineshaft) pumps, **2.1-2.2:** 1, 8f.
Barrel pumps *See* Can pumps
Baseline, **9.6.5:** 1
Baseplates (horizontal centrifugal pumps), **1.3:** 78
defined, **1.3:** 79
exterior edges, **1.3:** 85
fasteners, **1.3:** 81, 84
free standing baseplate, **1.3:** 79, 79f.
functional requirements, **1.3:** 79
grout holes, **1.3:** 84
grouted baseplate, **1.3:** 79, 79f., 85
high-energy pump, **1.3:** 79
lifting base assembly, **1.3:** 85
motor mounting pads, **1.3:** 80t., 81, 81f.
mounting blocks, **1.3:** 79, 85, 85f.
mounting pads, **1.3:** 79, 81f.
mounting surface flatness, **1.3:** 80t., 81, 81f.
mounting surface height, **1.3:** 80t., 81, 81f.
rigidity, **1.3:** 84
shims, **1.3:** 79f., 79, 81
stress levels, **1.3:** 81–84
sub base, **1.3:** 79f., 79, 85
superstructure, **1.3:** 79f., 79
support and anchoring, **1.3:** 86, 86f.
tolerancing, **1.3:** 80, 80t.
torsional stiffness, **1.3:** 86, 86f.
Bearing, **3.1-3.5:** 4, **9.1-9.5:** 3

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Bearing failure mode causes and indicators, **9.6.5:** 18, 21t.
Bearing life, **9.6.3:** 2
Bearing lubrication
  comparison of stabilization temperature with manufacturer's standards, **1.4:** 12
  measurement of operating temperature, **1.4:** 11, 12f.
  rolling element bearings, **1.4:** 11
  sleeve and tilting pad bearings, **1.4:** 11
  sleeve bearings, **1.4:** 12
  temperature vs. time, **1.4:** 12
Bearing materials, **4.1-4.6:** 15
Bearing wear monitoring, **9.6.5:** 14
  acoustic detection, **9.6.5:** 15
  bearing materials and characteristics, **9.6.5:** 14
  carbon bearing wear characteristics, **9.6.5:** 14
  contact detection, **9.6.5:** 15
  contact or continuity switch, **9.6.5:** 15
  control limits, **9.6.5:** 15
  frequency, **9.6.5:** 15
  indicators, **9.6.5:** 24
  means, **9.6.5:** 14
  power monitor, **9.6.5:** 15
  silicon carbide bearing wear characteristics, **9.6.5:** 14
  temperature probe, **9.6.5:** 15
  vibration sensor, **9.6.5:** 15
  wear detection methods, **9.6.5:** 14
Bearings
  adjusted rating life, **1.3:** 74, 75
  axial load, **1.3:** 74
  basic dynamic radial load rating, **1.3:** 74
  basic rating life, **1.3:** 74
  dynamic equivalent radial load, **1.3:** 74
  external, **5.1-5.6:** 19
  grease, **1.3:** 65
  housing closures, **1.3:** 70
  impeller mounted between, **1.3:** 58, 72f.
  impeller overhung from, **1.3:** 58, 70, 71f.
  internal, **5.1-5.6:** 18
  labyrinths, **1.3:** 70
  life, **1.3:** 74
  lubrication, **1.3:** 65–67
  oil lubrication, **1.3:** 65
  operating temperature, **1.3:** 75
  product lubrication, **1.3:** 66t., 67
  radial load, **1.3:** 74
  rating life, **1.3:** 74
  reference and source material, **5.1-5.6:** 38
  reliability, **1.3:** 74
  rolling element, **1.3:** 64, 64t.
  sleeve, **1.3:** 64
  types, **1.3:** 64
BEP *See* Best efficiency point
Best efficiency point, **1.1-1.2:** 58, **1.3:** 56, **1.6:** 1, **2.1-2.2:** 22, **2.3:** 17, **2.6:** 1, **9.6.1:** 2, **9.6.3:** 1, **11.6:** 3
Body, **3.1-3.5:** 4, **9.1-9.5:** 3
Boiler circulating pumps, **1.3:** 10
Boiler feed booster pumps, **1.3:** 9
Boiler feed pumps, **1.3:** 8
Bolt-proof load, **5.1-5.6:** 15
Booster service, **1.3:** 1, **2.3:** 1
Bowl assembly efficiency, **2.1-2.2:** 23, **2.6:** 7
  calculation, **2.6:** 16
Bowl assembly input power, **2.1-2.2:** 23, **2.6:** 7
Bowl assembly output power, **2.6:** 7
Bowl assembly performance test, **2.6:** 11, 11f.
Bowl assembly total head, **2.1-2.2:** 22, **2.6:** 6
  calculation, **2.6:** 15
  measurement, **2.6:** 29f., 29
Brass
  leaded red, **9.1-9.5:** 20
  yellow, **9.1-9.5:** 20
Bronze
  all bronze pumps, **9.1-9.5:** 16, 17
  aluminum, **9.1-9.5:** 21
  leaded nickel bronze, **9.1-9.5:** 21
  silicone, **9.1-9.5:** 20
  specific composition bronze pumps, **9.1-9.5:** 16, 17
  tin, **9.1-9.5:** 20
Bronze fitted pumps, **9.1-9.5:** 16, 17
Building services pumping systems, **9.6.1:** 9
Bull ring packing, **6.1-6.5:** 63, 63f.
Burst disc (rupture), **9.1-9.5:** 3
Bushings, **1.4:** 6
Bypass, **1.4:** 13
Bypass piping, **9.1-9.5:** 3

Calibrated electric meters and transformers, **1.6:** 31
Can intakes
  closed bottom can, **9.8:** 13, 13f.
  design considerations, **9.8:** 11
  open bottom can intakes, **9.8:** 12, 12f.
Can pumps, **2.3:** 1, 3f.
Can pumps *See* Barrel or can (lineshaft) pumps
Canned motor pumps, **5.1-5.6:** 1
  canned motor temperature, **5.1-5.6:** 26
  close coupled end suction, **5.1-5.6:** 1, 3f.
  close coupled in-line, **5.1-5.6:** 1, 4f.
  defined, **5.1-5.6:** 13
  driver sizing, **5.1-5.6:** 25
  eddy currents, **5.1-5.6:** 13
  horizontal mounting base, **5.1-5.6:** 21
  induction motor, **5.1-5.6:** 13
  integral motors, **5.1-5.6:** 19
  location and foundation, **5.1-5.6:** 32
  locked rotor torque, **5.1-5.6:** 13

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Canned motor pumps (continued)
maintenance, **5.1-5.6:** 35
motor insulation, **5.1-5.6:** 13
motor winding integrity test, **5.1-5.6:** 40
motor winding temperature test, **5.1-5.6:** 40
separated pump and motor, **5.1-5.6:** 1, 5f.
starting torque, **5.1-5.6:** 13
submerged mounting, **5.1-5.6:** 21
vertical submerged canned motor pump, **5.1-5.6:** 1, 6f.
Canvas packing, **8.1-8.5:** 17
Capacity, **1.1-1.2:** 55, **1.6:** 3
Capacity *See* Pump rate of flow
Capacity *See also* Rate of flow (capacity)
Carbon, **9.1-9.5:** 26
Carbon and low alloy steels, **9.1-9.5:** 19
Carbon steel, **9.1-9.5:** 19
Casing, **3.1-3.5:** 4, **5.1-5.6:** 18
Casing rotation, **1.1-1.2:** 26
Casing types, **1.3:** 76
Casing working pressure, **1.1-1.2:** 60
Cavitation, **3.1-3.5:** 23, **9.6.1:** 3, 6, 10
damage factors, **9.6.1:** 4
Cavitation erosion resistance of, **9.1-9.5:** 26, 28f.
Centerline mounted pumps
separately coupled single stage, **1.1-1.2:** 41f.
separately coupled single stage (top suction), **1.1-1.2:** 43f.
separately coupled single stage–pump on base plate, **1.1-1.2:** 42f.
separately coupled single stage–pump on base plate (top suction), **1.1-1.2:** 44f.
Centerline support pumps, **1.1-1.2:** 12f.
Centipoises, **3.1-3.5:** 19
Centistokes, **3.1-3.5:** 19
Central stations, **2.3:** 7
Centrifugal and vertical pumps
sealed, **9.6.5:** 1
sealless, **9.6.5:** 1
Centrifugal pump materials, **9.1-9.5:** 16
Centrifugal pumps, **1.4:** 1
affinity laws, **11.6:** 28
defined, **1.1-1.2:** 1
horizontal pump installation, **1.4:** 2–8
maintenance, **1.4:** 15
nomenclature (alphabetical listing), **1.1-1.2:** 27t.–35t.
nomenclature (numerical listing), **1.1-1.2:** 35t.–38t.
operation, **1.4:** 10–15
size, **1.1-1.2:** 25
vertical volute pump installation, **1.4:** 8–10
Ceramics, **4.1-4.6:** 8, **5.1-5.6:** 13, **9.1-9.5:** 26
Check valve, **9.1-9.5:** 3
Chemical packings, **8.1-8.5:** 17
Chemical process pumps, **9.6.1:** 6
Chemical pump, **1.3:** 1
Chromates, **9.1-9.5:** 11
Chromium coatings, **9.1-9.5:** 23
Chromium (ferric) stainless steel, **9.1-9.5:** 20
Chromium-nickel (austenitic) stainless steel, **9.1-9.5:** 19
CIMA *See* Construction Industry Manufactures Association
Circular casings, **1.3:** 60, 60f.
Circular plan wet pits, **9.8:** 18, 18f., 19f.
Circular pump stations (clear liquid)
dimensioning, **9.8:** 6
floor clearance, **9.8:** 6
inflow pipe, **9.8:** 7
inlet bell clearance, **9.8:** 7
inlet bell or volute diameter, **9.8:** 7
sump diameter, **9.8:** 7, 7f., 8f.
wall clearance, **9.8:** 6
Circulation plans, **5.1-5.6:** 21, 23, 24, 27f.–31f.
Circumferential piston pumps, **3.1-3.5:** 1f., 3f., 3
Clean liquids, **5.1-5.6:** 24
Cleaning, **3.1-3.5:** 33
Close coupled (defined), **5.1-5.6:** 12, **4.1-4.6:** 7
Close coupled–vane type magnetic drive pump, **4.1-4.6:** 1, 2f.
Closed feedwater cycle, **1.3:** 6, 7f., **2.3:** 9f., 9
Closed lineshafts, **2.3:** 43
Closed suction tests, **2.6:** 5, 5f., 6, 6f.
CMP *See* Canned motor pump
Coating systems, **9.1-9.5:** 22, 23–24
Cobalt alloys, **9.1-9.5:** 23
Cobalt-chromium boron alloy, **9.1-9.5:** 23
Cobalt-chromium-tungsten alloy, **9.1-9.5:** 23
Coercive force, **4.1-4.6:** 7
Column, piping, **9.1-9.5:** 3
Compound gauge, **9.1-9.5:** 3
Computers and accessories (precautions), **5.1-5.6:** 32
Computers and computer storage and magnets, **4.1-4.6:** 19
Condensate pumps, **1.3:** 9, **2.3:** 9
Condenser circulating water pumps, **1.3:** 9, **2.3:** 10
Condition points, **1.1-1.2:** 58, **2.1-2.2:** 22
Confined wet well design, **9.8:** 19, 20f.
Constant speed pumps, **9.8:** 58, 59t., 60t.
Construction, **2.1-2.2:** 3, 6f.–12f.
parts listing, **2.1-2.2:** 14t.–18t.
Construction Industry Manufactures Association, **1.3:** 13
Containment
bolt-proof load, **5.1-5.6:** 15
driven component liner, **5.1-5.6:** 14
expectations, **5.1-5.6:** 23
maximum working pressure, **5.1-5.6:** 15
monitoring equipment, **5.1-5.6:** 15
secondary, **5.1-5.6:** 15

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

suction pressure, **5.1-5.6:** 15
Containment shell, **4.1-4.6:** 7, 12, **5.1-5.6:** 14, 17
air in, **4.1-4.6:** 20
draining, **4.1-4.6:** 21
materials, **4.1-4.6:** 15
Continuous service, **1.3:** 42, **2.3:** 17
Contractors Pump Bureau, **1.3:** 13
Control limits, **9.6.5:** 2
Controlled volume pump materials, **9.1-9.5:** 18
Controls and alarms, **2.4:** 8
Cooling liquid flow, **4.1-4.6:** 12
path, **4.1-4.6:** 7
Cooling towers, **9.6.1:** 7
Copper and copper alloys, **9.1-9.5:** 20
Copper-nickel alloys, **9.1-9.5:** 21
Correction factor K, **3.1-3.5:** 41, 42t.
Corrosion, **5.1-5.6:** 20, **9.1-9.5:** 11, 12
allowance for metallic centrifugal pumps, **1.3:** 76
in crevices, **9.1-9.5:** 15
galvanic, **9.1-9.5:** 13, 14
in pulp and paper applications, **1.3:** 16
severe, **9.1-9.5:** 15
Corrosion failure mode causes and indicators, **9.6.5:** 19t.
Corrosion monitoring, **9.6.5:** 5
control limits, **9.6.5:** 6
by electrical resistance, **9.6.5:** 5
frequency, **9.6.5:** 6
indicators, **9.6.5:** 23
by linear polarization resistance, **9.6.5:** 6
means, **9.6.5:** 5
by ultrasonic thickness measurement, **9.6.5:** 6
by visual/dimensional inspection, **9.6.5:** 5
Corrosive properties of liquid, **9.6.1:** 4
Cost evaluation, **4.1-4.6:** 16
Coupling failure mode causes and indicators, **9.6.5:** 19t.
Couplings, **2.1-2.2:** 13f., **3.1-3.5:** 36, 38f., **4.1-4.6:** 1
dimensions, **2.1-2.2:** 13f.
disk, **1.3:** 68
elastomer, **1.3:** 68
flexible, **1.3:** 67
gear, **1.3:** 67
limited end float, **1.3:** 67
offset, **1.3:** 67
selection, **4.1-4.6:** 12
speed limitations, **1.3:** 68
Cover, **3.1-3.5:** 4
Cracking pressure, **3.1-3.5:** 4
Credit cards (precautions), **5.1-5.6:** 32
Credit cards and magnets, **4.1-4.6:** 19
Critical carrying velocity, **6.1-6.5:** 27, **9.1-9.5:** 5
Critical speed
See Dry critical speed
See Lateral critical speed
Cross-sectional drawings, **2.1-2.2:** 3, 6f.–12f.
Cup type pistons
composition cups, **6.1-6.5:** 64, 65f.
installation, **6.1-6.5:** 64–65
synthetic rubber cups, **6.1-6.5:** 64, 65f.
Curie temperature, **4.1-4.6:** 7, **5.1-5.6:** 14
Cyclic service, **1.3:** 42, **2.3:** 17

D *See* Displacement
D *See also* Pump displacement
D slide valves, **8.1-8.5:** 4, 6f.
Data packs, **9.1-9.5:** 61
Data sheet, **4.1-4.6:** 18f.
Data sheet for pump selection or design, **3.1-3.5:** 29, 30f.–32f.
Datum, **1.1-1.2:** 55, **1.6:** 3, **2.1-2.2:** 19, 21f., **2.6:** 3, 4f., **3.1-3.5:** 16, **3.6:** 4, **6.6:** 4, **11.6:** 3
Datum elevation, **1.1-1.2:** 55f., **1.6:** 3
horizontal pumps, **11.6:** 3, 4f.
horizontal units, **1.6:** 3, 4f.
vertical double suction pumps, **1.6:** 3, 4f.
vertical pumps, **11.6:** 4, 4f.
vertical single suction pumps, **1.6:** 3, 4f.
Dead weight tester, **9.1-9.5:** 3
Deceleration devices, **1.3:** 77, **2.3:** 45
Decontamination of returned products, **9.1-9.5:** 61
Decoupling, **4.1-4.6:** 7, **5.1-5.6:** 35
defined, **5.1-5.6:** 14
Deep well (lineshaft) pumps, **2.1-2.2:** 1, 6f.
Definitions, **3.1-3.5:** 4–5, **4.1-4.6:** 7–10, **5.1-5.6:** 12, **6.1-6.5:** 20–28, **8.1-8.5:** 7, **9.1-9.5:** 3–6
symbols, **9.8:** 38
terminology, **9.8:** 35
Deflection, **3.1-3.5:** 40
Δp *See* Differential pressure
Δp_{max} *See* Maximum differential pressure
Demagnetization, **4.1-4.6:** 7, 20, **5.1-5.6:** 14
Dephase, **4.1-4.6:** 7
Design guidelines, **8.1-8.5:** 12
Design review, **9.6.5:** 16
frequency, **9.6.5:** 17
hydraulic application review, **9.6.5:** 16
indicators, **9.6.5:** 24
installation review, **9.6.5:** 17
mechanical application review, **9.6.5:** 16
operating procedures review, **9.6.5:** 17
procedure, **9.6.5:** 16
Dewatering service, **2.3:** 4
Dichromates, **9.1-9.5:** 11
Differential pressure, **3.1-3.5:** 17, **3.6:** 5, 11
Diffusers, **1.3:** 76
Dilatant fluids, **3.1-3.5:** 22
Dimensional designations, **1.1-1.2:** 39–46
Dimensionally interchangeable pump, **1.1-1.2:** 25, **2.1-2.2:** 3

Copyright © 2002 By Hydraulic Institute, All Rights Reserved

Direct acting (steam) pump materials, **9.1-9.5:** 18
Direct acting (steam) pumps
  defined, **8.1-8.5:** 1
  double-acting pump, **8.1-8.5:** 1
  duplex pump, **8.1-8.5:** 2
  horizontal pump, **8.1-8.5:** 1
  inspection, **8.1-8.5:** 22
  nomenclature, **8.1-8.5:** 3
  piston pump, **8.1-8.5:** 1f., 2
  simplex pump, **8.1-8.5:** 2, 2f.
  types, **8.1-8.5:** 1, 1f.
  typical services, **8.1-8.5:** 12
  vertical pump, **8.1-8.5:** 1
Direction of rotation, **3.1-3.5:** 5
Dirty liquids, **5.1-5.6:** 24
Discharge, **3.1-3.5:** 33
  insufficient, **2.4:** 15
  lack of, **2.4:** 15
Discharge flow, **5.1-5.6:** 36
Discharge piping, **2.4:** 4, **6.1-6.5:** 45, 46f.
  See also Piping, Suction piping
  air release valves, **2.4:** 5
  lining up, **2.4:** 3
  reducers, **2.4:** 4f., 5
  siphons, **2.4:** 6
  supports, anchors, and joints, **2.4:** 4
  valves, **2.4:** 5, 6f.
Discharge port, **3.1-3.5:** 4, **9.1-9.5:** 4
Discharge pressure, **8.1-8.5:** 7
Discharge recirculation, **1.3:** 43
Discharge valve position, **1.4:** 12
Disk couplings, **1.3:** 68
Displacement, **3.1-3.5:** 14, **3.6:** 2, **6.1-6.5:** 20, **8.1-8.5:** 7
Displacement type meters, **6.6:** 13
Dissolved gases, **3.1-3.5:** 19, 21f.
Double suction pump specific speed, **1.3:** 32, 35f., 36f
Double volute casing *See* Dual volute casing
Double-acting pump, **6.1-6.5:** 1, 2f., 3f.
Dowelling, **1.4:** 13
Draining, **5.1-5.6:** 18
Drains, **8.1-8.5:** 23
Drive (steam) cylinder, **8.1-8.5:** 4
Drive (steam) end, **8.1-8.5:** 3, 5f.
  lubrication, **8.1-8.5:** 15, 23
Drive (steam) piston, **8.1-8.5:** 4
Drive characteristics, **4.1-4.6:** 17
Drive shaft, **1.3:** 67
Drive specification, **3.1-3.5:** 24
Driven component liner, **5.1-5.6:** 14
Driver mounting, **3.1-3.5:** 34
Driver sizing, **5.1-5.6:** 25
Drivers, **1.3:** 76, **2.3:** 45
  deceleration devices, **1.3:** 77, **2.3:** 45
  electric motors, **1.3:** 77, **2.3:** 45
  engines, **1.3:** 77
  gears, **2.3:** 45
  magnetic, **1.3:** 77
  mounting and alignment, **2.4:** 6
  non-reverse ratchets, **2.3:** 46
  pre-lubrication, **2.4:** 8
  pump-to-driver shafting, **2.3:** 46
  steam turbine, **1.3:** 77
  thrust bearings, **2.3:** 46
  variable speed, **1.3:** 77, **2.3:** 45
Dry critical speed, **9.6.4:** 2
Dry vacuum test, **1.6:** 25
Dual volute casing, **1.3:** 58, 59f., 76
  K versus rate of flow, **1.3:** 58, 59f.
Ductile iron, **9.1-9.5:** 18
Duplex pump, **6.1-6.5:** 2
Duplex stainless steels, **9.1-9.5:** 20
Duplicate performance pump, **1.1-1.2:** 25, **2.1-2.2:** 3
Duplicate pump, **2.1-2.2:** 3
Duty cycle, **3.1-3.5:** 24
Dynamic analysis report, **9.6.4:** 4, 5
Dynamic balance, **5.1-5.6:** 20
Dynamic balancing, **1.1-1.2:** 61
Dynamometers, **1.6:** 30, **3.6:** 18, **9.1-9.5:** 3
  calibration, **1.6:** 31

Earthquake-resistance requirements, **2.4:** 14
Eccentric reducers, **2.4:** 4, 4f.
Economic consequences of failure, **9.6.5:** 2
Eddy currents, **4.1-4.6:** 7, **5.1-5.6:** 13, 17
  drive, **4.1-4.6:** 7, **5.1-5.6:** 14
  drive coupling, **9.1-9.5:** 3
  losses, **4.1-4.6:** 7, **5.1-5.6:** 14
  magnetic coupling, **4.1-4.6:** 11
Effective particle diameter, **6.1-6.5:** 27, **9.1-9.5:** 5
Efficiency, **1.3:** 43, **2.6:** 7, **5.1-5.6:** 26, **6.1-6.5:** 23, **11.6:** 6
  best efficiency point (BEP), **1.3:** 56
  calculation, **2.6:** 15
  and high suction specific speed, **1.3:** 53
  and impeller diameter trim, **1.3:** 53
  and mechanical losses, **1.3:** 53
  optimum, **1.3:** 49
  prediction charts, **1.3:** 49, 50f., 51f., 52f.
  prediction method for centrifugal pumps, **1.3:** 49–57
  and pump type, **1.3:** 56, 56t.
  and slurries, **1.3:** 56
  and solids size, **1.3:** 56
  and surface finish, **1.3:** 53, 54f.
  tolerance at specified flow rate, **11.6:** 9, **11.6:** 10t.
  and viscosity, **1.3:** 53
  and wear ring clearances, **1.3:** 53, 55f.
Elastomer couplings, **1.3:** 68
Elastomeric polymers, **9.1-9.5:** 24
Electric driver input power, **1.6:** 7, **2.6:** 7

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Electric motor input power, **1.1-1.2:** 58, **2.1-2.2:** 23
Electric motors, **1.3:** 77, **2.3:** 45
Electric power pumps, **9.6.1:** 7
Electrolytes, **9.1-9.5:** 12
Electronic instruments and magnets, **4.1-4.6:** 19
Electronic methods of speed measurement, **6.6:** 18
Elevation head, **1.1-1.2:** 55, **1.6:** 4, **2.1-2.2:** 19, **2.6:** 3, 4, **3.6:** 4, **6.1-6.5:** 22, **6.6:** 4, **8.1-8.5:** 9, **11.6:** 4
Elevation pressure, **3.1-3.5:** 16, **3.6:** 4, **6.1-6.5:** 22, **6.6:** 4, **8.1-8.5:** 9
Encapsulation, **4.1-4.6:** 7
End plate, **3.1-3.5:** 4, **9.1-9.5:** 3
End suction pumps, **1.1-1.2:** 4f.
  submersible, **1.1-1.2:** 5f.
End suction slurry pumps, **9.6.2:** 16
Engines, **1.3:** 77
Entrained air, **2.4:** 3, **4.1-4.6:** 14, 20, **9.8:** 1
Entrained gases, **3.1-3.5:** 19, 20f., **4.1-4.6:** 14, 20
Entrained, non-condensable gas, **5.1-5.6:** 26
Environmental consequences of failure, **9.6.5:** 2
Environmental considerations, **5.1-5.6:** 24
Equipment mounting drilling dimensions, **1.3:** 87, 87f.
Erosion, **9.1-9.5:** 15
  cavitation erosion resistance of materials, **9.1-9.5:** 26, 28f.
Erosion failure mode causes and indicators, **9.6.5:** 19t.
η *See* Efficiency
η_{ba} *See* Bowl assembly efficiency
η_{mot} *See* Submersible motor efficiency
η_{OA} *See* Overall efficiency
η_p *See* Pump efficiency
η_p *See* Pump hydraulic efficiency
η_p *See* Pump mechanical efficiency
η_v *See* Pump volumetric efficiency
η_v *See* Volumetric efficiency
η_V *See* Volumetric efficiency
Excessive radial thrust, **1.3:** 43
Explosive atmosphere around magnets, **4.1-4.6:** 19
External bearings, **5.1-5.6:** 19
External couplings and guards, **5.1-5.6:** 21
External flush, **5.1-5.6:** 25
External gear and bearing screw pump on base plate, **3.1-3.5:** 10f.
External gear pumps
  on base plate, **3.1-3.5:** 9f.
  flanged ports, **3.1-3.5:** 8f.
  threaded ports, **3.1-3.5:** 8f.

Fabrics, **9.1-9.5:** 26
Face type seals, **3.1-3.5:** 5
Face-mounted motor dimensions, **1.1-1.2:** 49t.
  type JM, **1.1-1.2:** 51t.
  type JM having rolling contact bearings, **1.1-1.2:** 50f.
  type JP, **1.1-1.2:** 52t.
  type JP having rolling contact bearings, **1.1-1.2:** 50f.
Failure mode causes and indicators, **9.6.5:** 1, 18t.
FEA *See* Finite element analysis
Ferrite, **4.1-4.6:** 8
Field test pressure, **1.1-1.2:** 60, **2.1-2.2:** 25
Field values
  between bearing, single and multistage, **9.6.4:** 17f.
  end suction foot mounted, **9.6.4:** 9f.
  end suction, centerline support, **9.6.4:** 13f.
  end suction, close coupled horizontal and vertical in-line, **9.6.4:** 11f.
  end suction, frame mounted, **9.6.4:** 12f.
  end suction, hard metal and rubber-lined horizontal and vertical, **9.6.4:** 16f.
  end suction, paper stock, **9.6.4:** 14f.
  end suction, solids handling, horizontal and vertical, **9.6.4:** 15f.
  vertical in-line, separately coupled, **9.6.4:** 10f.
  vertical turbine, mixed flow and propeller type, **9.6.4:** 18f.
  vertical turbine, short set pumps, assembled for shipment by the manufacturer, **9.6.4:** 19f.
Filter, **5.1-5.6:** 13
Finite element analysis, **9.6.4:** 3, 5, 7
Fire pumps, **1.3:** 10, **2.3:** 11
First critical speed, **9.6.4:** 1f., 1, 4
First mode shape, **9.6.4:** 4
Fittings, **8.1-8.5:** 14
Flammability, **5.1-5.6:** 24
Flammable liquids or vapors, **8.1-8.5:** 14
Flange loads, **3.1-3.5:** 40t., 41
  correction factor K, **3.1-3.5:** 41, 42t.
Flanges, **8.1-8.5:** 14
Flexible couplings, **1.3:** 67, **3.1-3.5:** 36, **9.1-9.5:** 3
Flexible member pumps, **3.1-3.5:** 1f., 2, 2f.
Flooded suction, **6.1-6.5:** 25, **8.1-8.5:** 10
Flow, **9.8:** 26
Flow monitoring *See* Rate of flow monitoring
Flow rate, **6.1-6.5:** 20, **11.6:** 3
Flow rate check, **1.4:** 13
Flow rate tolerance at specified total head, **11.6:** 9, 10t.
Fluid drive, **9.1-9.5:** 3
Fluidborne noise, **3.1-3.5:** 27, 28
Fluids, **3.1-3.5:** 4, 33
  dilatant, **3.1-3.5:** 22
  miscellaneous properties, **3.1-3.5:** 24
  Newtonian, **3.1-3.5:** 19
  non-Newtonian, **3.1-3.5:** 22
  plastic, **3.1-3.5:** 22
  pseudo-plastic, **3.1-3.5:** 22
  rheopectic, **3.1-3.5:** 22
  thixotropic, **3.1-3.5:** 22
  time-independent non-Newtonian, **3.1-3.5:** 22
Flushing and filling, **2.4:** 9
Flux, **4.1-4.6:** 7
  density, **4.1-4.6:** 7

Copyright © 2002 By Hydraulic Institute, All Rights Reserved

Foot valves, **3.1-3.5:** 41, **9.1-9.5:** 3
Force and mass requirements, **1.4:** 1
Formed suction intakes, **9.8:** 3, 6f.
  application standards, **9.8:** 4
  dimensions, **9.8:** 3
Foundation, **2.3:** 45, **3.1-3.5:** 34, **6.1-6.5:** 55
  bolts, **1.4:** 1, 2f., **3.1-3.5:** 34, 34f., **6.1-6.5:** 56, 56f.
  requirements, **1.4:** 1, **2.4:** 2
  typical bolt design, **2.4:** 2f.
Frame mounted pumps
  ANSI B73.1, **1.1-1.2:** 13f.
  lined, **1.1-1.2:** 11f.
  separately coupled single stage–mixed flow, **1.1-1.2:** 21f.
  separately coupled single stage–self-priming, **1.1-1.2:** 24f.
  separately coupled–single stage, **1.1-1.2:** 10f., 39f.
  separately coupled–single stage (vertically mounted), **1.1-1.2:** 47f.
  separately coupled–single stage–pump on base plate, **1.1-1.2:** 40f.
Francis vane, **1.1-1.2:** 3f., 3
Free-surface vortices, **9.8:** 1, 26, 26f.
Frequency-responsive devices, **1.6:** 31, **6.6:** 18
Friction characteristic, **6.1-6.5:** 27, **9.1-9.5:** 5
Friction factor, **3.6:** 17
Friction head, **1.1-1.2:** 57, **2.1-2.2:** 22
Friction loss pressure, **6.1-6.5:** 23
Full-flow bypass pressure, **3.1-3.5:** 5
Fully suspended solids, **9.1-9.5:** 5

Galvanic corrosion, **9.1-9.5:** 13
  minimizing, **9.1-9.5:** 14
Galvanic series, **9.1-9.5:** 13
Gap, **4.1-4.6:** 7
Gap *See* Air gap, Liquid gap, Total gap
Gas, **2.3:** 21
  effect on performance, **2.3:** 21, 21f.
Gas *See* Liquids with vapor or gas
Gas bubbles, **9.8:** 1
Gas content, **9.6.1:** 4
Gaskets, **5.1-5.6:** 18, **8.1-8.5:** 15
Gauge head, **1.1-1.2:** 55, **1.6:** 4, **2.1-2.2:** 19, **2.6:** 3, **11.6:** 4
Gauge pressure, **3.6:** 4, **6.1-6.5:** 22, **6.6:** 4, **8.1-8.5:** 9
Gauss, **4.1-4.6:** 8
Gear couplings, **1.3:** 67
Gear pumps, **3.1-3.5:** 1f., 2, 3f., 6f.–10f.
Gears, **2.3:** 45
General purpose service, **1.3:** 13–15
Gilbert, **4.1-4.6:** 8
Gland, **3.1-3.5:** 5
Gland follower, **3.1-3.5:** 5
Gland, packing, **9.1-9.5:** 3
Glossary, **9.8:** 35
Graphic level recorders, **9.1-9.5:** 50
Graphite, **5.1-5.6:** 13, **9.1-9.5:** 26
Gray cast iron, **9.1-9.5:** 18
Grouting, **2.4:** 3, **3.1-3.5:** 35, 36f.
  horizontal pumps, **1.4:** 5
  vertical volute pumps, **1.4:** 8

h *See* Head
H *See* Total head
h *See* Head
h_{atm} *See* Atmospheric head
h_d *See* Total discharge head
h_g *See* Gauge head
h_s *See* Total suction head
h_v *See* Velocity head
H *See* Total head
h_{acc} *See* Acceleration head
h_{atm} *See* Atmospheric head
H_{ba} *See* Bowl assembly total head
h_d *See* Pump total discharge head
h_d *See* Total discharge head
h_f *See* Friction head
h_g *See* Gauge head
h_s *See* Total suction head
h_v *See* Velocity head
Halide, **9.1-9.5:** 11
Handling equipment, **1.4:** 1
Hands and fingers (precautions), **5.1-5.6:** 32
Hardware terms, **9.1-9.5:** 3
Hazardous chemicals, **9.1-9.5:** 61
Hazardous materials, **5.1-5.6:** 32
Head, **1.1-1.2:** 55, **1.6:** 3, **2.1-2.2:** 19, **2.6:** 3, **3.1-3.5:** 4, **11.6:** 4
  atmospheric, **1.1-1.2:** 57
  elevation, **1.1-1.2:** 55
  friction, **1.1-1.2:** 57
  gauge, **1.1-1.2:** 55
  loop manifold connecting pressure taps, **1.6:** 30f.
  measurement, **1.6:** 29, **2.6:** 27–31
  measurement by gauge/valve arrangement, **2.6:** 28f., 28
  measurement by means of pressure gauges, **1.6:** 30
  measurement by multiple tap connections, **2.6:** 28, 28f.
  measurement by pressure gauges, **2.6:** 29
  measurement by single tap connection, **2.6:** 28, 28f.
  measurement with bourdon gauge below atmospheric pressure, **2.6:** 30, 30f.
  measurement with fluid gauge below atmospheric pressure, **2.6:** 30, 30f.
  net positive suction head available, **1.1-1.2:** 58
  net positive suction head required, **1.1-1.2:** 58
  pressure tap location for level A tests, **1.6:** 29, 29f.
  pressure tap location for level B tests, **1.6:** 29, 30f.
  single tap connection, **1.6:** 29f.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

total, **1.1-1.2:** 57, 59
total discharge, **1.1-1.2:** 57
total suction (closed suction), **1.1-1.2:** 57
total suction (open suction), **1.1-1.2:** 57
velocity, **1.1-1.2:** 55
Head rate of flow curve
centrifugal pumps, **9.6.3:** 4
vertical pumps, **9.6.3:** 4, 4f.
Head type rate meters, **6.6:** 13, 14f.
pressure tap opening, **6.6:** 14, 14f.
Heat exchanger, **9.1-9.5:** 3
Heater drain pumps, **1.3:** 10, **2.3:** 11
Hermetic integrity test, **4.1-4.6:** 24, **5.1-5.6:** 39
Heterogeneous mixture, **6.1-6.5:** 27, **9.1-9.5:** 5
High alloy steels, **9.1-9.5:** 19
High copper alloys, **9.1-9.5:** 20
High silicon cast irons, **9.1-9.5:** 19
High temperature, **5.1-5.6:** 24
High viscosity, **4.1-4.6:** 13, **5.1-5.6:** 25
High-energy pumps, **1.1-1.2:** 59, 59f., 60f., **2.1-2.2:** 23, 24f.
Hollow/solid shaft driver, **2.1-2.2:** 2, 6f., 9f., 10f., 11f.
Homogeneous flow, **6.1-6.5:** 27, **9.1-9.5:** 5
Homogeneous mixture, **6.1-6.5:** 27, **9.1-9.5:** 5
Horizontal end suction pumps
adjustment factors, **9.6.2:** 4, 9t.
allowable combination nozzle loads, **9.6.2:** 6t., 7t.
allowable individual nozzle loads, **9.6.2:** 5t.
alternate pump mounting, **9.6.2:** 3
driver/pump coupling alignment, **9.6.2:** 2
grouted nonmetal baseplate, **9.6.2:** 4
internal pump distortion, **9.6.2:** 2
material specifications, **9.6.2:** 8t.
nomenclature, **9.6.2:** 1, 2f.
nozzle load adjustment factors, **9.6.2:** 3
nozzle loads, **9.6.2:** 1, 5t., 6t., 7t.
nozzle stress, **9.6.2:** 2
pressure-temperature, **9.6.2:** 2
pump hold down bolts, **9.6.2:** 2
pump mounting, **9.6.2:** 2
spring-mounted metal baseplate, **9.6.2:** 4
stilt-mounted metal baseplate, **9.6.2:** 3
temperature and material adjustment factors, **9.6.2:** 4
ungrouted metal baseplate, **9.6.2:** 3
ungrouted nonmetal baseplate, **9.6.2:** 4
Horizontal mounting base, **5.1-5.6:** 21
Horizontal pump, **6.1-6.5:** 1f., 1
Horizontal pump installation
alignment, **1.4:** 2
alignment of gear type couplings, **1.4:** 4, 5f.
alignment of spacer type couplings, **1.4:** 5, 5f.
alignment of special couplings, **1.4:** 5
angular alignment, **1.4:** 3, 3f.
controls and alarms, **1.4:** 8
coupling guard, **1.4:** 5
dial indicator method of alignment, **1.4:** 4, 4f.
final alignment, **1.4:** 6
final alignment check, **1.4:** 6
full pump alignment, **1.4:** 6
grouting, **1.4:** 5
laser method of alignment, **1.4:** 4
leveling pump and driver, **1.4:** 2, 2f.
parallel alignment, **1.4:** 3, 3f.
pre-run lubrication, **1.4:** 7
shaft and coupling alignment, **1.4:** 3
straightedge method of alignment, **1.4:** 3
stuffing-box bushings, **1.4:** 6
stuffing-box mechanical seals, **1.4:** 6
stuffing-box packing, **1.4:** 5
stuffing-box steps, **1.4:** 5
suction and discharge pipes, **1.4:** 7
thermal expansion and alignment, **1.4:** 7
v-belt drive, **1.4:** 5
Horsepower limit, **9.6.3:** 3
Hot oil pump, **1.3:** 1
Housing, **3.1-3.5:** 4
HP and HPH vertical solid-shaft motor dimensions, **1.1-1.2:** 53f., 53t., 54t.
HPRT *See* Hydraulic power recover turbines
Hydraulic action, **8.1-8.5:** 15
Hydraulic disturbances, **9.6.4:** 24
Hydraulic drag, **4.1-4.6:** 8
Hydraulic failure mode causes and indicators, **9.6.5:** 20t.
Hydraulic hammer, **1.3:** 22
See also Water hammer analysis
Hydraulic load balance, **5.1-5.6:** 13
Hydraulic parasitic losses, **4.1-4.6:** 8
Hydraulic piston packing, **8.1-8.5:** 17
applications, **8.1-8.5:** 17
fitting, **8.1-8.5:** 18, 18f.
joint types, **8.1-8.5:** 17, 18f.
Hydraulic power recover turbines, **2.3:** 12
Hydraulic pressure pump, **1.3:** 14
Hydraulic resonance *See* Resonance
Hydraulic sizing, **5.1-5.6:** 25
Hydraulic slip, **4.1-4.6:** 10
Hydraulic turbines, pumps used as *See* Pumps used as hydraulic turbines
Hydrocarbon physical properties, **6.1-6.5:** 50t., 51t.
Hydrostatic test, **1.6:** 7, **2.6:** 1, **3.6:** 13
assembled pump, **6.6:** 10
assembled pumps, **3.6:** 13
components, **3.6:** 13, **6.6:** 10
duration, **6.6:** 10
objective, **1.6:** 7, **2.6:** 8, **6.6:** 10
parameters, **1.6:** 8, **2.6:** 8, **6.6:** 10
procedure, **1.6:** 8, **2.6:** 8, **6.6:** 11
records, **1.6:** 8, **2.6:** 9, **3.6:** 13, **6.6:** 11

Copyright © 2002 By Hydraulic Institute, All Rights Reserved

Hydrostatic test (continued)
temperature, **6.6:** 10
test liquid, **6.6:** 10
Hydrostatic test pressure, **4.1-4.6:** 8
Hydrostatic tests. *See* Submersible pump hydrostatic test
Hysteresis, **4.1-4.6:** 8

Identical performance and dimensional pump, **1.1-1.2:** 25
Identical pump, **2.1-2.2:** 3
Impeller balancing, **1.1-1.2:** 60
Impeller between bearings, **1.1-1.2:** 1f., 2
separately coupled–multistage axial (horizontal) split case, **1.1-1.2:** 18f.
separately coupled–multistage radial (vertical) split case, **1.1-1.2:** 19f.
separately coupled–multistage radial (vertical) split–double casing, **1.1-1.2:** 20f.
separately coupled–single stage axial (horizontal) split case, **1.1-1.2:** 46f.
separately coupled–single stage axial (horizontal) split case pump on base plate, **1.1-1.2:** 45f.
separately coupled–single stage–axial (horizontal) split case, **1.1-1.2:** 16f.
separately coupled–single stage–radial (vertical) split case, **1.1-1.2:** 17f.
Impeller designs, **1.1-1.2:** 2
axial flow, **1.1-1.2:** 4, 4f.
Francis vane, **1.1-1.2:** 3f., 3
impeller between bearing type, **1.1-1.2:** 1f., 2
mixed flow, **1.1-1.2:** 3, 3f.
radial flow, **1.1-1.2:** 3, 3f.
specific speed, **1.1-1.2:** 3f.
suction specific speed, **1.1-1.2:** 3f.
Impeller eye diameter, **9.6.1:** 3, **9.6.1:** 4
Impeller material, **9.6.1:** 4
Impeller vanes
incidence angle, **9.6.1:** 2
overlap, **9.6.1:** 2f., **9.6.1:** 2
Impellers, **1.3:** 57, 75
See also Overhung impeller pumps
axial flow, **2.1-2.2:** 3, 11f.
axial thrust for volute pump, **1.3:** 60–63
with back ring, **1.3:** 62f.
balancing, **2.1-2.2:** 25
and bearing arrangements, **1.3:** 58
diameter change and pump performance, **2.3:** 16, 16f.
double suction, **1.3:** 75
dynamic balancing, **2.1-2.2:** 25
enclosed, **1.3:** 76, **2.3:** 44
enclosed with plain back shroud, **1.3:** 61f.
with inducers, **2.3:** 44, 44f.
mixed flow, **2.1-2.2:** 3, 10f.
modified radial flow, **2.1-2.2:** 3, 10f.
mounted between bearings, **1.3:** 58, 72f.
open, **1.3:** 76
open (axial flow), **2.3:** 44
overhung, **1.3:** 58, 70, 71f.
predicting pump performance after diameter change, **1.3:** 48, 49f.
pressure distribution on enclosed impeller shrouds, **1.3:** 60f.
profiles, **2.1-2.2:** 2, 5f.
pump characteristic curves, **2.1-2.2:** 5f.
radial flow, **2.1-2.2:** 3
semi-open, **1.3:** 76, **2.3:** 44
single plane balancing, **2.1-2.2:** 25
single suction, **1.3:** 75
specific speed, **2.1-2.2:** 2
static balancing, **2.1-2.2:** 25
top suction, **1.3:** 20, 21f.
two plane balancing, **2.1-2.2:** 25
types, **1.3:** 75, **2.3:** 44
various configurations and axial thrust, **2.3:** 38, 38f., 39f., 40f.
venting the eye of, **1.3:** 20, 21f.
wear ring arrangements, **2.1-2.2:** 12f.
Indicators, **9.6.5:** 1, 22
Induced eddy currents, **5.1-5.6:** 17
Inducers, **1.3:** 20, 57, 57f., **2.3:** 44, 44f.
Induction motor, **5.1-5.6:** 13
Industrial plant, **2.3:** 7
Industrial pumps, **9.6.1:** 9
Inert gas sniffer test, **4.1-4.6:** 24
Inlet, **3.1-3.5:** 4, 33, **9.1-9.5:** 3
Inlet bell design diameter, **9.8:** 21t., 28, 30f., 31f.
Inlet boosters, **1.3:** 20
Inlet geometry, **9.6.1:** 2
Inlet piping geometry, **9.6.1:** 2
Inlet port, **9.1-9.5:** 3
Inlet pressure, **3.1-3.5:** 17, **3.6:** 5
Inlet system, **6.1-6.5:** 38–40
booster pumps, **6.1-6.5:** 40
connection of piping sections, **6.1-6.5:** 39f.
foot valve, **6.1-6.5:** 40
high points in piping system, **6.1-6.5:** 39
inlet line valve, **6.1-6.5:** 40
inlet piping, **6.1-6.5:** 40
inlet piping diameters, **6.1-6.5:** 39, 40f.
inlet pressure gauge, **6.1-6.5:** 40
liquid source features, **6.1-6.5:** 38
multiple-pump installations, **6.1-6.5:** 39
pulsation dampener, **6.1-6.5:** 41
screens or strainers, **6.1-6.5:** 40, 40f.
suction system relationships, **6.1-6.5:** 41, 42f., 43f.
suction tanks, **6.1-6.5:** 38f., 41
In-line pumps, **1.1-1.2:** 7f.
flexible coupling, **1.1-1.2:** 8f.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

rigid coupling, **1.1-1.2:** 9f.
Inner magnet assembly, **4.1-4.6:** 8, 12
Inner magnet ring, **5.1-5.6:** 14
Inside-adjustable lost-motion valve gear, **8.1-8.5:** 6f.
Inside-fixed lost-motion valve gear, **8.1-8.5:** 6f.
Inspection, **5.1-5.6:** 32, **6.1-6.5:** 65–66
Inspection (pre-installation), **2.4:** 1
Inspection (shipment), **3.1-3.5:** 33
Inspection frequency, **4.1-4.6:** 21
Installation, **2.4:** 2, **3.1-3.5:** 33, **4.1-4.6:** 19, **6.1-6.5:** 56, **8.1-8.5:** 14
See also Maintenance, Operation, Troubleshooting
accessory equipment, **3.1-3.5:** 41–44
adjustment factors for alternate designs, **3.1-3.5:** 42t.
alignment, **3.1-3.5:** 36, 37f., **5.1-5.6:** 33
auxiliary connections and monitoring devices, **5.1-5.6:** 33
bearings, **6.1-6.5:** 60
cleaning, **3.1-3.5:** 33
coupling alignment, **5.1-5.6:** 33
couplings, **3.1-3.5:** 36, 38f.
drive alignment, **6.1-6.5:** 57
driver mounting, **3.1-3.5:** 34
and entrained air, **2.4:** 3
flanges and fittings, **6.1-6.5:** 57
flexible coupling, **6.1-6.5:** 58
foot valves, **3.1-3.5:** 41
forces and moments, **6.1-6.5:** 57
foundation, **3.1-3.5:** 34
foundation bolts, **3.1-3.5:** 34, 34f.
gaskets, **6.1-6.5:** 58
gear drive, **6.1-6.5:** 58
grouting, **2.4:** 3, **3.1-3.5:** 35, 36f.
handling equipment, **2.4:** 1
horizontal pumps, **1.4:** 2–8
inlet piping, **3.1-3.5:** 39
jacket piping, **3.1-3.5:** 39
leveling, **3.1-3.5:** 35, 36f.
leveling the unit, **5.1-5.6:** 33, **6.1-6.5:** 56
limiting forces and moments for steel pumps, **3.1-3.5:** 39, 40t., 42t.
lining up pump discharge, **2.4:** 3
location, **3.1-3.5:** 33
location and foundation, **5.1-5.6:** 32
lubrication, **6.1-6.5:** 60
nozzle loads and criteria, **3.1-3.5:** 39, 40t., 42t.
outlet piping, **3.1-3.5:** 39
pipe dope and tape, **6.1-6.5:** 58
pipe-to-pump alignment, **3.1-3.5:** 39f., 39
piping, **3.1-3.5:** 38, **5.1-5.6:** 33, **6.1-6.5:** 56
piston rod packing, **6.1-6.5:** 60–64
pit dimensional checks, **2.4:** 3
priming, **6.1-6.5:** 57
procedure, **5.1-5.6:** 32
protective devices, **3.1-3.5:** 43
pump leveling and plumbness, **2.4:** 3, 3f.
pump location, **2.4:** 3
relief valve set pressure, **6.1-6.5:** 57, 58t.
relief valves, **3.1-3.5:** 43
rotation check, **3.1-3.5:** 35
strainers, **3.1-3.5:** 42
tools, **1.4:** 1, **2.4:** 1
V-belt drive, **6.1-6.5:** 59, 59t., 60t., 60f.
v-belts and sheaves, **3.1-3.5:** 38, 38f.
vertical volute pump, **1.4:** 8–10
well inspection, **2.4:** 2
wells, **2.4:** 2, 2f.
Instrument calibration intervals, **3.6:** 20, 21f.
Instrumentation
calibration interval, **6.6:** 18, 19t.
fluctuation, **6.6:** 6
performance test, **6.6:** 6
Instrumentation options, **5.1-5.6:** 22
Intake designs, **1.3:** 57
alternative, **9.8:** 1
design objectives, **9.8:** 1
general information, **9.8:** 1
Intake structures
basin entrance conditions, **9.8:** 58
can intakes, **9.8:** 11
circular plan wet pits, **9.8:** 18, 18f., 19f.
circular pump stations (clear liquids), **9.8:** 5
for clear liquids, **9.8:** 1
confined wet well design, **9.8:** 19, 20f.
formed suction intakes, **9.8:** 3, 6f.
model tests, **9.8:** 22
rectangular intakes, **9.8:** 1, 3f., 4t., 5t.
rectangular wet wells, **9.8:** 19
remedial measures, **9.8:** 42
for solids-bearing liquids, **9.8:** 15
submersible vertical turbine pump intakes, **9.8:** 11, 14
suction tanks, **9.8:** 9
trench-type intakes (clear liquids), **9.8:** 7, 8f., 9f.
trench-type wet wells, **9.8:** 16f., 17
unconfined intakes, **9.8:** 14
Intake system design, **2.3:** 46
Integral motors, **5.1-5.6:** 19
Integrity tests. *See* Submersible motor integrity tests
Interchangeable pump, **1.1-1.2:** 25, **2.1-2.2:** 3
Intermediate input power, **3.6:** 6
Intermediate mechanism efficiency, **3.6:** 6
Intermittent service, **1.3:** 42, **2.3:** 17
Internal bearings, **5.1-5.6:** 18
Internal gear pumps
close coupled, **3.1-3.5:** 7f.
flange mounting, **3.1-3.5:** 6f.
foot mounting, **3.1-3.5:** 6f.
frame mounting, **3.1-3.5:** 7f.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Internal mechanical contact, **9.6.3:** 2
Internal sleeve bearings, **4.1-4.6:** 12
Intrinsic induction, **4.1-4.6:** 8
Iron
  abrasion resistant cast irons, **9.1-9.5:** 19
  all iron pumps, **9.1-9.5:** 16, 17
  austenitic gray cast iron, **9.1-9.5:** 18
  ductile, **9.1-9.5:** 18
  gray cast iron, **9.1-9.5:** 18
  high silicon cast irons, **9.1-9.5:** 19
  malleable cast iron, **9.1-9.5:** 18
  nickel-chromium-iron alloys, **9.1-9.5:** 21
Irrigation service, **1.3:** 4, **2.3:** 6

Jacketed pump, **3.1-3.5:** 5, **9.1-9.5:** 3
Joint bolting, **5.1-5.6:** 18

K *See* Correction factor K
Kinetic pumps, **9.1-9.5:** 1, 2f.
  impeller between bearing type, **1.1-1.2:** 1f., 2
  overhung impeller types, **1.1-1.2:** 1f., 2, 4f.–15f., 21f., 24f.
  regenerative turbine type, **1.1-1.2:** 1, 1f., 2
  special variations, **1.1-1.2:** 2
  types, **1.1-1.2:** 1f., 1

l_s *See* Static suction lift
L *See* Stroke
Lantern ring, **3.1-3.5:** 5, **9.1-9.5:** 3
Lateral critical speed, **9.6.4:** 1, 1f.
  calculations, **9.6.4:** 1
Lateral dynamic analysis, **9.6.4:** 3
Lead and lead alloys, **9.1-9.5:** 23
Leaded nickel bronze (nickel silvers), **9.1-9.5:** 21
Leaded red brass, **9.1-9.5:** 20
Leak check, **1.4:** 13, **2.4:** 11
Leak detection, **9.6.5:** 6
  control limits, **9.6.5:** 7
  double-walled protection, **9.6.5:** 7
  by flow increase, **9.6.5:** 7
  frequency, **9.6.5:** 7
  indicators, **9.6.5:** 23
  means, **9.6.5:** 6
  by sniffer inspection, **9.6.5:** 6
  by visual inspection, **9.6.5:** 6
Leakage detection
  by flow increase, **9.6.5:** 7
  by pressure buildup, **9.6.5:** 7
  by sniffer inspection, **9.6.5:** 7
  by visual inspection, **9.6.5:** 7
Leather, **9.1-9.5:** 26
Legal requirements, **5.1-5.6:** 24
Letter designations, **1.1-1.2:** 39–46
Leveling, **3.1-3.5:** 35, 36f.
Life cycle cost analysis, **4.1-4.6:** 16
Limited end float couplings, **1.3:** 67
Lineshafts, **2.3:** 43
Lip seal, **9.1-9.5:** 3
Liquid
  classification, **4.1-4.6:** 14
  gap, **4.1-4.6:** 7
  lubricating, **4.1-4.6:** 14
  non-lubricating, **4.1-4.6:** 14
  prevention of operation without liquid flow, **4.1-4.6:** 19
  properties, **4.1-4.6:** 13
  pumped liquid characteristics, **4.1-4.6:** 17
  shear sensitivity, **4.1-4.6:** 15
  vapor pressure, **4.1-4.6:** 14
Liquid bypass, **6.1-6.5:** 35, 36f.
Liquid end, **5.1-5.6:** 12, **8.1-8.5:** 1f., 3
  cylinder liner, **6.1-6.5:** 5, 11f.
  gland, **6.1-6.5:** 7, 7f.
  lantern ring (seal cage), **6.1-6.5:** 7, 7f.
  liquid cylinder, **6.1-6.5:** 5, 5f.
  manifolds, **6.1-6.5:** 5, 5f.
  packing, **6.1-6.5:** 7, 7f.
  parts, **6.1-6.5:** 5–8, 9f., 10f., 11f., 12t.
  piston, **6.1-6.5:** 5, 6f.
  plunger, **6.1-6.5:** 3f., 6, 7f.
  stuffing box, **6.1-6.5:** 7, 7f.
  upper crosshead, **6.1-6.5:** 8, 8f.
  valve assembly, **6.1-6.5:** 8, 8f.
  valve chest cover, **6.1-6.5:** 5, 11f.
  valve plate (check valve), **6.1-6.5:** 5, 11f.
Liquid expansion factor, **6.1-6.5:** 50, 51f.
Liquid gap, **5.1-5.6:** 12
Liquid level, **9.8:** 26
Liquid velocity in casing throat, **9.6.3:** 3
Liquids, **3.1-3.5:** 4, 33
  chemical symbols, **9.1-9.5:** 11
  clean, **5.1-5.6:** 24
  common polymer materials for, **9.1-9.5:** 37, 38t.
  containment shells, **5.1-5.6:** 17
  dirty, **5.1-5.6:** 24
  effects of temperature and concentration, **9.1-9.5:** 11
  entrained gases in, **3.1-3.5:** 19, 20f.
  high temperature, **9.1-9.5:** 11, 12
  identification and properties, **3.1-3.5:** 18
  low temperature, **9.1-9.5:** 11, 12
  material selection for maximum continuous temperature of various liquids, **9.1-9.5:** 39, 40t.–44t., 45t.–49t.
  materials commonly used for pumping, **9.1-9.5:** 27, 29t.–37t.
  specific gravity, **9.1-9.5:** 11
  that solidify, **5.1-5.6:** 24
  toxicity ratings, **5.1-5.6:** 23
  types, **3.1-3.5:** 18
  volatile, **5.1-5.6:** 24

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Liquids with vapor or gas, **1.3:** 19–21
effect of gas on performance, **1.3:** 19, 20f.
inducers (inlet boosters), **1.3:** 20
special designs for, **1.3:** 20, 21f.
top suction impeller, **1.3:** 20, 21f.
venting the eye of the impeller, **1.3:** 20, 21f.
Lobe pumps, **3.1-3.5:** 1f., 2, 2f., 12f.
Location, **3.1-3.5:** 33
Location of unit, **1.4:** 2
Locked rotor torque, **5.1-5.6:** 13
Locked-rotor torque ratings, **6.1-6.5:** 38, 39t.
Long-term storage, **1.4:** 1
Losses, **2.3:** 33
Low alloy steels, **9.1-9.5:** 19
Low viscosity, **4.1-4.6:** 13
Low-energy pumps, **1.1-1.2:** 59, 59f., 60f.
Lubricant analysis, **9.6.5:** 9
control limits, **9.6.5:** 11
evaluating wear rates, **9.6.5:** 10
indicators, **9.6.5:** 23
measuring contamination of lubricant, **9.6.5:** 10
measuring inorganic contamination, **9.6.5:** 10
measuring lubricant degradation, **9.6.5:** 10
measuring metal particles from wear, **9.6.5:** 9
measuring organic contamination, **9.6.5:** 10
monitoring frequency, **9.6.5:** 11
sampling techniques, **9.6.5:** 11
Lubricating liquid, **4.1-4.6:** 14
Lubrication, **3.1-3.5:** 44, **8.1-8.5:** 15
Lubrication and cooling, **5.1-5.6:** 12

Magnet tape and magnets, **4.1-4.6:** 19
Magnetic couplings, **4.1-4.6:** 8, **5.1-5.6:** 14, 19
Magnetic drive and driver sizing, **5.1-5.6:** 25
Magnetic drive configurations, **4.1-4.6:** 1, 2f., 3f., 4f.
Magnetic drive pump, **5.1-5.6:** 2
alignment, **5.1-5.6:** 33
alnico, **5.1-5.6:** 14
close coupled, **5.1-5.6:** 2, 8f.
coupling alignment, **5.1-5.6:** 33
Curie temperature, **5.1-5.6:** 14
decoupling, **5.1-5.6:** 14, 35
defined, **5.1-5.6:** 13
demagnetization, **5.1-5.6:** 14
eddy current drive, **5.1-5.6:** 14
eddy current losses, **5.1-5.6:** 14
external bearings, **5.1-5.6:** 19
external couplings and guards, **5.1-5.6:** 21
inner magnet ring, **5.1-5.6:** 14
location and foundation, **5.1-5.6:** 32
magnetic couplings, **5.1-5.6:** 14, 19
magnetic drive and driver sizing, **5.1-5.6:** 25
magnets, **5.1-5.6:** 14
maintenance, **5.1-5.6:** 35
mounting base, **5.1-5.6:** 21
neodymium, **5.1-5.6:** 14
outer magnet ring, **5.1-5.6:** 14
pole (N-S), **5.1-5.6:** 14
precautions, **5.1-5.6:** 32
rows of magnets, **5.1-5.6:** 14
samarium cobalt, **5.1-5.6:** 14
separately coupled, **5.1-5.6:** 2, 7f.
shipping precautions, **5.1-5.6:** 32
slip, **5.1-5.6:** 14
vertical submerged, **5.1-5.6:** 2, 9f.
Magnetic drives, **1.3:** 77
Magnetic Material Producers Association, **4.1-4.6:** 23
Magnetic materials, **4.1-4.6:** 8
Magnetic slip, **4.1-4.6:** 9
Magnets, **5.1-5.6:** 14
assembly, **4.1-4.6:** 12
assembly caution, **4.1-4.6:** 21
cautions, **4.1-4.6:** 19, 21, 22
component temperature, **5.1-5.6:** 26
demagnetization, **4.1-4.6:** 20
handling cautions, **4.1-4.6:** 22
humidity effects, **4.1-4.6:** 21
installation and safety considerations, **4.1-4.6:** 19
permanent, **4.1-4.6:** 8
shipping, **4.1-4.6:** 19
temperature limits, **4.1-4.6:** 20, **5.1-5.6:** 26
Main drive (steam) slide valve, **8.1-8.5:** 4
Main drive (steam) valves, **8.1-8.5:** 4, 6f.
setting (duplex pumps), **8.1-8.5:** 22
setting (simplex pumps), **8.1-8.5:** 23
Maintenance, **2.4:** 14, **4.1-4.6:** 21–22, **5.1-5.6:** 32, 35
access, **1.4:** 1, **2.4:** 2
canned motor, **5.1-5.6:** 35
close running fits, **5.1-5.6:** 35
examination of wear patterns, **5.1-5.6:** 36
excessive power consumption, **1.4:** 16
inspections, **5.1-5.6:** 35
insufficient discharge flow or pressure, **1.4:** 16
little or no discharge flow, **1.4:** 16
loss of suction, **1.4:** 16
magnet assembly, **5.1-5.6:** 35
mechanical seals, **3.1-3.5:** 46
noise, **1.4:** 15
packing, **3.1-3.5:** 46
parts replacements, **2.4:** 14
preventive, **3.1-3.5:** 45
spare parts, **3.1-3.5:** 46
troubleshooting, **1.4:** 15, **2.4:** 15
wear plates, **1.4:** 15
wear rings, **1.4:** 15, **2.4:** 14
Maintenance inspection, **9.6.5:** 12
characteristics to consider, **9.6.5:** 12
coupling flexible elements inspection, **9.6.5:** 12
erosion inspection, **9.6.5:** 13
frequency, **9.6.5:** 13

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Maintenance inspection (continued)
hydraulic performance, **9.6.5:** 13
indicators, **9.6.5:** 24
key/keyways inspection, **9.6.5:** 12
shaft bending fatigue inspection, **9.6.5:** 12
shaft torsional fatigue inspection, **9.6.5:** 13
torsional overload inspection, **9.6.5:** 13
Malfunction causes and remedies, **3.1-3.5:** 46, 47t.–49t., **6.1-6.5:** 66, 66t.–68t., **8.1-8.5:** 23, 24t.
Malleable cast iron, **9.1-9.5:** 18
Manufacturer's erecting engineer, **1.4:** 1
Manufacturer's instructions, **1.4:** 1, **2.4:** 1
Manufacturer's service personnel, **2.4:** 1
Material Safety Data Sheets, **9.1-9.5:** 61
Materials, **4.1-4.6:** 15, **5.1-5.6:** 20
abrasion resistant cast irons, **9.1-9.5:** 19
adhesives, **9.1-9.5:** 26
aluminum and aluminum alloys, **9.1-9.5:** 22
aluminum bronze, **9.1-9.5:** 21
austenitic ductile iron, **9.1-9.5:** 19
austenitic gray cast iron, **9.1-9.5:** 18
carbon, **9.1-9.5:** 26
carbon and low alloy steels, **9.1-9.5:** 19
carbon steel, **9.1-9.5:** 19
cavitation erosion resistance of, **9.1-9.5:** 26, 28f.
centrifugal pumps, **9.1-9.5:** 16
ceramics, **9.1-9.5:** 26
chemical and physical properties, **9.1-9.5:** 12
chromium coatings, **9.1-9.5:** 23
chromium (ferric) stainless steel, **9.1-9.5:** 20
chromium-nickel (austenitic) stainless steel, **9.1-9.5:** 19
coating systems, **9.1-9.5:** 22, 23–24
cobalt alloys, **9.1-9.5:** 23
cobalt-chromium-tungsten alloy, **9.1-9.5:** 23
common polymer for various liquids, **9.1-9.5:** 37, 38t.
controlled volume pumps, **9.1-9.5:** 18
copper and copper alloys, **9.1-9.5:** 20
copper-nickel alloys, **9.1-9.5:** 21
and crevice corrosion, **9.1-9.5:** 15
direct acting (steam) pumps, **9.1-9.5:** 18
ductile iron, **9.1-9.5:** 18
duplex stainless steels, **9.1-9.5:** 20
elastomeric polymers, **9.1-9.5:** 24
fabrics, **9.1-9.5:** 26
factors affecting selection, **9.1-9.5:** 11–16
galling resistance, **9.1-9.5:** 15
and galvanic corrosion, **9.1-9.5:** 13, 14
and galvanic series, **9.1-9.5:** 13
and general design, **9.1-9.5:** 12
general designations by pump type, **9.1-9.5:** 16–18
graphite, **9.1-9.5:** 26
gray cast iron, **9.1-9.5:** 18
high alloy steels, **9.1-9.5:** 19
high copper alloys, **9.1-9.5:** 20
high silicon cast irons, **9.1-9.5:** 19
lead and lead alloys, **9.1-9.5:** 23
leaded nickel bronze (nickel silvers), **9.1-9.5:** 21
leaded red brass, **9.1-9.5:** 20
leather, **9.1-9.5:** 26
and liquid temperature, **9.1-9.5:** 12, 39, 40t.–44t., 45t.–49t.
and liquids, **9.1-9.5:** 11
low alloy steels, **9.1-9.5:** 19
malleable cast iron, **9.1-9.5:** 18
and mechanical situation in pumping, **9.1-9.5:** 15
microstructure of metals, **9.1-9.5:** 15
nickel alloys, **9.1-9.5:** 21
nickel copper alloys, **9.1-9.5:** 21
nickel or cobalt-chromium boron alloy, **9.1-9.5:** 23
nickel-chromium-iron alloys, **9.1-9.5:** 21
nickel-molybdenum alloys, **9.1-9.5:** 21
nickel-molybdenum-chromium alloys, **9.1-9.5:** 21
non-metal, **9.1-9.5:** 24–26
optimizing life cost, **9.1-9.5:** 12
power pumps, **9.1-9.5:** 18
reciprocating pumps, **9.1-9.5:** 18
reinforced fibers, **9.1-9.5:** 26
rigid polymers and composites, **9.1-9.5:** 25
rotary pumps, **9.1-9.5:** 17
sealants, **9.1-9.5:** 26
selection, **5.1-5.6:** 25
and severe corrosion or abrasion, **9.1-9.5:** 15
silicon bronze, **9.1-9.5:** 20
and thermal or hydraulic shock, **9.1-9.5:** 16
thermoplastics, **9.1-9.5:** 25
thermosetting polymers, **9.1-9.5:** 25
tin bronze, **9.1-9.5:** 20
tin-base bearing metals, **9.1-9.5:** 23
titanium alloys, **9.1-9.5:** 23
used for pumping various liquids, **9.1-9.5:** 27, 29t.–37t.
and velocity effects, **9.1-9.5:** 16
vertical pumps, **9.1-9.5:** 16
for wetted pump parts, **9.1-9.5:** 11
yellow brass, **9.1-9.5:** 20
zinc and zinc alloys, **9.1-9.5:** 23
zirconium, **9.1-9.5:** 23
Maximum allowable casing working pressure, **1.1-1.2:** 60, **2.1-2.2:** 23
Maximum allowable inlet working pressure, **3.1-3.5:** 17, **3.6:** 5
Maximum allowable working pressure, **3.1-3.5:** 17, **3.6:** 5
Maximum differential pressure, **3.1-3.5:** 17, **3.6:** 5
Maximum discharge pressure, **2.1-2.2:** 25
Maximum suction pressure, **1.1-1.2:** 58, 60, **2.1-2.2:** 22
Maximum working pressure, **4.1-4.6:** 9, **5.1-5.6:** 15
Maxwell, **4.1-4.6:** 8

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

MDP *See* Magnetic drive pump
Measurement of airborne sound *See* Airborne sound measurement
Mechanical integrity test, **5.1-5.6:** 40
Mechanical seal chamber, **9.1-9.5:** 4
Mechanical seal gland, **9.1-9.5:** 4
Mechanical seals, **1.3:** 68, **1.4:** 6, **3.1-3.5:** 5, 46, **9.1-9.5:** 3
  applications, **1.3:** 68
  classifications, **1.3:** 68, 69f.
  typical schematics, **1.3:** 68
Mechanical test, **1.6:** 23, **2.6:** 1, 22
  acceptance levels, **1.6:** 24, **2.6:** 23
  instrumentation, **1.6:** 23, **2.6:** 23
  objective, **1.6:** 23, **2.6:** 22
  operating conditions, **1.6:** 23, **2.6:** 23
  procedure, **1.6:** 23, **2.6:** 23
  records, **1.6:** 24, **2.6:** 24
  setup, **1.6:** 23, **2.6:** 22
  temperature instruments, **1.6:** 23
  vibration instruments, **1.6:** 23
Metallic-type piston packing, **8.1-8.5:** 19
  application, **8.1-8.5:** 19
  clearance, **8.1-8.5:** 19
  joints, **8.1-8.5:** 18f., 19
  material, **8.1-8.5:** 19
  maximum temperature for ring materials, **8.1-8.5:** 19t.
Metals
  galling resistance, **9.1-9.5:** 15
  microstructure, **9.1-9.5:** 15
Metering efficiency, **3.6:** 2
Metric units, **9.1-9.5:** 7
  conversion factors, **9.1-9.5:** 8t.–10t.
  rounded equivalents, **9.1-9.5:** 7t.
Microphone locations (airborne sound measurement), **9.1-9.5:** 50
  axially split case centrifugal pump, **9.1-9.5:** 55f.
  axially split case multistage centrifugal pump, **9.1-9.5:** 57f.
  double case centrifugal pump, **9.1-9.5:** 56f.
  horizontal end suction centrifugal pump, **9.1-9.5:** 54f.
  horizontal reciprocating pump, **9.1-9.5:** 57f.
  horizontal rotary gear pump, **9.1-9.5:** 59f.
  horizontal rotary screw pump, **9.1-9.5:** 59f.
  primary, **9.1-9.5:** 51
  vertical in-line centrifugal pump, **9.1-9.5:** 55f.
  vertical reciprocating pump, **9.1-9.5:** 58f.
  vertical rotary pump, **9.1-9.5:** 60f.
Microphone systems, **9.1-9.5:** 50
Mine dewatering, **1.3:** 4
Minimum flow, **1.3:** 43
Minimum spares, **1.1-1.2:** 27
Miscellaneous mechanical problems, **9.6.4:** 24
Mixed flow impellers, **2.1-2.2:** 3, 10f.
Mixed flow pumps, **1.1-1.2:** 3, 3f.
Model tests, **1.6:** 32, **2.6:** 32, **9.8:** 22
  acceptance criteria, **9.8:** 28
  equations, **2.6:** 33–34
  flow, **9.8:** 26
  free-surface vortices, **9.8:** 26, 26f.
  at increased head, **1.6:** 34, **2.6:** 34
  instrumentation and measuring techniques, **9.8:** 26
  liquid level, **9.8:** 26
  model scope, **9.8:** 25
  objectives, **9.8:** 23
  pre-swirl, **9.8:** 27
  procedure, **1.6:** 32, **2.6:** 32–34
  report preparation, **9.8:** 28
  similitude and scale selection, **9.8:** 24
  sub-surface vortices, **9.8:** 26f., 27
  swirl in the suction pipe, **9.8:** 27
  swirl meters, **9.8:** 27, 27f.
  test plan, **9.8:** 28
  velocity profiles, **9.8:** 27
Modified radial flow impellers, **2.1-2.2:** 3, 10f.
Molded ring packings, **8.1-8.5:** 17
Monitoring
  baseline, **9.6.5:** 1
  failure mode indicators, **9.6.5:** 1, 18–21
  frequency, **9.6.5:** 1–2
  indicators, **9.6.5:** 22–24
Monitoring devices, **4.1-4.6:** 16, 20
Monitoring equipment, **5.1-5.6:** 15
Motor dimensions
  face-mounted, **1.1-1.2:** 49t.
  HP and HPH vertical solid-shaft, **1.1-1.2:** 53f., 53t., 54t.
  type JM, **1.1-1.2:** 51t.
  type JM having rolling contact bearings, **1.1-1.2:** 50f.
  type JP, **1.1-1.2:** 52t.
  type JP having rolling contact bearings, **1.1-1.2:** 50f.
Motor efficiency, **3.6:** 6, 19
Motor insulation, **5.1-5.6:** 13
  temperature limits, **5.1-5.6:** 26
Motor power, **3.6:** 19
Motor winding integrity test, **5.1-5.6:** 40
Motor winding temperature test, **5.1-5.6:** 40
Mounting
  base, **5.1-5.6:** 21
  horizontal mounting base, **5.1-5.6:** 21
  submerged, **5.1-5.6:** 21
  vertical, **5.1-5.6:** 21
Mounting, above and below floor discharge, **2.1-2.2:** 2, 9f., 11f.
MSDS *See* Material Safety Data Sheets
Mud pump, **9.1-9.5:** 4
Multiple screw pump, **3.1-3.5:** 11f.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Multiplex pump, **6.1-6.5:** 2
Multistage pumps, **9.6.1:** 4
Multi-volute casings, **1.3:** 76

n *See* Speed, **11.6:** 3
Natural frequency, **9.6.4:** 6, 7
  and resonance, **9.6.4:** 23
Negative thrust, **4.1-4.6:** 9
Neodymium, **4.1-4.6:** 8, **5.1-5.6:** 14
Net positive inlet pressure, **6.6:** 5
  See also Net positive suction head available, NPSHA test
Net positive inlet pressure available, **3.1-3.5:** 17, **3.6:** 5, **6.1-6.5:** 25
Net positive inlet pressure required, **3.1-3.5:** 17, 23, **3.6:** 1, 5, **6.1-6.5:** 25, **6.6:** 5, **8.1-8.5:** 9
  See also Net positive suction head required, NPSHR
  acceptable deviation of quantities, **3.6:** 15
  test, **3.6:** 15
  and viscosity, **3.1-3.5:** 23
Net positive suction head, **1.3:** 38–4
  allowable, **9.6.3:** 3
  available, **1.1-1.2:** 58
  insufficient, **1.3:** 43
  margin, **2.3:** 21
  margin considerations, **1.3:** 39
  NPSHA corrections for temperature and elevation, **1.3:** 38
  reduction, **1.3:** 39, 40f., 41f.
  reduction for liquids other than hydrocarbons or water, **1.3:** 40f., 41f., 42
  required, **1.1-1.2:** 58
  requirements for pumps handling hydrocarbon liquids and water at elevated temperatures, **1.3:** 39, 40f., 41f., **2.3:** 22, 23f., 24f.
Net positive suction head available, **1.6:** 6, **2.1-2.2:** 22, **2.3:** 19, 7, **6.1-6.5:** 25, **6.6:** 5, **8.1-8.5:** 10, **9.6.1:** 1, 1f., **11.6:** 5
  calculation on a dry-pit pump, **11.6:** 30
  calculation on a wet-pit pump, **11.6:** 30
  correction to rated speed, **6.6:** 10
  corrections for temperature and elevation, **2.3:** 20
Net positive suction head margin
  See NPSH margin
Net positive suction head required, **1.6:** 1, 7, **2.1-2.2:** 22, **2.6:** 7, **6.1-6.5:** 25, **6.6:** 5, **8.1-8.5:** 9, **9.6.1:** 1, 1f., **11.6:** 5
  See also NPSHR test
  correction to rated speed, **6.6:** 10
Net positive suction head required test, **2.6:** 1, 18
  arrangements, **2.6:** 18–20
  at constant rate of flow, **2.6:** 21f., 20, 20f.
  correction to rated speed from test speed, **2.6:** 21
  experimental deviation from the square law, **2.6:** 21
  objective, **2.6:** 18
  procedure, **2.6:** 20, 20f., 21f.
  records, **2.6:** 22
  report, **2.6:** 22
  test suction conditions, **2.6:** 22
  at varying rate of flow, **2.6:** 20, 21f.
Net positive suction head test. *See* Submersible pump NPSH test
Newtonian fluids, **3.1-3.5:** 19
Nickel alloys, **9.1-9.5:** 21
Nickel copper alloys, **9.1-9.5:** 21
Nickel or cobalt-chromium boron alloy, **9.1-9.5:** 23
Nickel-chromium-iron alloys, **9.1-9.5:** 21
Nickel-molybdenum alloys, **9.1-9.5:** 21
Nickel-molybdenum-chromium alloys, **9.1-9.5:** 21
NIST, **9.1-9.5:** 50
Noise, **1.4:** 15, **2.4:** 12, **9.6.3:** 2
  hydraulic resonance in piping, **2.4:** 13
Noise levels, **1.3:** 57, **2.3:** 18, **3.1-3.5:** 27–29
Nomenclature, **4.1-4.6:** 5t.–6t., **8.1-8.5:** 3, **9.8:** 38
  alphabetical listing, **1.1-1.2:** 27t.–35t.
  numerical listing, **1.1-1.2:** 35t.–38t.
Non-clog pumps, **1.3:** 14
Non-homogeneous flow, **6.1-6.5:** 27, **9.1-9.5:** 5
Non-lubricating liquid, **4.1-4.6:** 14
Non-Newtonian fluids, **3.1-3.5:** 22
Nonreverse ratchets, **2.3:** 46, **2.4:** 8
Non-settling slurry, **6.1-6.5:** 27, **9.1-9.5:** 5
Normal condition point, **1.1-1.2:** 58, **1.6:** 1, **2.1-2.2:** 22, **2.6:** 1, **6.6:** 1
Nozzle loads
  axial split case pumps, **9.6.2:** 15
  end suction slurry pumps, **9.6.2:** 16
  horizontal end suction pumps, **9.6.2:** 1
  vertical turbine short set pumps, **9.6.2:** 17
  vertical-in-line pumps, **9.6.2:** 10
Nozzle stress, **3.1-3.5:** 41
Nozzles, **6.6:** 14, 15t., **9.1-9.5:** 4
NPIPA *See* Net positive inlet pressure available
NPIPR *See* Net positive inlet pressure required
NPSH margin, **9.6.1:** 1, 10
  building services pumping systems, **9.6.1:** 9
  chemical process pumps, **9.6.1:** 6
  cooling towers, **9.6.1:** 7
  defined
  electric power pumps, **9.6.1:** 7
  general industrial pumps, **9.6.1:** 9
  guidelines, **9.6.1:** 4, 5t.
  nuclear power pumps, **9.6.1:** 7
  petroleum process pumps, **9.6.1:** 6
  pipeline pumps, **9.6.1:** 10
  pulp and paper pumps, **9.6.1:** 9
  ratio, **9.6.1:** 1
  slurry service pumps, **9.6.1:** 9
  and vertical turbine pumps, **9.6.1:** 6
  water/wastewater pumps, **9.6.1:** 8

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

waterflood (injection) pumps, **9.6.1:** 10
NPSH *See* Net positive suction head
NPSHA margin, **9.6.3:** 3, 3f.
NPSHA *See also* Net positive suction head available
NPSHA *See* Net positive suction head allowable
NPSHA. *See* Net positive suction head available
NPSHR *See* Net positive suction head required
NPSHR *See also* Net positive suction head required test
NPSHR test, **1.6:** 19
arrangements, **1.6:** 19, 19f., 20f.
closed tank supply, **6.6:** 11, 12f.
constant level supply, **6.6:** 11
correction to rated speed, **6.6:** 12
data presentation, **6.6:** 12, 13f.
equipment arrangements, **6.6:** 11, 11f., 12f.
level control with deep sump supply, **1.6:** 20f., 20
minimizing water aeration, **6.6:** 12
objective, **1.6:** 19, **6.6:** 11
procedure, **1.6:** 20, **6.6:** 12
with rate of flow held constant, **1.6:** 21, 21f.
records, **1.6:** 22
report, **1.6:** 23
suction conditions, **1.6:** 22
with suction head held constant, **1.6:** 21, 21f.
sump supply, **6.6:** 11f., 11
suppression type with constant level sump, **1.6:** 19f., 19
test liquid, **6.6:** 11
tolerance parameters, **6.6:** 12
vacuum and/or heat control with closed loop, **1.6:** 20f., 20
NPSHR. *See* Net positive suction head required
NST *See* Turbine specific speed
Nuclear power pumps, **9.6.1:** 7

Octave-band analyzers, **9.1-9.5:** 50
Octave-band sound pressure levels, **9.1-9.5:** 50, 51, 52
Oersted, **4.1-4.6:** 9
Off design rating procedures, **4.1-4.6:** 15
Offset couplings, **1.3:** 67
Oil lubricated pumps, **2.3:** 44
Oil seal, **9.1-9.5:** 4
Open/enclosed impeller, **2.1-2.2:** 2, 6f., 12f.
Open/enclosed lineshaft, **2.1-2.2:** 2, 6f.
Open feedwater cycle, **1.3:** 7, 7f., **2.3:** 9, 10f.
Open lineshafts, **2.3:** 43
Open suction tests, **2.6:** 4, 4f., 6
Operating principles, **4.1-4.6:** 11
Operating range, **2.3:** 17, 17f.
Operation, **4.1-4.6:** 19–21, **5.1-5.6:** 32
bearing lubrication, **1.4:** 11
bypass, **1.4:** 13
checking speed, rate of flow, pressure, power, vibration and leaks, **2.4:** 11
decoupling, **5.1-5.6:** 35
draw-down in wells, **2.4:** 11
filling, **1.4:** 10
flushing, **1.4:** 10
lubrication, **3.1-3.5:** 44
minimum flow, **1.4:** 13
parallel, **1.4:** 14, **2.4:** 12
precautions, **5.1-5.6:** 34
pre-filling, **1.4:** 11
pre-startup, **3.1-3.5:** 44
prevention without liquid flow, **4.1-4.6:** 19
priming, **1.4:** 10
range, **5.1-5.6:** 25
reduced flow, **1.4:** 13
reverse runaway speed, **1.4:** 14, 14f., **2.4:** 12, 13f.
series, **1.4:** 14, **2.4:** 12
shutdown, **3.1-3.5:** 45
starting, **5.1-5.6:** 34
start-up, **1.4:** 12, **2.4:** 10, **3.1-3.5:** 44
stopping, **2.4:** 12
system preparation, **1.4:** 10, **2.4:** 9
valve setting, **2.4:** 10
vibration, **5.1-5.6:** 35
water hammer, **1.4:** 13, **2.4:** 11
Operation and maintenance, **8.1-8.5:** 14
Orifice, **9.1-9.5:** 4
Outer magnet ring, **5.1-5.6:** 14
Outer magnetic assembly, **4.1-4.6:** 9, 12
Outlet, **3.1-3.5:** 4, 33, **9.1-9.5:** 4
Outlet port, **9.1-9.5:** 4
Outlet pressure, **3.1-3.5:** 16, **3.6:** 4, 9
Outside-adjustable lost-motion valve, **8.1-8.5:** 6f.
Overall efficiency, **1.1-1.2:** 58, **1.6:** 7, **2.1-2.2:** 23f., **2.6:** 8, **3.6:** 6, **6.6:** 5, **11.6:** 6
calculation, **2.6:** 16
Overhung impeller
separately coupled single stage–frame mounted, **1.1-1.2:** 39f.
Overhung impeller pumps, **1.1-1.2:** 1f., 2, **1.3:** 70
close couple single stage-diffuser style–end suction–submersible, **1.1-1.2:** 5f.
close coupled single stage–end suction, **1.1-1.2:** 4f.
close coupled single stage–in-line, **1.1-1.2:** 7f.
close coupled single stage–submersible, **1.1-1.2:** 6f.
close coupled–single stage–end suction, **1.1-1.2:** 47f.
separately coupled single stage–axial flow–horizontal, **1.1-1.2:** 15f.
separately coupled single stage–centerline mounted, **1.1-1.2:** 41f.
separately coupled single stage–centerline mounted (top suction), **1.1-1.2:** 43f.
separately coupled single stage–centerline mounted pump on base plate (top suction), **1.1-1.2:** 44f.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Overhung impeller pumps (continued)
separately coupled single stage–centerline mounted–pump on base plate, **1.1-1.2:** 42f.
separately coupled single stage–centerline support–API 610, **1.1-1.2:** 12f.
separately coupled single stage–frame mounted, **1.1-1.2:** 10f.
separately coupled single stage–frame mounted–ANSI B73.1, **1.1-1.2:** 13f.
separately coupled single stage–frame mounted–lined pump, **1.1-1.2:** 11f.
separately coupled single stage–frame mounted–mixed flow, **1.1-1.2:** 21f.
separately coupled single stage–frame mounted–self-priming, **1.1-1.2:** 24f.
separately coupled single stage–in-line–flexible coupling, **1.1-1.2:** 8f.
separately coupled single stage–in-line–rigid coupling, **1.1-1.2:** 9f.
separately coupled single stage–wet pit volute, **1.1-1.2:** 14f.
separately coupled–single stage–frame mounted (vertically mounted), **1.1-1.2:** 47f.
separately coupled–single stage–frame mounted–pump on base plate, **1.1-1.2:** 40f.

p *See* Pressure
P *See* Power
P_{mot} *See* Submersible motor input power
P_p *See* Pump input power
P_w *See* Pump output power
p_{acc} *See* Acceleration pressure
p_b *See* Barometric pressure
P_{ba} *See* Bowl assembly input power
p_d *See* Discharge pressure
p_d *See* Outlet pressure
p_d *See* Total discharge pressure
p_d *See* Working pressure
p_f *See* Friction loss pressure
p_g *See* Gauge pressure
p_H *See* Total differential pressure
P_{mot} *See* Electric driver input power
P_{mot} *See* Electric motor input power
p_{mot} *See* Total input power
P_p *See* Pump input power
p_s *See* Inlet pressure
p_s *See* Suction pressure
p_s *See* Total suction pressure
p_v *See* Velocity pressure
P_w *See* Pump output power
P_{wba} *See* Bowl assembly output power
p_z *See* Elevation pressure
Pacemakers (precautions), **5.1-5.6:** 32
Pacemakers and magnets, **4.1-4.6:** 19
Packed stuffing-box, **1.3:** 69, 69f., 70f.

Packing, **3.1-3.5:** 5, 46, **9.1-9.5:** 4
allowance for expansion, **8.1-8.5:** 16
basis of recommendations, **8.1-8.5:** 17
canvas, **8.1-8.5:** 17
chemical, **8.1-8.5:** 17
clearance, **8.1-8.5:** 18
drip, **8.1-8.5:** 17
fitting, **8.1-8.5:** 18, 18f.
gland adjustment, **8.1-8.5:** 16
hydraulic packing, **8.1-8.5:** 17
installation, **8.1-8.5:** 15
lubrication, **8.1-8.5:** 17
molded ring, **8.1-8.5:** 17
soaking, **8.1-8.5:** 18
swelling, **8.1-8.5:** 18
Packing box, **9.1-9.5:** 4
Packing gland, **9.1-9.5:** 4
Paper stock, **1.3:** 15
See also Pulp and paper applications
Parallel misalignment, **3.1-3.5:** 36, 37, 37f.
Parallel operation, **1.4:** 14, **2.4:** 12
Parallel operation and rate of flow, **2.3:** 17, 17f.
Parasitic losses, **5.1-5.6:** 12
Partially suspended solids, **9.1-9.5:** 5
Particles, **4.1-4.6:** 14
Parts, **2.1-2.2:** 3, 6f.–12f.
alphabetical listing, **2.1-2.2:** 14t.–18t.
maintenance review, **4.1-4.6:** 21
names of, **4.1-4.6:** 5t.–6t.
Parts replacements, **2.4:** 14
PATs *See* Pumps as turbines
Percent accumulation, **3.1-3.5:** 5
Percent overpressure, **3.1-3.5:** 5
Percent regulation, **3.1-3.5:** 5
Percent solids by volume, **6.1-6.5:** 27, **9.1-9.5:** 5
Percent solids by weight, **6.1-6.5:** 27, **9.1-9.5:** 5
Performance and selection criteria, **1.3:** 21
Performance test, **1.6:** 9, **2.6:** 1, 9, **6.6:** 1
acceptable deviation of dependent test quantities from specified values, **3.6:** 7
acceptable deviation of independent test quantities from specified values, **3.6:** 6
acceptable instrument fluctuation, **6.6:** 6
acceptance, **3.6:** 6, **6.6:** 5
acceptance criteria, **2.6:** 9
acceptance levels, **1.6:** 9
acceptance tolerances, **1.6:** 9, **2.6:** 9
acceptance values, **6.6:** 6
accuracy, **3.6:** 7
bowl assembly, **2.6:** 11, 11f.
calculation of bowl assembly efficiency, **2.6:** 16
calculation of bowl assembly total head, **2.6:** 15
calculation of efficiency, **2.6:** 15, **6.6:** 9
calculation of inlet or suction pressure, **6.6:** 9
calculation of input power, **6.6:** 9

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

calculation of outlet or discharge pressure, **6.6:** 9
calculation of output power, **6.6:** 9
calculation of overall efficiency, **2.6:** 16
calculation of pump efficiency, **2.6:** 16
calculation of pump input power, **2.6:** 15
calculation of total differential pressure, **6.6:** 9
calculation of total discharge head, **2.6:** 13
calculation of total head, **2.6:** 15
calculations, **1.6:** 15, **3.6:** 11, **6.6:** 9
calculations of pump output power, **2.6:** 15
calculations of total suction head, **2.6:** 13
calibration interval for instruments, **1.6:** 11, 12t.
correcting for solids in suspension, **2.6:** 18
correcting for specific weight variations, **2.6:** 18
correcting for speed variations, **2.6:** 17
correcting for viscosity variations, **2.6:** 18
correction for solids in suspension, **1.6:** 19
correction for temperature variations, **1.6:** 18
correction for viscosity, **6.6:** 10
correction for viscosity variations, **1.6:** 19
correction to rated speed, **1.6:** 17, **6.6:** 10
data requirements, **1.6:** 13, **2.6:** 13, 14f.
data sheet, **6.6:** 7, 8f.
differential pressure formulas, **3.6:** 11
efficiency calculation, **1.6:** 16
efficiency formulas, **3.6:** 11
fluctuation, **3.6:** 7
fluctuation and accuracy, **2.6:** 11t.
at increased speed, **1.6:** 17, **2.6:** 17
inlet conditions, **3.6:** 8
input power calculation, **1.6:** 15
input power formulas, **3.6:** 11
instrument calibration interval, **2.6:** 9, 10t.
instrument fluctuation and accuracy, **2.6:** 10
instrumentation, **1.6:** 11, **2.6:** 9, **3.6:** 7, 20, 21t., **6.6:** 6
instrumentation accuracy, **1.6:** 11
instrumentation fluctuation, **1.6:** 11
key conditions, **3.6:** 8
Level A acceptance, **3.6:** 6
level A acceptance, **1.6:** 9
Level B acceptance, **3.6:** 6
level B acceptance, **1.6:** 9
liquid conditions, **3.6:** 9
at non-rated conditions, **2.6:** 16–18
open or closed tank, **1.6:** 13f.
at other than rated speed, **1.6:** 16
outlet pressure, **3.6:** 9
output power calculation, **1.6:** 15
output power formulas, **3.6:** 11
plotting of results, **3.6:** 12, 12f.
plotting results, **1.6:** 16, 16f., **2.6:** 16, 16f., **6.6:** 9, 10f.
power correction (formula), **3.6:** 11, 12f.
procedure, **3.6:** 9, **6.6:** 7
pump (closed loop), **2.6:** 11, 12f.
pump (closed suction), **2.6:** 11, 12f.
pump (general), **2.6:** 12
rate of flow correction (formula), **3.6:** 11
records, **1.6:** 15, **2.6:** 13, **3.6:** 10, **6.6:** 9
at reduced speed, **1.6:** 16, **2.6:** 16
report, **1.6:** 19, **2.6:** 18, **6.6:** 10
sample data sheet, **1.6:** 14
setup, **1.6:** 11, **2.6:** 11–8f., **6.6:** 6, 7f.
for specific weight variations, **1.6:** 18
speed, **3.6:** 9
with suction lift, **1.6:** 11f.
tabulation sheet, **3.6:** 10t.
and temperature variations, **2.6:** 17
terminology, **6.6:** 1–5
total discharge head calculation, **1.6:** 15
total head calculation, **1.6:** 15
total suction head calculation, **1.6:** 15
Type I, **3.6:** 10, **6.6:** 6
Type II, **3.6:** 10, **6.6:** 6
Type III, **6.6:** 6
Type III and IV, **3.6:** 7, 11
Type III and IV reports, **3.6:** 12, 14f.
witnessing, **1.6:** 9, **2.6:** 9, **3.6:** 6
witnessing of, **6.6:** 5
Performance. *See also* Submersible pump performance test
calculation based on change in pump impeller diameter, **11.6:** 29
calculation based on change in pump speed, **11.6:** 29
calculation of ranges based on level A and level B acceptance criteria tolerances, **11.6:** 31
Peripheral velocity, **9.6.1:** 2
Permeability (magnetic), **4.1-4.6:** 9
Permeance, **4.1-4.6:** 9
Petroleum process pumps, **9.6.1:** 6
Phenolic piston rings, **8.1-8.5:** 19
application, **8.1-8.5:** 19
clearance, **8.1-8.5:** 20
forms, **8.1-8.5:** 20
maximum concentration of chemicals, **8.1-8.5:** 19t.
Pilot-operated relief valve, **9.1-9.5:** 4
Pipe dope, **8.1-8.5:** 15
Pipe tape, **8.1-8.5:** 15
Pipeline pumps, **9.6.1:** 10
Piping, **2.3:** 45, **3.1-3.5:** 38, **5.1-5.6:** 33
See also Discharge piping, Suction piping
hydraulic resonance, **2.4:** 13
inlet, **3.1-3.5:** 39
jacket, **3.1-3.5:** 39
nozzle loads and criteria (limiting forces and moments), **3.1-3.5:** 39, 40t., 42t.
outlet, **3.1-3.5:** 39
pipe-to-pump alignment, **3.1-3.5:** 39f., 39

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Piston cups, **8.1-8.5:** 21f.
assembling, **8.1-8.5:** 21f.
composition, **8.1-8.5:** 20
inspection, **8.1-8.5:** 22
installation, **8.1-8.5:** 21
list of liquids and materials suitable for, **8.1-8.5:** 20
nut tightening, **8.1-8.5:** 21, 22f.
synthetic rubber, **8.1-8.5:** 21
Piston pumps, **3.1-3.5:** 1f., 2, 2f., **6.1-6.5:** 1, **6.1-6.5:** 2f.
cup type pistons, **6.1-6.5:** 64
typical service, **6.1-6.5:** 53–54
Piston rod load, **6.1-6.5:** 23
Piston rod packing
drip, **8.1-8.5:** 17
installation, **8.1-8.5:** 15
Piston rod packing installation, **6.1-6.5:** 60
allowance for expansion of packing, **6.1-6.5:** 61
chemical packings, **6.1-6.5:** 62
drip, **6.1-6.5:** 61
gland adjustment, **6.1-6.5:** 61
hydraulic piston packing, **6.1-6.5:** 62–63
lubrication of packing, **6.1-6.5:** 62
metallic piston-ring-type packing, **6.1-6.5:** 63–64
molded ring packings, **6.1-6.5:** 61
phenolic piston ring packing, **6.1-6.5:** 64, 64t.
piston packing, **6.1-6.5:** 62–64
Piston type, **8.1-8.5:** 1f., 3
Piston valves, **8.1-8.5:** 4, 6f.
Pit dimensional checks, **2.4:** 3
Pitot tubes, **6.6:** 15
Plastic fluids, **3.1-3.5:** 22
Plunger load, **6.1-6.5:** 23
Plunger or piston speed, **6.1-6.5:** 20
Plunger or piston velocity, **8.1-8.5:** 7
Plunger packing installation, **8.1-8.5:** 15
See also Piston rod packing installation
Plunger pumps, **6.1-6.5:** 1f., 1, 2f., 3f.
typical service, **6.1-6.5:** 53–54
Plunger type, **8.1-8.5:** 3
Poise, **3.1-3.5:** 19
Poles (N-S), **4.1-4.6:** 9, **5.1-5.6:** 14
Polymers
composites, **9.1-9.5:** 16, 25
elastomeric, **9.1-9.5:** 24
material selection for maximum continuous temperature of various liquids, **9.1-9.5:** 39, 40t.–44t., 45t.–49t.
rigid, **9.1-9.5:** 16, 25
thermosetting, **9.1-9.5:** 25
used in pump construction, **9.1-9.5:** 38
for various liquids, **9.1-9.5:** 37, 38t.
Popping pressure, **3.1-3.5:** 4
POR *See* Preferred operating region
Positive displacement pumps, **9.1-9.5:** 1, 2f.
Positive thrust, **4.1-4.6:** 9
Power, **1.1-1.2:** 58, **1.6:** 7, **2.1-2.2:** 23, **2.6:** 7, **3.1-3.5:** 18, **3.6:** 5, **6.1-6.5:** 23, **6.6:** 5, **11.6:** 5
checking, **2.4:** 11
consumption too high, **2.4:** 16
correction to rated speed, **6.6:** 10
input to motors, **3.6:** 19
measurement, **1.6:** 30, **2.6:** 31–32
measurements, **6.6:** 17
Power check, **1.4:** 13
Power consumption, excessive, **5.1-5.6:** 37
Power drive end, **5.1-5.6:** 12
Power end
connecting rod, **6.1-6.5:** 13, 14f.
crankpin bearing, **6.1-6.5:** 13, 14f.
crankshaft, **6.1-6.5:** 13, 13f.
crosshead extension (plunger extension), **6.1-6.5:** 14, 15f.
frame extension, **6.1-6.5:** 14, 15f.
main bearing, **6.1-6.5:** 13, 13f., 14f.
parts, **6.1-6.5:** 13–14, 15f.–18f., 19t.
power crosshead, **6.1-6.5:** 13, 14f.
power frame, **6.1-6.5:** 13, 13f.
wrist pin, **6.1-6.5:** 14, 15f.
wrist pin bearing, **6.1-6.5:** 14, 15f.
Power measurements, **3.6:** 18–19
Power monitoring, **9.6.5:** 3
control limits, **9.6.5:** 3
frequency, **9.6.5:** 3
indicators, **9.6.5:** 22
means, **9.6.5:** 3
Power plant pumps, **2.3:** 9
Power pump materials, **9.1-9.5:** 18
Precautions, **4.1-4.6:** 11, **5.1-5.6:** 34
hazardous materials, **5.1-5.6:** 32
with magnets, **5.1-5.6:** 32
starting, **5.1-5.6:** 34
Preferred measurement units, **9.1-9.5:** 7
conversion factors, **9.1-9.5:** 8t.–10t.
rounded equivalents, **9.1-9.5:** 7t.
Preferred operating region, **9.6.3:** 1
vertical pumps, **9.6.3:** 1
Pre-installation, **2.4:** 1
foundation bolts, **1.4:** 1, 2f.
foundation requirements, **1.4:** 1, **2.4:** 2, 2f.
handling equipment, **1.4:** 1
handling equipment for installation, **2.4:** 1
inspection, **2.4:** 1
installation tools, **1.4:** 1
location of unit, **1.4:** 2
long-term storage, **1.4:** 1, **2.4:** 1
maintenance and repair access, **1.4:** 1, **2.4:** 2
manufacturer's erecting engineer, **1.4:** 1

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

manufacturer's instructions, **1.4:** 1, **2.4:** 1
manufacturer's service personnel, **2.4:** 1
protection against elements and environment, **1.4:** 1
receiving inspection, **1.4:** 1
short-term storage, **1.4:** 1, **2.4:** 1
site preparation, **1.4:** 1, **2.4:** 1
suction and discharge pipes, **1.4:** 2
tools for installation, **2.4:** 1
unloading, **2.4:** 1
Pre-installation hydrotest, **9.6.5:** 15
axially split case pumps, **9.6.5:** 16
control limits, **9.6.5:** 16
double suction pumps, **9.6.5:** 16
frequency, **9.6.5:** 16
indicators, **9.6.5:** 24
means, **9.6.5:** 15
vertical double casing can type pumps, **9.6.5:** 16
warnings, **9.6.5:** 15
Pre-lubrication
controls and alarms, **2.4:** 8
drivers, **2.4:** 8
lube filtration types, **2.4:** 8
nonreverse ratchets, **2.4:** 8
pumps, **2.4:** 8
submersible pumps, **2.4:** 9
Pressure, **3.1-3.5:** 16, **3.6:** 2, **6.1-6.5:** 20–4, **8.1-8.5:** 7
calculation of inlet or suction pressure, **6.6:** 9
checking, **2.4:** 11
insufficient, **2.4:** 16, **5.1-5.6:** 36
measurement, **3.6:** 16–15
measurement by gauges, **3.6:** 18, 18f., **6.6:** 16, 17f., 17
measurement by other methods, **6.6:** 17
multiple measurement orifices, **3.6:** 17, 18f.
tap location, **6.6:** 15, 16t.
tap openings, **3.6:** 17, 17f., **6.6:** 14, 14f.
Pressure boundary leakage failure mode causes and indicators, **9.6.5:** 20t.
Pressure check, **1.4:** 13
Pressure monitoring, **9.6.5:** 7
control limits, **9.6.5:** 8
frequency, **9.6.5:** 8
indicators, **9.6.5:** 23
means, **9.6.5:** 8
Pressure pulsation, **3.1-3.5:** 27
Pressure rating, **5.1-5.6:** 25
Pressure tap opening, **2.6:** 25f.
Pre-startup, **3.1-3.5:** 44
Pre-swirl, **9.8:** 1, 27
Preventive maintenance, **3.1-3.5:** 45
Priming, **1.4:** 10, **2.4:** 9, **8.1-8.5:** 14
by ejector or exhauster, **1.4:** 10, **2.4:** 9
with foot valve, **1.4:** 10
by vacuum pumps, **1.4:** 10, **2.4:** 10
Priming time test, **1.6:** 24
conversion factor, **1.6:** 25
determination of maximum developed vacuum by means of dry vacuum test, **1.6:** 25
of self-priming pumps, **1.6:** 24
suction line, **1.6:** 24, 25f.
Probe locations, **9.6.4:** 8
between bearing, single and multistage, **9.6.4:** 17f.
end suction foot mounted, **9.6.4:** 9f.
end suction, centerline support, **9.6.4:** 13f.
end suction, close coupled horizontal and vertical in-line, **9.6.4:** 11f.
end suction, frame mounted, **9.6.4:** 12f.
end suction, hard metal and rubber-lined horizontal and vertical, **9.6.4:** 16f.
end suction, paper stock, **9.6.4:** 14f.
end suction, solids handling, horizontal and vertical, **9.6.4:** 15f.
vertical in-line, separately coupled, **9.6.4:** 10f.
vertical turbine, mixed flow and propeller type, **9.6.4:** 18f.
vertical turbine, short set pumps, assembled for shipment by the manufacturer, **9.6.4:** 19f.
Process service, **1.3:** 1, **2.3:** 3
Product lubricated bearings, **5.1-5.6:** 13
Propeller pumps *See* Axial flow pumps
Protection against seepage or flood, **8.1-8.5:** 14
Protective devices, **3.1-3.5:** 43
Proximity probes, **9.6.5:** 9
Pseudo-plastic fluids, **3.1-3.5:** 22
Pulp and paper applications, **1.3:** 15, **9.6.1:** 9
corrosion, **1.3:** 16
hydraulic performance correction, **1.3:** 16
paper stock and consistency categories, **1.3:** 16
Pulsation dampener, **9.1-9.5:** 4
Pump displacement, **6.6:** 3, **8.1-8.5:** 7
Pump duty cycle, **9.6.1:** 4
Pump efficiency, **1.1-1.2:** 59, **1.6:** 7, **2.1-2.2:** 23, **2.6:** 7, **3.1-3.5:** 18, **3.6:** 6, 11, **6.1-6.5:** 23, **6.6:** 5, **8.1-8.5:** 10
calculation, **1.6:** 16, **2.6:** 16, **6.6:** 9
Pump hydraulic efficiency, **11.6:** 6
Pump input power, **1.1-1.2:** 58, **1.6:** 7, **2.1-2.2:** 23, **2.6:** 7, **3.1-3.5:** 18, **3.6:** 5, 11, **6.1-6.5:** 23, **6.6:** 5, **11.6:** 5
calculation, **1.6:** 15, **2.6:** 15, **6.6:** 9
measurements, **1.6:** 30, **2.6:** 31, **3.6:** 18, **6.6:** 17
and viscosity, **3.1-3.5:** 23
Pump location, **8.1-8.5:** 14
Pump mechanical efficiency, **3.1-3.5:** 18, **6.1-6.5:** 23
Pump output power, **1.1-1.2:** 58, **1.6:** 7, **2.1-2.2:** 23, **2.6:** 7, **3.1-3.5:** 18, **3.6:** 6, 11, **6.1-6.5:** 23, **6.6:** 5, **11.6:** 6
calculation, **1.6:** 15, **2.6:** 15, **6.6:** 9

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Pump performance tests
closed loop, **2.6:** 11, 12f.
closed suction, **2.6:** 11, 12f.
general, **2.6:** 12
Pump pressures, **1.1-1.2:** 60, **2.1-2.2:** 23–25
Pump rate of flow, **8.1-8.5:** 7
Pump selection, **5.1-5.6:** 25
Pump selection criteria
axial thrust calculation, **2.3:** 41f., 41, 42
axial thrust terminology, **2.3:** 40
axial thrust vs. rate of flow, **2.3:** 42, 43f.
axial thrust with various impeller and shaft configurations, **2.3:** 38, 38f., 39f., 40f.
continuous service, **2.3:** 17
cyclic service, **2.3:** 17
handling slurry liquids, **2.3:** 36
handling viscous liquids, **2.3:** 22, 25f., 26f., 27f., 28f.
impeller diameter change and pump performance, **2.3:** 16, 16f.
intermittent service, **2.3:** 17
liquids with vapor or gas, **2.3:** 21, 21f.
losses, **2.3:** 33
net positive suction head available, **2.3:** 19
noise levels, **2.3:** 18
non-settling slurries, **2.3:** 36, 38f.
NPSH margin, **2.3:** 21
NPSH requirements for pumps handling hydrocarbon liquids and water at elevated temperatures, **2.3:** 22, 23f., 24f.
operating range, **2.3:** 17, 17f.
operation away from best efficiency point, **2.3:** 17
parallel operation and rate of flow, **2.3:** 17, 17f.
pump versus system curves, **2.3:** 13, 14f.
pumping system requirements, **2.3:** 13
and reduced rates of flow, **2.3:** 18
reverse runaway speed, **2.3:** 14
rotating speed limitations, **2.3:** 33, 34f., 35f.
and runout conditions, **2.3:** 18
series operation and rate of flow, **2.3:** 17, 17f.
settling slurries, **2.3:** 36, 38f.
slurries and construction materials, **2.3:** 36f.
slurries and performance changes, **2.3:** 36, 37f.
slurries and rotative speed, **2.3:** 38
speed torque curves, **2.3:** 15
start-up and shut-down analysis, **2.3:** 15
submergence, **2.3:** 19
suction conditions, **2.3:** 18
suction specific speed, **2.3:** 32
system pressure limitation, **2.3:** 14
water hammer analysis, **2.3:** 14
Pump shaft rotation, **2.1-2.2:** 3
Pump size, **9.6.1:** 4
Pump speeds, **8.1-8.5:** 12
Pump suction piping, **9.8:** 20, 21f., 21t., 22f., 23f.
Pump torque, **3.1-3.5:** 18, **6.1-6.5:** 23
characteristics, **6.1-6.5:** 34
requirements, **6.1-6.5:** 35
Pump total discharge head, **2.1-2.2:** 21
Pump versus system curve, **1.3:** 21, 21f., **2.3:** 13, 14f.
Pump vibration, **1.4:** 17, **2.4:** 12
Pump volumetric efficiency, **3.1-3.5:** 14, **3.6:** 2
Pumping chamber, **3.1-3.5:** 4
Pumping system requirements, **2.3:** 13
Pumping water level, **2.3:** 5
Pumps
characteristics, **4.1-4.6:** 17
constant speed pumping, **9.8:** 58, 59t., 60t.
decontamination of returned products, **9.1-9.5:** 61
defined, **9.1-9.5:** 4
hardware terms, **9.1-9.5:** 3
hydraulic phenomena adversely affecting, **9.8:** 1
kinetic, **9.1-9.5:** 1, 2f.
materials, **4.1-4.6:** 15
positive displacement, **9.1-9.5:** 1, 2f.
ratings, **4.1-4.6:** 17
slurry application terms, **9.1-9.5:** 5
sump volumes, **9.8:** 54
types of, **9.1-9.5:** 1, 2f.
variable speed, **9.8:** 58
Pumps as turbines, **2.3:** 11, 12f., 13f.
Pumps operating in parallel, **1.3:** 42, 42f.
Pumps operating in series, **1.3:** 42, 42f.
Pumps used as hydraulic turbines, **1.3:** 11
total available exhaust head (TAEH), **1.3:** 12
total required exhaust head (TREH), **1.3:** 12
turbine performance characteristics, **1.3:** 11, 11f., 12f.
turbine specific speed, **1.3:** 11
PWL *See* Pumping water level

Q *See* Flow rate
Q *See* Rate of flow
Q *See also* Pump rate of flow

Radial flow impellers, **2.1-2.2:** 3
Radial flow pumps, **1.1-1.2:** 3, 3f.
separately coupled single stage–(vertical) split case, **1.1-1.2:** 17f.
separately coupled–mulitstage–(vertical) split case, **1.1-1.2:** 19f.
separately coupled–mulitstage–(vertical) split–double casing, **1.1-1.2:** 20f.
Radial load, **5.1-5.6:** 13
Radial seal, **3.1-3.5:** 5, **9.1-9.5:** 4
Radial thrust
calculation for volute pumps, **1.3:** 58
excessive, **1.3:** 43

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Rate of flow, **1.1-1.2:** 55, **1.6:** 3, **3.1-3.5:** 14, **3.6:** 2, **6.1-6.5:** 20, **6.6:** 4, **8.1-8.5:** 7
checking, **2.4:** 11
correction formula, **3.6:** 11
correction to rated speed, **6.6:** 10
measurement, **3.6:** 15, **6.6:** 13
measurement by displacement type meters, **6.6:** 13
measurement by head type rate meters, **1.6:** 26, **6.6:** 13, 14f.
measurement by nozzles, **1.6:** 27, **6.6:** 14, 15t.
measurement by other methods, **1.6:** 29, **3.6:** 16, **6.6:** 15
measurement by pitot tubes, **6.6:** 15
measurement by thin square-edged orifice plate, **1.6:** 27, **6.6:** 14
measurement by venturi meter, **1.6:** 26, **6.6:** 14
measurement by volume, **1.6:** 25, **3.6:** 16, **6.6:** 13
measurement by weight, **1.6:** 25, **3.6:** 16, **6.6:** 13
measuring system requirements, **1.6:** 25
and parallel operation, **2.3:** 17, 17f.
pressure tap openings, **1.6:** 26, 26f.
pressure tap openings for head type rate meter measurements, **6.6:** 14, 14f.
reduced, **2.3:** 18
and series operation, **2.3:** 17, 17f.
straight pipe requirements associated with nozzle meters, **1.6:** 27, 28t.
straight pipe requirements associated with orifice plate meters, **1.6:** 28t.
straight pipe requirements associated with venturi meters, **1.6:** 26, 27t.
types, **1.6:** 25
Rate of flow (capacity), **2.1-2.2:** 19
defined, **2.6:** 3
measurement, **2.6:** 24
measurement by head type rate meters, **2.6:** 24, 25f.
measurement by nozzles, **2.6:** 25, 26t., 27t.
measurement by other methods, **2.6:** 27
measurement by pitot tubes, **2.6:** 27
measurement by thin, square-edged orifice plate, **2.6:** 25, 26t., 27t.
measurement by venturi meter, **2.6:** 25, 26t.
measurement by volume, **2.6:** 24
measurement by weight, **2.6:** 24
measurement by weirs, **2.6:** 25
Rate of flow monitoring, **9.6.5:** 11
control limits, **9.6.5:** 11
frequency, **9.6.5:** 11
indicators, **9.6.5:** 24
measuring rate of flow, **9.6.5:** 11
Rated (specified) condition point, **11.6:** 3
Rated condition point, **1.1-1.2:** 58, **1.6:** 1, **2.1-2.2:** 22, **2.6:** 1, **3.6:** 2, **6.6:** 1
Receiver-pulsation dampener, **9.1-9.5:** 4
Receiving inspection, **1.4:** 1
Reciprocating power pumps, **6.1-6.5:** 1
cup type pistons, **6.1-6.5:** 64
discharge piping, **6.1-6.5:** 45
foundation, **6.1-6.5:** 55
foundation bolts, **6.1-6.5:** 56, 56f.
inlet system, **6.1-6.5:** 38–45
inspection, **6.1-6.5:** 65–66
installation, **6.1-6.5:** 56–60
liquid end, **6.1-6.5:** 5–8, 9f., 10f., 11f., 12t.
location, **6.1-6.5:** 55
malfunctions, cause and remedies, **6.1-6.5:** 66t.–68t.
power end, **6.1-6.5:** 13–14, 15f.–18f., 19t.
pre-installation considerations, **6.1-6.5:** 55–56
protection against seepage or flood, **6.1-6.5:** 55
right and left hand shaft extension, **6.1-6.5:** 2–5
servicing space, **6.1-6.5:** 55
speeds, **6.1-6.5:** 29–34
starting, **6.1-6.5:** 34–38
storage, **6.1-6.5:** 55
types and nomenclature, **6.1-6.5:** 1
typical services, **6.1-6.5:** 29
Reciprocating power types, **6.1-6.5:** 1f.
Reciprocating pump materials, **9.1-9.5:** 18
Recirculation, **1.3:** 43
Recommended minimum spares, **1.1-1.2:** 27
Rectangular intakes
approach flow patterns, **9.8:** 1
design sequence, **9.8:** 5t.
dimensioning, **9.8:** 2
open vs. partitioned structures, **9.8:** 2
trash racks and screens, **9.8:** 2
Rectangular wet wells, **9.8:** 19
Reducers, **2.4:** 4, 4f., 5
Reed frequency, **9.6.4:** 6
See also Natural frequency
Reference materials, **4.1-4.6:** 23
References, **5.1-5.6:** 38
Regenerative turbine pumps, **1.1-1.2:** 1f., 1, 2, **1.4:** 1
impeller between bearings–two stage, **1.1-1.2:** 23f.
peripheral single stage, **1.1-1.2:** 22f.
side channel single stage, **1.1-1.2:** 22f.
Reinforced fibers, **9.1-9.5:** 26
Relief valve, **8.1-8.5:** 15, **9.1-9.5:** 4
Relief valves, **3.1-3.5:** 4, 43, **6.1-6.5:** 45
Reluctance, **4.1-4.6:** 9
Remedial measures, **9.8:** 42
approach flow patterns, **9.8:** 42, 43f., 44f., 45f.
cross-flow, **9.8:** 45, 46f.
expansion of concentrated flows, **9.8:** 46, 47f., 48f., 49f.
pump inlet disturbances, **9.8:** 48, 49f., 51f.
suction tank inlets, **9.8:** 50, 52f.
Repair access, **2.4:** 2
Reseating pressure, **3.1-3.5:** 5

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Resonance, **9.6.4:** 23
in piping, **9.6.4:** 24
Resonant frequency, **9.6.4:** 6
Return materials authorization number, **9.1-9.5:** 61
Reverse runaway speed, **1.3:** 22, **1.4:** 14, 14f., **2.3:** 14, **2.4:** 12, 13f.
Revolution counter, **9.1-9.5:** 4
Revolution counter and timer method, **1.6:** 31
Revolution counter and timer method of speed measurement, **6.6:** 18
Rheopectic fluids, **3.1-3.5:** 22
Right and left hand designations, **8.1-8.5:** 3
Rigid polymers and composites, **9.1-9.5:** 25
parts, **9.1-9.5:** 16
Rigidity, **9.6.4:** 24
RMA *See* Return materials authorization number
Rolling element bearings, **1.3:** 64, 64t.
Rotary pump materials, **9.1-9.5:** 17
Rotary pumps
data sheet for selection or design of, **3.1-3.5:** 29, 30f.–32f.
noise levels, **3.1-3.5:** 27–29
specified conditions chart, **3.1-3.5:** 24f.
types, **3.1-3.5:** 1, 1f.
typical operating conditions, **3.1-3.5:** 14
Rotary speed measurement, **3.6:** 19
Rotating assembly, **3.1-3.5:** 4
multistage, axially split, single or double suction centrifugal pumps, **1.1-1.2:** 26
single stage, axially (horizontally) split, single or double suction centrifugal pump, **1.1-1.2:** 25
Rotating speed limitations, **2.3:** 33, 34f., 35f.
Rotation, **1.4:** 13, **2.1-2.2:** 3
Rotation check, **3.1-3.5:** 35
Rotation of casing, **1.1-1.2:** 26
Rotation of pumps, **1.1-1.2:** 26, 26f.
Rotational inertia, **9.6.4:** 4, 5
Rotor, **3.1-3.5:** 4, **9.1-9.5:** 4
Rotor balancing, **9.6.4:** 20
allowable residual unbalance in pump impellers, **9.6.4:** 21f., 22f.
maximum looseness between balancing arbor and impeller, **9.6.4:** 23
Rotor lateral vibration, **9.6.4:** 1
See also Lateral critical speed
Rotor torsional vibration, **9.6.4:** 1
Rows of magnets, **5.1-5.6:** 14
RPM *See* Speed monitoring
Runout conditions, **2.3:** 18
Rupture, **9.1-9.5:** 3

s *See* Specific gravity
S *See* Slip
S *See* Suction specific speed
Safety, **6.1-6.5:** 55, **8.1-8.5:** 14
characteristics, **4.1-4.6:** 17
mechanical, **4.1-4.6:** 16
secondary containment, **4.1-4.6:** 16
secondary control, **4.1-4.6:** 16
with magnets, **4.1-4.6:** 19
Safety considerations, **5.1-5.6:** 23, **9.6.5:** 2
Saltation, **6.1-6.5:** 27, **9.1-9.5:** 6
Samarium cobalt, **4.1-4.6:** 8, **5.1-5.6:** 14
Sanitary pump, **1.3:** 14
Screw pumps, **3.1-3.5:** 1f., 3f., 3
Seal cage, **3.1-3.5:** 5
Seal chamber, **3.1-3.5:** 5, 13f., **9.1-9.5:** 4
Seal leakage failure mode causes and indicators, **9.6.5:** 18t.
Seal piping, **9.1-9.5:** 4
Sealants, **9.1-9.5:** 26
Sealing by impregnation, **9.1-9.5:** 12
Sealless (defined), **4.1-4.6:** 11, **9.1-9.5:** 4
Sealless centrifugal pumps
advantages, **5.1-5.6:** 23
alternative designs, **5.1-5.6:** 16
application guidelines, **5.1-5.6:** 23–26
defined, **5.1-5.6:** 12
design, **5.1-5.6:** 16–23
items to be avoided, **5.1-5.6:** 21
limitations, **5.1-5.6:** 23
nomenclature, **5.1-5.6:** 2, 10t.
reference and source material, **5.1-5.6:** 38
safety considerations, **5.1-5.6:** 23
special considerations, **5.1-5.6:** 16
types, **5.1-5.6:** 2f.
uses, **5.1-5.6:** 1
Sealless pumps
bearing wear monitoring (plain bearings), **9.6.5:** 14
failure mode causes and indicators, **9.6.5:** 21t.
temperature monitoring, **9.6.5:** 4
Sealless rotary pumps, **4.1-4.6:** 1
overview, **4.1-4.6:** 11
Second critical speed, **9.6.4:** 1f., 1
Secondary containment, **4.1-4.6:** 9, 16, **5.1-5.6:** 15
system, **4.1-4.6:** 9
Secondary control, **4.1-4.6:** 9, 16
system, **4.1-4.6:** 9
Seismic analysis, **2.4:** 14
Self-priming pumps, **1.3:** 13, 14f., 15f.
Separately coupled (defined), **4.1-4.6:** 9, **5.1-5.6:** 12
Separately coupled internal gear magnetic drive pump with secondary control, **4.1-4.6:** 1, 3f.
Separately coupled screw type magnetic drive pump, **4.1-4.6:** 1, 4f.
Series operation, **1.4:** 14, **2.4:** 12
Series operation and rate of flow, **2.3:** 17, 17f.
Servicing space, **8.1-8.5:** 14

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Set pressure, **3.1-3.5:** 4
Settling slurry, **6.1-6.5:** 28, **9.1-9.5:** 6
Settling velocity, **6.1-6.5:** 28, **9.1-9.5:** 6
Severity level, **9.6.5:** 1–2
Sewage pumps, **1.3:** 14
Shaft breakage mode causes and indicators, **9.6.5:** 19t.
Shaft deflection, **1.3:** 70
Shaft fatigue failure, **9.6.3:** 3
Shaft position monitoring, **9.6.5:** 11
  frequency, **9.6.5:** 11
  indicators, **9.6.5:** 24
  proximity probes, **9.6.5:** 11
Shaft seal life, **9.6.3:** 2
Shaft seals
  alternative, **1.3:** 70
  mechanical seals, **1.3:** 68, 69f.
  packed stuffing-box, **1.3:** 69, 69f.
Shafting, **2.3:** 43
  pump-to-driver, **2.3:** 46
Shear pin relief valve, **9.1-9.5:** 4
Shear rate, **3.1-3.5:** 19
Shear stress, **3.1-3.5:** 19
Shipment inspection, **3.1-3.5:** 33
Shipping of magnets, **4.1-4.6:** 19
Short-term storage, **1.4:** 1
Shut off, **1.1-1.2:** 58, **1.6:** 1, **2.6:** 1, **11.6:** 3
Shutdown, **1.3:** 22, **3.1-3.5:** 45
Shut-down analysis, **2.3:** 15
Shutdown limit (defined), **9.6.5:** 2
Shutoff, **2.1-2.2:** 22
Silicon bronze, **9.1-9.5:** 20
Silicon carbide, **5.1-5.6:** 13
Simplex pump, **6.1-6.5:** 2f., 2, 3f.
Single plane balancing, **1.1-1.2:** 60
Single suction pump specific speed, **1.3:** 32, 33f., 34f.
Single volute casing, **1.3:** 58, 58f., 76
  K versus rate of flow, **1.3:** 58, 59f.
Single-acting pump, **6.1-6.5:** 1f., 1, 2f.
Site preparation, **2.4:** 1
  foundation bolts, **1.4:** 1, 2f.
  foundation requirements, **1.4:** 1
  location of unit, **1.4:** 2
  maintenance access, **1.4:** 1
  protection against elements and environment, **1.4:** 1
  suction and discharge pipes, **1.4:** 2
Sleeve bearings, **1.3:** 64, **9.1-9.5:** 4
Slip, **3.1-3.5:** 14, **3.6:** 2, **5.1-5.6:** 14, **6.1-6.5:** 20, **6.6:** 4, **8.1-8.5:** 7
  hydraulic, **4.1-4.6:** 10
  magnetic, **4.1-4.6:** 9
  and slurries, **3.1-3.5:** 26
  and viscosity, **3.1-3.5:** 23
Sluice gates, **9.8:** 60
Slurries, **2.3:** 36, **3.1-3.5:** 24
  apparent viscosity vs. shear rate, **3.1-3.5:** 25, 26f.
  carrier liquids, **3.1-3.5:** 24
  characteristics, **3.1-3.5:** 24
  clearance provision for particle size, **3.1-3.5:** 26
  concentration of solids in, **3.1-3.5:** 25
  and construction materials, **2.3:** 36
  construction materials for, **3.1-3.5:** 27
  corrosion effect on wear, **3.1-3.5:** 27
  flow velocity, **3.1-3.5:** 26
  hardness of solids in, **3.1-3.5:** 25, 25f.
  non-settling, **2.3:** 36, 38f.
  operating sequences, **3.1-3.5:** 27
  and performance changes, **2.3:** 36, 37f., **3.1-3.5:** 26, 26f.
  pressure relief provision, **3.1-3.5:** 27
  pump design for, **3.1-3.5:** 27
  and rotative speed, **2.3:** 38
  sealing against, **3.1-3.5:** 27
  settling, **2.3:** 36, 38f.
  settling characteristics, **3.1-3.5:** 25
  shear rate effect on friction power, **3.1-3.5:** 26
  shear rate effect on slip, **3.1-3.5:** 26
  size of solids in, **3.1-3.5:** 25
  speed effect on wear, **3.1-3.5:** 27
  speed effects, **3.1-3.5:** 26
  testing and modeling for, **3.1-3.5:** 27
  wear, **3.1-3.5:** 27
Slurry, **6.1-6.5:** 27, **9.1-9.5:** 6
Slurry application terms, **9.1-9.5:** 5
Slurry service, **1.3:** 17–19
  materials of construction for slurry pumps, **1.3:** 17
  non-settling slurries, **1.3:** 17, 19f.
  relationship between concentration and specific gravity for aqueous slurries, **1.3:** 17, 18f.
  rotational speed of slurry pumps, **1.3:** 19
  settling slurries, **1.3:** 17, 19f.
Slurry service pumps, **9.6.1:** 9
Slush pump, **9.1-9.5:** 4
Smothering gland, **9.1-9.5:** 5
SO *See* Shut off
Soft start drivers, **6.1-6.5:** 37
Solids/abrasives in liquid, **9.6.1:** 4
Soluble chloride, **9.1-9.5:** 11
Sound level meters, **9.1-9.5:** 50
Source material, **5.1-5.6:** 38
Spacer type couplings, **3.1-3.5:** 37
Spare parts, **1.1-1.2:** 27, **3.1-3.5:** 46
Specific composition bronze pumps, **9.1-9.5:** 16, 17
Specific gravity, **3.1-3.5:** 23, **3.6:** 6, **4.1-4.6:** 14, **9.6.1:** 2
Specific heat, **4.1-4.6:** 14
Specific speed, **1.1-1.2:** 2, 3f., 59, **2.1-2.2:** 2
Specific weight, **3.6:** 6

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Specifications, **4.1-4 .6:** 17, 18f.
Specified condition point, **1.1-1.2:** 58, **1.6:** 1, **2.1-2.2:** 22, **2.6:** 1, **3.6:** 2
Speed, **1.1-1.2:** 55, **1.6:** 3, **2.1-2.2:** 19, **2.6:** 3, **3.1-3.5:** 14, **3.6:** 2, 9, **6.1-6.5:** 20, **8.1-8.5:** 7, **11.6:** 3
  See also Reverse runaway speed
  checking, **2.4:** 11
  measurement, **1.6:** 31, **2.6:** 32
  and viscosity, **3.1-3.5:** 23
Speed check, **1.4:** 13
Speed measurement, **6.6:** 18
Speed monitoring, **9.6.5:** 13
  constant speed systems, **9.6.5:** 14
  control limits, **9.6.5:** 14
  by electric counter, **9.6.5:** 14
  frequency, **9.6.5:** 14
  indicators, **9.6.5:** 24
  methods, **9.6.5:** 14
  by revolution counter, **9.6.5:** 14
  by strobe light, **9.6.5:** 14
  by tachometer, **9.6.5:** 14
  variable speed systems, **9.6.5:** 14
Speeds
  and application details, **6.1-6.5:** 33
  basic speed ratings and formulas, **6.1-6.5:** 29–33
  factors affecting operating speed, **6.1-6.5:** 33
  high, **6.1-6.5:** 34
  and liquid characteristics, **6.1-6.5:** 33
  medium, **6.1-6.5:** 33
  and pump design, **6.1-6.5:** 33
  slow, **6.1-6.5:** 34
  and type of duty, **6.1-6.5:** 33
Speed-torque curves, **1.4:** 13, **2.3:** 15, **2.4:** 10
Square root law, **9.1-9.5:** 6
Stainless steel fitted pumps, **9.1-9.5:** 16
Standards-setting organizations, **11.6:** 32
Start, **1.4:** 12
Starting, **5.1-5.6:** 34, **6.1-6.5:** 34
  with liquid bypass, **6.1-6.5:** 35, 36f.
  without liquid bypass, **6.1-6.5:** 35
  pump torque characteristics, **6.1-6.5:** 34
  pump torque requirements, **6.1-6.5:** 35
  soft start drivers, **6.1-6.5:** 37
  torque, **5.1-5.6:** 13
Start-to-discharge pressure, **3.1-3.5:** 4
Start-up, **2.4:** 10, **3.1-3.5:** 44
  across-the-line, **2.4:** 10
  caution, **2.4:** 10
  discharge valve position, **1.4:** 12
  dowelling, **1.4:** 13
  final alignment check, **1.4:** 13
  flow rate check, **1.4:** 13
  leak check, **1.4:** 13
  misalignment causes, **1.4:** 13
  motor, **1.4:** 13
  power check, **1.4:** 13
  pressure check, **1.4:** 13
  reduced voltage, **2.4:** 10
  rotation, **1.4:** 13
  speed check, **1.4:** 13
  speed-torque curves, **1.4:** 13, **2.4:** 10
  valve setting, **2.4:** 11
  vibration check, **1.4:** 13
  with closed discharge valve, **1.3:** 22
  with open discharge valve, **1.3:** 22
Start-up analysis, **2.3:** 15
Static balancing, **1.1-1.2:** 60
Static suction lift, **1.1-1.2:** 58, **2.1-2.2:** 22, **6.1-6.5:** 25, **8.1-8.5:** 10
Static water level, **2.3:** 5
Stator, **3.1-3.5:** 4
Steam electric power plants, **1.3:** 4, 5f., **2.3:** 6
  boiler circulating pumps, **1.3:** 10
  boiler feed booster pumps, **1.3:** 9
  boiler feed pumps, **1.3:** 8
  closed feedwater cycle, **1.3:** 6, 7f.
  condensate pumps, **1.3:** 9
  condenser circulating pumps, **1.3:** 9
  heater drain pumps, **1.3:** 10
  open feedwater cycle, **1.3:** 7, 7f.
  pumps, **1.3:** 8
  steam power cycle, **1.3:** 4, 5f.
Steam jacket, **9.1-9.5:** 5
Steam power cycle, **1.3:** 4, 5f., **2.3:** 7f., 7
Steam power plants, **2.3:** 6, 8f.
  closed feedwater cycle, **2.3:** 9f., 9
  condensate pumps, **2.3:** 9
  condenser circulating water pumps, **2.3:** 10
  heater drain pumps, **2.3:** 11
  open feedwater cycle, **2.3:** 9, 10f.
  power plant pumps, **2.3:** 9
  steam electric power plants, **2.3:** 6
  steam power cycle, **2.3:** 7f., 7
Steam turbine drivers, **1.3:** 77
Steel
  all stainless steel pumps, **9.1-9.5:** 16, 17
  carbon and low alloy steels, **9.1-9.5:** 19
  chromium (ferric) stainless steel, **9.1-9.5:** 20
  chromium-nickel (austenitic) stainless steel, **9.1-9.5:** 19
  duplex stainless steels, **9.1-9.5:** 20
  high alloy steels, **9.1-9.5:** 19
  stainless steel fitted pumps, **9.1-9.5:** 16
Stoke, **3.1-3.5:** 19
Stop valve, **9.1-9.5:** 5
Stopping, **2.4:** 12
Storage, **1.4:** 1, **3.1-3.5:** 33, **8.1-8.5:** 14
Storage (pre-installation), **2.4:** 1

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Strain gauge type torque measuring devices, **1.6:** 30, 31
Strainers, **3.1-3.5:** 42, **5.1-5.6:** 13
Stripping applications, **4.1-4.6:** 15
Stroboscopes, **1.6:** 31, **6.6:** 18
Stroke, **6.1-6.5:** 20, **6.6:** 3, **8.1-8.5:** 7
Structure dynamic analysis, **9.6.4:** 7
Structure lateral vibration, **9.6.4:** 1, 6
  vertical dry pit pumps, **9.6.4:** 6
  vertical wet pit pumps, **9.6.4:** 6
Structureborne noise, **3.1-3.5:** 28
Stuffing box, **3.1-3.5:** 5, 13f., **9.1-9.5:** 5
  area, **1.1-1.2:** 48f.
  bushings, **1.4:** 6, **9.1-9.5:** 5
  mechanical seals, **1.4:** 6, **2.4:** 7
  packing, **1.4:** 5, **2.4:** 7, 7f.
Submerged mounting, **5.1-5.6:** 21
Submerged suction, **1.1-1.2:** 58, **2.1-2.2:** 22, **6.1-6.5:** 24, **8.1-8.5:** 10
Submerged vortices, **9.8:** 1
Submergence, **1.1-1.2:** 57, **2.3:** 19
Submergence required for minimizing surface vortices, **9.8:** 29, 33f., 34f.
Submersible motor efficiency, **11.6:** 6
Submersible motor input power, **11.6:** 6
Submersible motor integrity tests
  electrical continuity and resistance test, **11.6:** 16
  electrical high-potential test, **11.6:** 17
  electrical megohmmeter resistance test, **11.6:** 17
  housing pressure test, **11.6:** 16, 16f.
  housing vacuum check, **11.6:** 16, 17f.
  objective, **11.6:** 15
  records, **11.6:** 17
  setup and procedure, **11.6:** 15
Submersible pump hydrostatic test
  acceptance criteria, **11.6:** 12
  objective, **11.6:** 10
  procedure, **11.6:** 11
  records, **11.6:** 12
  setup, **11.6:** 11, 11f.
Submersible pump NPSH test, **11.6:** 12
  acceptance criteria, **11.6:** 15
  closed-loop dry pit setup, **11.6:** 13f., 13
  closed-loop wet pit setup, **11.6:** 13, 14f.
  with flow rate held constant, **11.6:** 14
  objective, **11.6:** 12
  procedure, **11.6:** 14
  records, **11.6:** 15
  setup, **11.6:** 12, 12f., 13f., 14f.
  with suction head held constant, **11.6:** 14, 15f.
  suction throttling setup, **11.6:** 12f., 12
  variable lift setup, **11.6:** 13f., 13
Submersible pump performance test
  acceptance criteria, **11.6:** 9
  dry pit setup, **11.6:** 7, 8f.
  efficiency tolerance at specified flow rate, **11.6:** 9, 10t.
  flow rate tolerance at specified total head, **11.6:** 9, 10t.
  objective, **11.6:** 7
  pretest data requirements, **11.6:** 10
  procedure, **11.6:** 8
  records, **11.6:** 10
  setup, **11.6:** 7, 7f., 8f.
  test curve, **11.6:** 10, 11f.
  total head tolerance at specified flow rate, **11.6:** 9t., 9
  wet pit setup, **11.6:** 7, 7f.
Submersible pump tests, **11.6:** 1
  flow-measuring systems, **11.6:** 19
  gauges in head measurement, **11.6:** 24, 26f.
  instrument calibration intervals, **11.6:** 18,: 21t.
  instrument fluctuation and inaccuracy, **11.6:** 18, 21t.
  model tests, **11.6:** 27
  noncontact type flow meters in rate of flow measurement, **11.6:** 24
  pressure differential meters in rate of flow measurement, **11.6:** 22, 22t., 23t.
  pressure tap location for head measurement, **11.6:** 24, 25f.
  pump input power measurement, **11.6:** 25
  rotary speed measurement, **11.6:** 26
  rotating type flow meters in rate of flow measurement, **11.6:** 22
  routine production tests, **11.6:** 1
  standards-setting organizations, **11.6:** 32
  subscripts, **11.6:** 3t.
  symbols, **11.6:** 2t.
  temperature measurement, **11.6:** 27
  terminology and definitions, **11.6:** 1
  test conditions, **11.6:** 1
  test types, **11.6:** 1
  weirs in rate of flow measurement, **11.6:** 22
  witnessing of tests, **11.6:** 1
Submersible pump vibration test
  acceptance criteria, **11.6:** 18
  objective, **11.6:** 18
  procedure, **11.6:** 18
  pump support, **11.6:** 18
  records, **11.6:** 18
  setup, **11.6:** 18
  vibration instrumentation (transducer), **11.6:** 18, 19f.
  vibration limits, **11.6:** 18, 20f.
Submersible pumps, **1.1-1.2:** 5f., 6f., **2.1-2.2:** 2, 7f.
  special considerations, **2.4:** 9
Submersible vertical turbine pump intakes, **9.8:** 11, 14
Subscripts, **1.1-1.2:** 57t., **1.3:** 3t., **1.6:** 3t., **2.1-2.2:** 19, 21t., **2.3:** 3t., **2.6:** 3t., **3.1-3.5:** 16t., **3.6:** 4t., **6.1-6.5:** 22t., **6.6:** 1, 3t., **8.1-8.5:** 9t.
Sub-surface vortices, **9.8:** 26f., 27

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Suction, **3.1-3.5:** 33
loss of, **2.4:** 16, **5.1-5.6:** 37
pressure, **5.1-5.6:** 15
Suction and discharge pipes, **1.4:** 2
expansion joints and couplings, **1.4:** 7
flat faced flanges, **1.4:** 7
pipe support and anchors, **1.4:** 7
requirements, **1.4:** 7, 8
Suction conditions, **1.1-1.2:** 58, **1.3:** 57, **2.1-2.2:** 22, **2.3:** 18, **6.1-6.5:** 24, **8.1-8.5:** 10
Suction energy, **9.6.1:** 10, 5
determination, **9.6.1:** 3, 3f.
factors, **9.6.1:** 2
Suction energy level, **9.6.1:** 1
Suction nozzle, **9.1-9.5:** 5
Suction piping, **2.4:** 4
See also Discharge piping, Piping
eccentric reducers, **2.4:** 4, 4f.
elbows, **2.4:** 5
reducers, **2.4:** 4, 4f., 5
requirements, **2.4:** 4
strainers, **2.4:** 5
supports, anchors, and joints, **2.4:** 4
tanks, **2.4:** 5
valves, **2.4:** 5
Suction port, **3.1-3.5:** 4, **9.1-9.5:** 3
Suction pressure, **1.1-1.2:** 60, **8.1-8.5:** 7
Suction pumps, **1.1-1.2:** 4f.
datum elevations, **1.1-1.2:** 55f.
submersible, **1.1-1.2:** 5f.
Suction recirculation, **1.3:** 43, **9.6.3:** 5
centrifugal pumps, **9.6.3:** 5, 5f., 6f., 7f.
large boiler feed pumps, **9.6.3:** 8
vertical turbine pumps, **9.6.3:** 8, 8t.
Suction specific speed, **1.1-1.2:** 3f., 3, **1.3:** 32, 33f., 34f., 35f., 36f., **2.3:** 32, **9.6.1:** 1, **9.6.3:** 5
Suction system relationships, **6.1-6.5:** 41, 42f., 43f.
Suction tanks, **9.8:** 9
minimum submergence, **9.8:** 10, 10f., 11f.
multiple inlets or outlets, **9.8:** 11
NPSH considerations, **9.8:** 11
simultaneous inflow and outflow, **9.8:** 11
Sump volume
calculating, **9.8:** 54
decreasing by pump alternation, **9.8:** 57
minimum sequence, **9.8:** 55
operational sequences, **9.8:** 55, 56f.
pump and system head curves, **9.8:** 55, 56f.
Surface vortices
required submergence for minimizing, **9.8:** 29, 33f., 34f.
Swirl, **9.8:** 1
in the suction pipe, **9.8:** 27
meters, **9.8:** 27, 27f.
SWL *See* Static water level
Symbols, **1.1-1.2:** 56t., **1.3:** 1, 2t., **1.6:** 2t., **2.1-2.2:** 19, 20t., **2.3:** 1, 2t., 3t., **2.6:** 2t., **3.1-3.5:** 15t., **3.6:** 3t., **6.1-6.5:** 21t., **6.6:** 1, 2t., **8.1-8.5:** 8t., **9.8:** 38
Synchronous drive, **4.1-4.6:** 10
Synchronous magnet coupling, **4.1-4.6:** 11
System piping, **2.3:** 45
System preparation, **2.4:** 9
filling, **1.4:** 10
flushing, **1.4:** 10
pre-filling, **1.4:** 11
priming, **1.4:** 10
System pressure limitation, **1.3:** 22, **2.3:** 14
System ratings, **4.1-4.6:** 17
System requirements, **1.3:** 21
double suction pump specific speed, **1.3:** 32, 35f., 36f.
effects of handling viscous liquids, **1.3:** 23, 24f., 25f., 26f., 27f.
net positive suction head, **1.3:** 38–42
NPSH margin considerations, **1.3:** 39
NPSH reduction, **1.3:** 39, 40f., 41f.
NPSH reduction for liquids other than hydrocarbons or water, **1.3:** 40f., 41f., 42
NPSH requirements for pumps handling hydrocarbon liquids and water at elevated temperatures, **1.3:** 39, 40f., 41f.
NPSHA corrections for temperature and elevation, **1.3:** 38
pump selection for a given head, rate of flow, and viscosity, **1.3:** 28
pump versus system curve, **1.3:** 21, 21f.
reverse runaway speed, **1.3:** 22
shut-down, **1.3:** 22
single suction pump specific speed, **1.3:** 32, 33f., 34f.
starting with closed discharge valve, **1.3:** 22
starting with open discharge valve, **1.3:** 22
start-up, **1.3:** 22
suction specific speed, **1.3:** 32, 33f., 34f., 35f., 36f.
system pressure limitation, **1.3:** 22
torque curves, **1.3:** 23, 23f.
viscous liquid calculations, **1.3:** 30t., 31, 32t.
viscous liquid performance correction chart limitations, **1.3:** 23
viscous liquid performance curves, **1.3:** 30f., 30, 31f.
viscous liquid performance when water performance is known, **1.3:** 29, 30f., 31f.
viscous liquid symbols and definitions, **1.3:** 28
water hammer, **1.3:** 22

t *See* Temperature
Tachometers, **1.6:** 31, **6.6:** 18, **9.1-9.5:** 5
TAEH *See* Total available exhaust head
Tail rod, **6.6:** 3

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Tape recorders, **9.1-9.5:** 50
Temperature, **3.1-3.5:** 18, **4.1-4.6:** 13
  bearing, **1.3:** 75
  correction, **3.6:** 13
  Curie, **4.1-4.6:** 7, **5.1-5.6:** 14
  effects on NPSH and drive section, **5.1-5.6:** 25
  high, **5.1-5.6:** 24
  in hydrostatic test, **6.6:** 10
  instruments, **1.6:** 32
  internal rise, **4.1-4.6:** 20
  limits, **1.4:** 12, **5.1-5.6:** 13
  limits of magnets, **4.1-4.6:** 20
  limits on end suction pumps, **1.3:** 78, 78t.
  measurement, **1.4:** 11, **1.6:** 32, **3.6:** 20, **6.6:** 18
  rise in drive section, **5.1-5.6:** 13
  vs. time, **1.4:** 12, 12f.
Temperature buildup, **1.3:** 43
Temperature measurement and instruments, **2.6:** 32
Temperature monitoring, **9.6.5:** 3
  control limits, **9.6.5:** 5
  frequency, **9.6.5:** 5
  indicators, **9.6.5:** 23
  liquid film bearing and seal faces temperatures, **9.6.5:** 4
  means, **9.6.5:** 4
  motor winding temperature, **9.6.5:** 4
  pumped liquid temperature rise, **9.6.5:** 4
  rolling element bearing temperatures, **9.6.5:** 4
  sealless pump liquid temperature, **9.6.5:** 4
  sealless pump temperature damage, **9.6.5:** 5
  temperature sensitive fluids, **9.6.5:** 4
Temperature rise, **1.3:** 43, **9.6.3:** 2
  calculation, **1.3:** 43, 44f., 45f.
  and minimum flow, **1.3:** 46
  and pump performance, **1.3:** 44, 44f.
Terminology, **1.3:** 1, 2t., **1.6:** 1, **2.6:** 1–8, **3.1-3.5:** 15t., **3.6:** 2–6, **4.1-4.6:** 7–35
  alphabetical listing, **1.1-1.2:** 27t.–35t.
  numerical listing, **1.1-1.2:** 35t.–38t.
Tests, **1.6:** 1, **4.1-4.6:** 24, **5.1-5.6:** 39
  conditions, **1.6:** 1
  explanation, **3.6:** 1
  hermetic integrity, **5.1-5.6:** 39
  hermetic integrity test, **4.1-4.6:** 24
  inert gas sniffer test, **4.1-4.6:** 24
  mechanical integrity, **5.1-5.6:** 40
  objectives, **1.6:** 1
  reports, **5.1-5.6:** 40
  scope, **1.6:** 1
  torque confirmation test, **4.1-4.6:** 24
  types, **3.6:** 1
  winding integrity, **5.1-5.6:** 40
  winding temperature, **5.1-5.6:** 40
Thermal effects on NPSH and drive section, **5.1-5.6:** 25
Thermodynamic properties, **9.6.1:** 2
Thermoplastics, **9.1-9.5:** 25
Thermosetting polymers, **9.1-9.5:** 25
Thin square-edged orifice plate, **6.6:** 14
Thixotropic fluids, **3.1-3.5:** 22
Thrust bearings, **2.3:** 46
Thrust reversal on impeller, **9.6.3:** 3
Tie-down fasteners, **3.1-3.5:** 40
Time-independent non-Newtonian fluids, **3.1-3.5:** 22
Timing gear, **3.1-3.5:** 4, **9.1-9.5:** 5
Tin bronze, **9.1-9.5:** 20
Tin-base bearing metals, **9.1-9.5:** 23
Titanium alloys, **9.1-9.5:** 23
Top suction impellers, **1.3:** 20, 21f.
Torque, **5.1-5.6:** 12
Torque confirmation test, **4.1-4.6:** 24
Torque curves, **1.3:** 23, 23f.
Torque shafts, **3.6:** 18
Torsional critical speed, **9.6.4:** 4, 4f.
  calculation, **9.6.4:** 5
Torsional dynamic analysis, **9.6.4:** 5
Torsional dynamometer, **9.1-9.5:** 5
Torsional stiffness, **9.6.4:** 4, 5
Total available exhaust head, **1.3:** 12, **2.3:** 13
Total differential pressure, **6.1-6.5:** 22, **6.6:** 4, **8.1-8.5:** 7
  calculation, **6.6:** 9
Total discharge head, **1.1-1.2:** 57, **1.6:** 5, **2.1-2.2:** 21, **2.6:** 5, 5f., **11.6:** 5
  calculations, **1.6:** 15, **2.6:** 13
Total discharge pressure, **6.1-6.5:** 20, **6.6:** 4
  calculation, **6.6:** 9
Total gap, **4.1-4.6:** 8, **5.1-5.6:** 12
Total head, **1.1-1.2:** 57, 59, **1.6:** 5, **2.1-2.2:** 21, **2.6:** 5, 6, **11.6:** 5
  calculation, **1.6:** 15, **2.6:** 15
  effects of compressibility of liquid on, **1.6:** 5
  measurement, closed suction above atmospheric pressure (can pump), **2.6:** 29, 29f.
  measurement, open suction above atmospheric pressure (wet pit), **2.6:** 30, 30f.
Total head tolerance at specified flow rate, **11.6:** 9t., 9
Total input power, **3.6:** 5, **6.6:** 5
Total required exhaust head, **1.3:** 12, **2.3:** 13
Total suction head, **1.6:** 4, **2.6:** 4, 5, 5f., **11.6:** 4
  calculation, **1.6:** 15
  calculations, **2.6:** 13
  closed suction, **2.1-2.2:** 19
  closed suction test, **1.1-1.2:** 57
  net positive suction head available, **1.1-1.2:** 58
  net positive suction head required, **1.1-1.2:** 58
  open suction, **1.1-1.2:** 57, **2.1-2.2:** 19
Total suction lift, **1.6:** 5, **6.1-6.5:** 25, **6.6:** 4, **8.1-8.5:** 10
Total suction pressure, **6.1-6.5:** 20, **6.6:** 4
  calculation, **6.6:** 9

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Toxic liquids or vapors, **8.1-8.5:** 14
Toxicity ratings, **5.1-5.6:** 23
Transfer pumping, **1.3:** 4
Transfer service, **2.3:** 4
Transition manholes, **9.8:** 59
Transmission dynamometers, **1.6:** 30, 31, **9.1-9.5:** 5
Trash pumps, **1.3:** 14
TREH *See* Total required exhaust head
Trench-type intakes, **9.8:** 7, 8f., 9f.
  approach velocity, **9.8:** 9
  centerline spacing, **9.8:** 9
  end wall clearance, **9.8:** 9
  floor clearance, **9.8:** 9
  inlet conduit elevation, **9.8:** 9
  orientation, **9.8:** 9
  width, **9.8:** 9
Trench-type wet wells, **9.8:** 16f., 17
Troubleshooting, **2.4:** 15, **5.1-5.6:** 36
  See Malfunctions, causes and remedies
  excessive power consumption, **1.4:** 16, **5.1-5.6:** 37
  insufficient discharge, **2.4:** 15
  insufficient discharge flow, **5.1-5.6:** 36
  insufficient discharge flow or pressure, **1.4:** 16
  insufficient pressure, **2.4:** 16, **5.1-5.6:** 36
  little or no discharge flow, **1.4:** 16
  loss of suction, **1.4:** 16, **2.4:** 16, **5.1-5.6:** 37
  no discharge, **2.4:** 15
  no discharge flow, **5.1-5.6:** 36
  power consumption too high, **2.4:** 16
Turbine specific speed, **1.3:** 11, **2.3:** 12
Turbines *See* Pumps as turbines
Two plane balancing, **1.1-1.2:** 61
Type I performance test, **6.6:** 6
Type II performance test, **6.6:** 6
Type III performance test, **6.6:** 6
Type JM motors, **1.1-1.2:** 51t.
  having rolling contact bearings, **1.1-1.2:** 50f.
Type JP motors, **1.1-1.2:** 52t.
  having rolling contact bearings, **1.1-1.2:** 50f.

Unbalance, **9.6.4:** 20
  allowable residual in impellers, **9.6.4:** 21f., 22f.
  maximum looseness between balancing arbor and impeller, **9.6.4:** 23
Unconfined intakes, **9.8:** 14
  cross-flow velocities and pump location, **9.8:** 15
  debris and screens, **9.8:** 15
  submergence, **9.8:** 15
Units (pumps complete with mounting bases), **3.1-3.5:** 33
Units of measure, **1.3:** 1, 2t., **2.3:** 1, 2t., 3t., **3.1-3.5:** 15t., **9.1-9.5:** 7
  conversion factors, **9.1-9.5:** 8t.–10t.
  rounded equivalents, **9.1-9.5:** 7t.
  viscosity, **3.1-3.5:** 19
Universal joint, **9.1-9.5:** 5
Unloading, **2.4:** 1
US Customary units, **9.1-9.5:** 7
  conversion factors, **9.1-9.5:** 8t.–10t.
  rounded equivalents, **9.1-9.5:** 7t.

v *See* Plunger or piston speed
v *See* Velocity
Vacuum breaker piping, **9.1-9.5:** 5
Valve gear, **8.1-8.5:** 4
Valve gear adjustments, **8.1-8.5:** 4, 6f.
Valve plate type, **8.1-8.5:** 3, 3f.
Valve pot type, **8.1-8.5:** 3, 4f.
Valve seat area, **6.1-6.5:** 24, 24f., 25f.
Valve setting, **2.4:** 10
  discharge valve position (high or medium head pumps), **2.4:** 11
  discharge valve position (mixed or axial flow pumps), **2.4:** 11
  reduced flow/minimum flow discharge bypass, **2.4:** 11
  at start-up, **2.4:** 11
  warning against closed valve operation, **2.4:** 10
Vane pumps, **3.1-3.5:** 1f., 1
Vane-in-rotor pumps, **3.1-3.5:** 1f., 2, 2f.
Vane-in-stator pumps, **3.1-3.5:** 1f., 2, 2f.
Vapor, **2.3:** 21
Vapor *See* Liquids with vapor or gas
Vapor pressure, **3.1-3.5:** 23
Variable speed drives, **1.3:** 77, **2.3:** 45
Variable speed pumps, **9.8:** 58
Variable viscosity, **4.1-4.6:** 14
Vegetable oils, **9.1-9.5:** 11
Velocity, **8.1-8.5:** 7, **9.8:** 1
Velocity head, **1.1-1.2:** 55, **1.6:** 4, **2.1-2.2:** 19, **2.6:** 4, **11.6:** 4
Velocity pressure, **3.1-3.5:** 16, **3.6:** 4, **6.1-6.5:** 22, **6.6:** 4, **8.1-8.5:** 9
Velocity profiles, **9.8:** 27
Vent piping, **9.1-9.5:** 5
Venting, **5.1-5.6:** 18
Venturi meter, **6.6:** 14, **9.1-9.5:** 5
Vertical diffuser pumps (excluded), **1.4:** 1
Vertical hollow shaft drivers, **2.4:** 6
Vertical mounting, **5.1-5.6:** 21
Vertical pump materials, **9.1-9.5:** 16
Vertical pump tests, **2.6:** 1
  conditions, **2.6:** 1
Vertical pumps, **2.4:** 1, **6.1-6.5:** 1, 2f.
  bearing and spacing types, **2.3:** 42
  classification by configuration, **2.1-2.2:** 2
  classification by impeller design, **2.1-2.2:** 2
  definition, **2.1-2.2:** 1
  drivers, **2.3:** 45
  enclosed lineshaft, **2.3:** 43

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

final alignment check, **2.4:** 8
flexibility of design, **2.3:** 1
foundation, **2.3:** 45
impeller types, **2.3:** 44
intake system design, **2.3:** 46
leveling and plumbness, **2.4:** 3, 3f.
locating, **2.4:** 3
lubrication systems, **2.3:** 43
open lineshaft, **2.3:** 43
operating, **2.4:** 9
pre-lubrication, **2.4:** 8
shafting, **2.3:** 43
system piping, **2.3:** 45
types, **2.1-2.2:** 1, 4f.
typical applications, **2.3:** 1–13
vibration, **2.4:** 12
Vertical solid shaft drivers, **2.4:** 6
Vertical solid-shaft motor dimensions (HP and HPH), **1.1-1.2:** 53f., 53t., 54t.
Vertical turbine pumps, **9.6.1:** 6
and inlet eye diameter, **9.6.1:** 4
and NPSH margin, **9.6.1:** 6
Vertical turbine short set pumps, **9.6.2:** 17
force analysis, **9.6.2:** 17
loading examples, **9.6.2:** 32
nozzle loads, **9.6.2:** 17, 18f., 19f.
terminology, **9.6.2:** 17
Vertical volute pump installation
alignment, **1.4:** 9
configurations, **1.4:** 8
couplings, **1.4:** 9, 10
discharge piping requirements, **1.4:** 8
flexible or line shaft configuration, **1.4:** 8
grouting, **1.4:** 8
in-line configuration, **1.4:** 8
mounting to support structure, **1.4:** 9
pump leveling and plumbness, **1.4:** 8
separately coupled configuration, **1.4:** 8
solid shaft coupling, **1.4:** 10
stuffing-box steps, **1.4:** 10
suction piping requirements, **1.4:** 8
v-belt drive, **1.4:** 10
wet pit configuration, **1.4:** 8
Vertical-in-line pumps
adjustment factors, **9.6.2:** 11, 14t.
flange stress, **9.6.2:** 10
material specifications, **9.6.2:** 13t.
nomenclature, **9.6.2:** 10, 10f.
nozzle loads, **9.6.2:** 10, 12t.
pressure-temperature, **9.6.2:** 10
Vibration, **1.4:** 17, **2.4:** 12, **5.1-5.6:** 35, **9.6.3:** 2
checking, **2.4:** 11
dynamics, **9.6.4:** 1
factors affecting, **9.6.4:** 20
field values, **9.6.4:** 8, 9f.–19f.
frequencies and methods of determination, **9.6.4:** 1
measurements, **9.6.4:** 7
probe locations, **9.6.4:** 8, 9f.–19f.
Vibration check, **1.4:** 13
Vibration monitoring, **9.6.5:** 8
bearing housing vibrations, **9.6.5:** 8
control limits, **9.6.5:** 9
frequency, **9.6.5:** 9
indicators, **9.6.5:** 22
means, **9.6.5:** 8
proximity probe, **9.6.5:** 9
shaft vibrations, **9.6.5:** 8
on vertical pumps, **9.6.5:** 9
Vibration test. *See* Submersible pump vibration test
Viscometers, **3.1-3.5:** 19
Viscosity, **3.1-3.5:** 19, **4.1-4.6:** 13, **5.1-5.6:** 25
apparent, **3.1-3.5:** 19
dynamic, **3.1-3.5:** 19
effect on pump and system performance, **3.1-3.5:** 23
high, **3.1-3.5:** 14, **4.1-4.6:** 13
kinematic, **3.1-3.5:** 19
low, **3.1-3.5:** 14, **4.1-4.6:** 13
units of measure, **3.1-3.5:** 19
variable, **4.1-4.6:** 14
Viscous input power, **1.3:** 30
Viscous liquids
calculations, **1.3:** 30t., 31, 32t.
correction chart limitations, **2.3:** 22
effects of handling, **1.3:** 23, 24f., 25f., 26f., 27f.
handling, **2.3:** 22, 25f., 26f., 27f., 28f.
performance correction chart limitations, **1.3:** 23
performance correction charts, **2.3:** 25f., 26f., 27f., 28f.
performance curves, **1.3:** 30f., 30, 31f.
performance when water performance is known, **1.3:** 29, 30f., 31f.
pump performance when performance on water is known, **2.3:** 30, 30f., 31t., 31f., 32t.
pump selection for given head and rate of flow, **2.3:** 25f., 26f., 26, 27f., 28f.
pump selection for given head, rate of flow, and viscosity, **1.3:** 28
symbols and definitions, **1.3:** 28, **2.3:** 22
Viscous response types, **3.1-3.5:** 19–22
VOCs *See* Volatile organic compounds
Volatile liquid pump, **1.3:** 3
Volatile liquids, **5.1-5.6:** 24
Volatile organic compounds, **9.6.5:** 6
Volume, **1.6:** 3, **2.6:** 1, **11.6:** 3
Volume units, **6.6:** 1
Volumetric efficiency, **6.1-6.5:** 23, **6.6:** 5
calculating for hydrocarbons, **6.1-6.5:** 47–53
calculating for water, **6.1-6.5:** 45–47, 48t., 49t.
water compressibility, **6.1-6.5:** 47, 48t., 49t.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

Volute pumps
  calculation for radial thrust, **1.3:** 58
  calculation of axial thrust for enclosed impellers, **1.3:** 60–63
  circular casings, **1.3:** 60, 60f.
  dual volute casing, **1.3:** 58, 59f.
  K versus rate of flow (double volute casing), **1.3:** 58, 59f.
  K versus rate of flow (single volute casing), **1.3:** 58, 59f.
  single volute casing, **1.3:** 58, 58f.
Vortices, **9.8:** 1
  free surface, **9.8:** 1, 26, 26f.
  required submergence for minimizing surface vortices, **9.8:** 29, 33f., 34f.
  submerged, **9.8:** 1
  sub-surface, **9.8:** 26f., 27

Wastewater, **9.1-9.5:** 61
Wastewater service pumps, **1.3:** 14
Watches and magnets, **4.1-4.6:** 19, **5.1-5.6:** 32
Water compressibility, **6.1-6.5:** 47, 48t., 49t.
Water hammer, **1.3:** 22, **1.4:** 13, **2.4:** 11
  analysis, **2.3:** 14
Water lubricated pumps, **2.3:** 44
Water/wastewater pumps, **9.6.1:** 8
Waterflood (injection) pumps, **9.6.1:** 10
Wear plates, **1.4:** 15
Wear rings, **1.4:** 15, **2.4:** 14
  arrangements, **2.1-2.2:** 12f.
Welding, **5.1-5.6:** 20, **9.1-9.5:** 12
Well pumping, **1.3:** 4
Well service, **2.3:** 5
Wells, **2.4:** 2, 2f.
  checking, **2.4:** 2
  draw-down, **2.4:** 11
Wet critical speed, **9.6.4:** 2
Wet pit pumps, **2.3:** 1
Wet pit, short setting or close-coupled (lineshaft) pumps, **2.1-2.2:** 1, 9f.
Wet pit volute pumps, **1.1-1.2:** 14f.
  total suction head, **1.1-1.2:** 57
Wet wells (solids-bearing liquids), **9.8:** 15
  cleaning procedures, **9.8:** 17
  confined inlets, **9.8:** 16
  trench-type, **9.8:** 16f.
  vertical transitions, **9.8:** 16
  wet well volume, **9.8:** 17
Winding temperature test, **5.1-5.6:** 40
Working pressure, **1.1-1.2:** 60, **2.1-2.2:** 23

Yellow brass, **9.1-9.5:** 20
Yield point, **3.1-3.5:** 22
Yield value, **9.1-9.5:** 6

Z *See* Elevation head
Z *See* Elevation pressure
Zinc and zinc alloys, **9.1-9.5:** 23
Zirconium, **9.1-9.5:** 23

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

This page intentionally blank.

Copyright © 2002 By Hydraulic Institute, All Rights Reserved.

| 2002